

**MEMS INS & GNSS INTEGRATED
NAVIGATION BY USING HEADING OF
AHRS-MAGNETOMETER DURING GNSS
SIGNAL OUTAGES**

**IMPROVED ONLINE HEADING
INFORMATION SOLUTIONS REALIZED
THROUGH GNSS/INS/MAGNETOMETER
INTEGRATED NAVIGATION SYSTEM**

A thesis submitted to the
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By:

Muhammad Iqbal

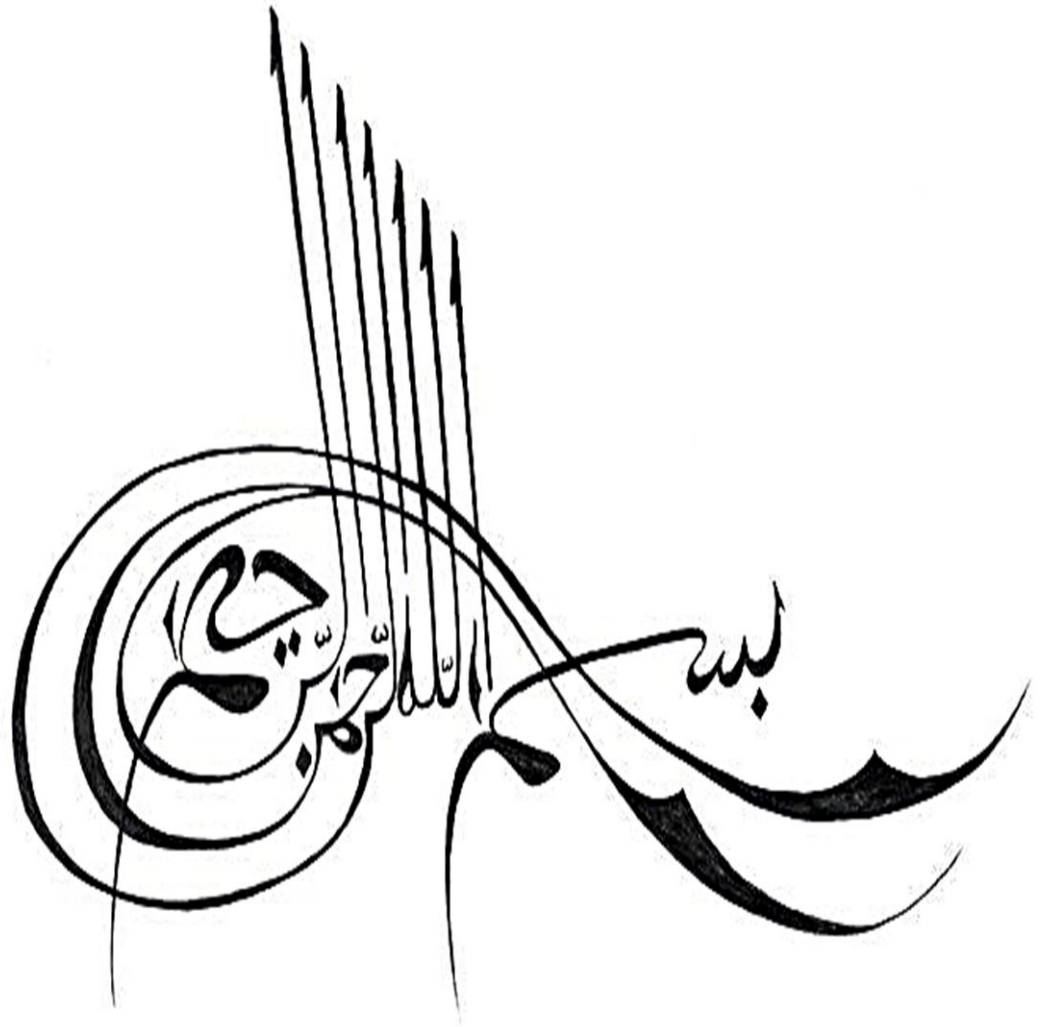
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March 2020**



In the name of Almighty Allah the most beneficent the most merciful

Institute of Space Technology
Department of Aeronautics & Astronautics



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ABSTRACT

These days, inertial navigation system (INS) manufactured by using MEMS technology, are used for many crucial applications. To obtain the optimal navigation solution they are integrated with GNSS to obtain the optimal navigation results, as the both navigation systems are complementary. However, MEMS based inertial navigation system (MEMS INS) have low cost, miniature in size and less weight but they have poor accuracy. Heading estimation for navigation solutions is one of the most important feature for moving objects. GNSS (Global Navigation Satellite System) is universally used navigation system and it is used extensively for advance navigation applications these day. GNSS can also be used for heading calculation during advanced navigation. Heading means “where an object is pointing with respect to true north on the horizontal plane” (Inertial Navigation System (INS) applications heading is also named as yaw). During navigation application, the GNSS may experience signal blockage due to interference, signal obstruction, congestion, multipath and spoofing. Therefore, for critical nature applications, stand-alone GNSS cannot serve the best during integrated navigation of MEMS INS and GNSS.

Since, Magnetometer is widely used low cost inertial sensor which can estimate the yaw from earth’s magnetic north by sensing the magnetic field of globe. In many inertial navigation applications, the magnetometer is one of the reliable sensor, which have a good heading observation. By using magnetometers for heading estimation during navigation applications can provide good accuracy. The external magnetic field & some additional sources of error can also affect the readings of magnetometer.

The calibration of magnetometer is challenging task. To calibrate it some precautionary measurements are required. To accurately calibrate the magnetometer a magnetic free environment is required. Different methods were devised for its calibration in lab environment. In this dissertation a fast and autonomous online calibration technique is proposed and vetted through experiments. This technique is fast and required little space for calibration. The parameters estimated from this calibration technique are used online for calibrated magnetometer components measurements. The heading is also estimated online by using these magnetic components. This heading is used during MEMS INS & GNSS integrated navigation when GNSS signals are not available due to blockage.

In this thesis, the navigation solution will be obtained by the integration of low-cost MEMS INS, GNSS and AHRS-Magnetometer. Magnetometer heading will be used during GNSS signal outages. The result will be an improved navigation system, even if the GNSS is not available for a short duration.

This work is carried out with the aspirations to develop a low-cost MEMS based navigation system which is augmented & added by GNSS and Magnetometer which can serve at its best during advanced navigation.

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Dedicated

To my parents

To my wife

To my children Muhammad and Ali.

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LIST OF ABBREVIATIONS

C/A	Coarse Acquisition
CDMA	Code Division Multiple Access
DGPS	Differential GPS
DOP	Dilution of Precision
EKF	Extended Kalman Filter
ENU	East, North, Up
ECEF	Earth Cantered Earth Fixed
GLONASS	Globalnaya Navigazionnaya Sputnikovaya Sistema (Russian Word)
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
GDOP	Geometric Dilution of Precision
GPST	GPS Time
IGRS	International Geodetic Reference System
HDOP	Horizontal Dilution of Precision
IERS	International Earth Rotation Service
IRNSS	Indian Regional Navigation Satellite System
ITRF	International Terrestrial Reference Frame
PPP	Precise Point Positioning
PPS	Precise Positioning Service
PRC	Pseudo-Range Corrections
PRN	Pseudo Random Noise
QZSS.	Quasi-Zenith Satellite System (Japan)

RTCM	Radio Technical Commission for Maritime Service
RTK	Real-Time Kinematic
RINEX	Receiver Independent Exchange Format
SPS	Standard Positioning Service
TDOP	Time Dilution of Precision
TAI	International Atomic Time
UERE	User Equivalent Range Error
US	United States
US-DOD	United States of America - Department of Defence
UT	Universal Time
VDOP	Vertical Dilution of Precision
ZUPT	Zero update

1. INTRODUCTION

MEMS based inertial navigation system is commonly used for navigation of land vehicles. They can give good results for short navigation applications as their errors are accumulated with time. To keep the errors in bound and to attain optimal navigation solution for land vehicles navigation MEMS-INS is integrated with GNSS. The accumulated errors of MEMS based navigation systems are mitigated by GNSS updates during integrated navigation. GNSS may suffer difficult environments like signal obstruction, multipath or jamming and the integrated solution obtained by using combined MEMS INS and GNSS remains in deft state. The reason of this phenomena is that GNSS signals are obstructed by building structures partially or completely. Consequently, to mitigate the MEMS INS errors by using GNSS updates no information being available. MEMS INS have poor performance for long endurance. If no updates are available from GNSS, then errors due to MEMS sensors will increase gradually and the reliability of navigation solution will be compromised. Heading is a most important parameter for navigation system. External heading information is required for optimal navigation. GNSS is also be used as external heading source. When GNSS experience the outages, the heading can be provided to navigation system. Some other reliable source is required for heading updates. Hence, Magnetometer is one of the reliable source for heading information. Therefore, Magnetometer is added in MEMS INS and GNSS integrated navigation.

The overview of research background is provided in this chapter. MEMS INS importance related detailed discussion is also provided. MEMS INS advantages and disadvantages are discussed. GNSS advantages and disadvantages are also discussed. The reason of GNSS integration with MEMS INS is described in detail. The limitations of integrated navigation system comprises on MEMS-INS and GNSS is described in

detail. The reason of GNSS signals outages are given and viable solution is also provided. The research methodology is described at the end of this chapter.

1.1 Background

Inertial navigation systems manufactured by using MEMS sensors are widely used for navigation purposes since early nineteen. The inertial sensors used in INS have been improved with passage of time. Since inertial navigation system is composed of gyroscopes and accelerometers. Gyroscope was invented in 1852 by a French physicist Leon Foucault. Initially mechanical gyroscopes were developed and used. After 1960 ring laser gyroscopes were developed and used in INS. After a decade fiber optics gyroscopes were manufactured and utilized in navigation systems. The idea of MEMS sensors was first proposed in 1960s. Analog Devices Corporation produced an accelerometer in 1991, where mechanical & electronic portions were integrated on the same chip. Late 1990's and early 2000's improved MEMS navigation systems were developed. These days, more accurate, light in weight, having low power consumption and low cost MEMS based inertial navigation systems are available. For short applications of navigation these systems are reliable. But for long endurance and long navigational applications they cannot serve as best as required. As the errors in inertial sensors accumulated and are not bound therefore, MEMS INS and GNSS based integrated navigation is best option for optimal navigation.

Due to current advancements in developments of low cost MEMS inertial sensors the manufacturing of light weight, small size low cost inertial navigation systems. This made the MEMS INS more attractive option for navigation applications like vehicular and pedestrian navigations. They become an attractive grade inertial measurements unit due to their property of having low cost, small size and short power consumptions.

However, some parameters are need to be refined and mitigated like their in-run biases, high noise and scale factors instability [40].

Like tactical grade inertial navigation system, MEMS based INS are self-sufficient. The environment like urban canyons do not degrade their performance. They do not dependent on external electromagnetic signals. Additionally, they are more accurate in the short term. It can supply data continuously with very high rate at several hundred Hz. They also provide accurate attitude information of the object on which they are installed.

MEMS inertial navigation system performance is degraded with time as the errors of sensors are not bounded. These errors are accumulated with time and for long distance navigation the error budget does not remain in required limit. To keep the errors in bound and in acceptable level regular updates like velocities, position and time are required from an external source. GNSS measurements like position, velocity & time are used to update the INS sensors outputs [37].

To obtain optimal navigation solution, MEMS inertial navigation system is integrated with satellite navigation system. In this type of integration combined navigation solution is utilized for optimal results. GNSS errors are bounded and it provide all weather navigation, position, velocity and time information to the users worldwide. These updates are uses for inertial systems error mitigation. Currently, GPS, GLONASS, BEIDOU & GALILEO are fully operational and more widespread GNSS systems They provide position, navigation, velocity & time information to user at globe and even in space. Sometimes these parameters are transformed in latitude, longitude, velocity & height etc.

During navigation of vehicle in urban or mountainous area, the GNSS signal are degraded due to signal obstruction from local topography and buildings. The GNSS signals may experience multipath, spoofing, jamming and signal blockage. As a result position, navigation or time estimations cannot be obtained by GNSS.

In order to get position, velocity or navigation and time from GNSS, a good visibility to satellites is required. The blockage of GNSS signals may produce multipath phenomena and weakens the satellite availability. Thus the navigation solution may not available and optimal integrated navigation solution cannot attained. Thus errors arose in MEMS INS cannot be bounded due to GNSS outages. Therefore, an accurate and continuous navigation in urban environment cannot be guaranteed by utilizing GNSS during MEMS INS and GNSS integrated navigation applications.

In these situations, MEMS INS-GNSS integrated navigation system must be augmented with other reliable system to ensure continuous and accurate navigation solution during GNSS signal outages [37].

An integrated navigation system comprises on MEMS INS & GNSS which is also added by AHRS-Magnetometer is able to provide navigation solution during GNSS gaps. Magnetometers provide additional heading information are often integrated with the IMU to produce the attitude and heading reference system (AHRS). Often, for manufacturing an AHRS, a three axis magnetometer is mounted in conjunction with an IMU. In this manner the roll and pitch of the INS can be used to orient the magnetometer data in the local level frame. From this method, the heading can be computed and used to update the system. The success of multi-sensor fusion technology has meant that manufacturers of AHRS are more commonly including magnetometers within the IMU triad [2].

Magnetometers are often used to control the direction of travel for navigation systems. Magnetometers require calibration of the soft and hard iron biases and constant field monitoring to detect magnetic interference. Most often magnetometers simply provide heading updates to the integrated navigation system. [22] Introduced a different technique of quasi-static magnetic field (QSF) measurements to update the entire attitude in addition to estimate the gyro biases. [11] Proposed a fast method for 3D Magnetometer calibration using little space for pedestrian applications.

This thesis explores the integration of MEMS-INS, GNSS & AHRS-Magnetometer for pedestrian navigation or vehicular navigation in urban environment. Magnetometer is integrated with MEMS INS & GNSS to achieve the heading information when GNSS signals are not available during pedestrian navigation or vehicular navigation and also in general land navigation.

1.2 Previous Work

A lot of research has been performed in the field of integrated navigation by using MEMS-INS and GNSS especially GPS. Subsequently, GPS is most common and extensively used global navigation system. During integrated navigation the GPS is not available for some time due to signal obstructions in urban areas. Therefore, MEMS INS stand alone cannot serve the best results for navigation in urban areas. Since the INS errors are not bounded and accumulate with passage of time. The position, velocity and attitude outputs provided by MEMS INS have some errors. These errors have to be minimized during navigation applications for accurate navigation. An aided navigation system can mitigate the INS errors for optimal navigation. Therefore, an aided system used for MEMS-INS and GNSS during integrated navigation is required to provide assisted navigation during GNSS outages.

High-end inertial sensors were used and investigated by several researchers (e.g. [42, 45, 48] etc.). During GPS outages, the use of tactical grade INS allows a sub-metric position accuracy level. It is confined to generally technical high-accuracy applications due to their price and size. Several novel researchers have investigated the integration of GNSS systems with MEMS based INS for the last few years. Which are characterized by low cost and low performance, for instance [35, 41, 43, 45].

All the researchers have focused their work on the use of GPS system in differential mode. Which involves the deployment of a base station not far from the area where the navigation takes place. In case of real-time application, a radio-modem linking the base and the rover must be included too. The need of the mentioned surplus equipment can be considered an important limitation of these works.

In [41, 43] it is proposed during GNSS outages, the use of motion constraints to bound the INS errors. The motion constraints are applied to vehicular navigation case. They are based on the assumptions that the vehicle does not slip and always remains in contact with the ground. So, the velocity of the vehicle in the plane perpendicular to the forward direction is approximately zero. In [45] with a medium accuracy IMU, the effectiveness of velocity constraints is demonstrated. In addition to the velocity constraints, [35] adopted a height constraint. Which based on the assumption that the height solution does not typically vary by more than few tens of meters in a small region. With an IMU in vehicular application, [39] demonstrated the motion constraint effectiveness.

1.3. Objectives

Main objective of this research work is to devise a method for magnetometer calibration in lab and find compensated parameters to find heading for navigation applications. This thesis also describes the magnetometer online calibration technique to find magnetometer parameters during online applications. The calibration parameters of magnetometer are found for each roll and pitch sectors. These parameters are transformed from navigation frame to body frame and are used to find heading for online navigation solutions. Magnetometer heading is utilized for optimal navigation during MEMS INS & GNSS integrated navigation when GNSS is not available due to signals obstruction. Kalman filter is used to update the navigation solution and to fuse the data. Simple loose coupling integration scheme is applied for integration of MEMS INS, GNSS & AHRS-Magnetometer. The scale factors and biases of inertial sensors are estimated and modeled using Kalman filter.

1.4 Research Methodology

The methodology of loose integration scheme for integration of MEMS INS, GNSS & Magnetometer is used. 15 state Kalman filter is used which is comprises on sensors biases, scale factors, roll, pitch, yaw, GNSS position & velocity. By adopting these techniques following performance will be analyzed. The main objectives of this thesis by adopting the research methodology is

- ◆ To analyze the MEMS-INS performance
- ◆ To evaluate the GNSS performance
- ◆ To analyze the performance of AHRS
- ◆ To analyze the performance of Magnetometer
- ◆ To perform online calibration of Magnetometer
- ◆ To calculate magnetometer heading by using its parameters
- ◆ To analyze the INS/GNSS based simple integrated performance
- ◆ To use the heading of Magnetometer during GNSS signals outages
- ◆ To analyze the performance of navigation solution of simple integrated, GNSS outage and compensated by AHRS-Magnetometer

2 LITERATURE OVERVIEW

This chapter discuss the literature overview about navigation, navigation systems, and types of navigation, GNSS, IMU/INS and main constituents of navigation systems.

2.1 Navigation

Navigation is a method or technique through which position, speed, attitude and orientation of an object is estimated.

Definition of navigation according to dictionary of Merriam Webster is “the science of getting ships, space crafts or aircraft from space to space and the method of determining position, course and distance travelled”.

According to your dictionary website, “Navigation is defined as the act or process of planning out a route or directing the travels of a boat, ship, car, person etc.”

Wikipedia pronounces that, “navigation focuses on controlling and monitoring the movement of a vehicle or craft from one point to another point. Four general categories of navigation are as follows.

- Navigation on land
- Oceanic navigation
- Aeronautic navigation
- Space navigation

Generally, navigation is science or art of moving the ships or watercraft from one place to another safely. These days, the Navigation term is used in broader sense. A system of navigation define as a system providing useful information to determine the position of a boat, car or airplane etc.

The word “navigation” contains 2 basic perceptions [17].

1. Estimation of a moving body's position and velocity with respect to a known reference
2. Planning and maintenance of a course from one location to another.

2.1.1 Navigation System

A computing system that aids in navigation is called a navigation system. The system which estimates orientation, velocity and position of moving body is called as navigation system. There are different types of navigation systems, which are as follows.

- Inertial navigation & guidance system
- Satellite navigation system
- Automotive navigation system
- Surgical navigation system
- Robotic mapping
- XNAV

Navigation system can determine the existing position of body by measuring changes in its velocity and acceleration. They determine automatically position and velocity. They are navigation aid. They can provide angular rate, acceleration and some or all of the attitude. They may be aboard the navigating body or may be located elsewhere e.g. GNSS [17]. Sometimes a combination of these two systems is used for navigation.

2.1.2 Navigation Solution

Navigation solution is the output of a navigation system. Generally, navigation solution represents the orientation, attitude and position or velocity in coordinate frame of the navigating body. Some navigation systems outputs latitude, longitude, height and heading as navigation solution. The orientation and attitude can be expressed as pitch, roll and

heading of navigating body. The navigating body can be an aircraft, a car, a ship or a person navigating with respect to some determined coordinate frame. For navigation application earth is common reference frame [17].

2.1.3 Types of Navigation

2.1.3.1 Position Fixing

Position fixing is a navigation technique in which position of object is fixed with current location. There are a number of position-fixing methods. One of a well-known method of position fixing is feature matching, which compares features at the current location. These features may be landmarks, way points, or terrain height through a map to determine the current position. This position fixing method is easier for humans than for a machine.

Position fixing can also be attained by assessing the ranges or bearing to known objects. Which is demonstrated for the two-dimensional case by Figure 2.1. Whereby X denotes the user unknown position. Where A & B are the known positions of two reference objects [17].

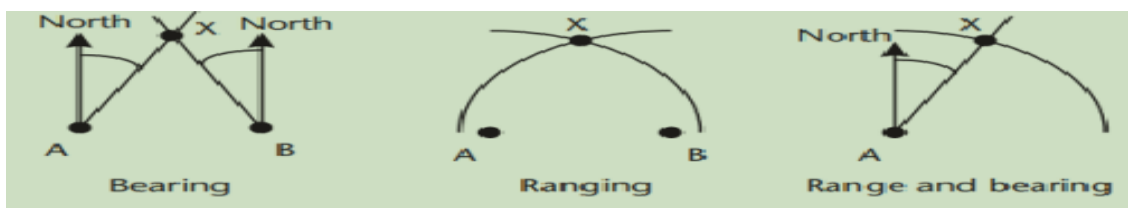


Fig 2.1 Ranging and bearing position fixing method

2.1.3.2 Dead Reckoning

Current position is computed during navigation applications using a previously determined position and estimated velocity & heading over elapsed time. Dead reckoning (possibly derived from “deduced reckoning”) assesses the change in position of moving object or it measures the velocity and integrates it. In order to attain

the current position of navigating body, dead reckoning is added to its previous position. The moving object's speed or traveled distance is assessed in the body coordinate frame. Therefore, a distinct attitude measurement is required to obtain the direction of travel in the subject reference frame [17].

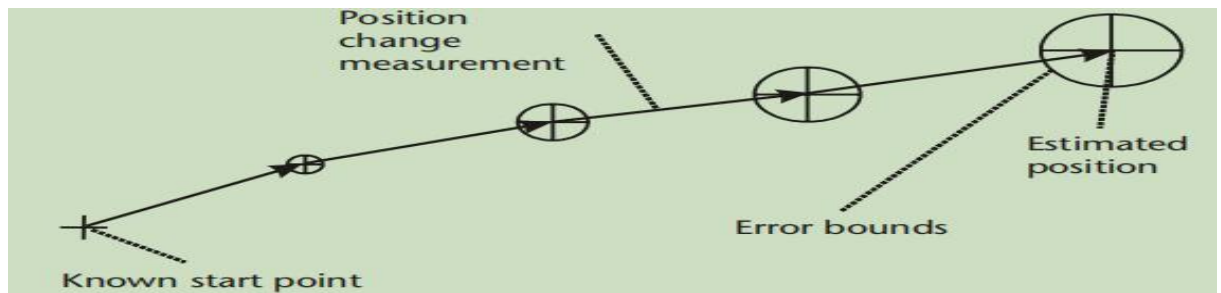


Fig 2.2 Dead reckoning

2.1.3.3 Pilotage

Visually recognizing landmarks to determine position and heading. Piloting or *pilotage* is a technique of navigation on water or in air by using fixed points of reference on the sea or on land. Generally, with reference to a nautical chart or aeronautical chart to attain a fix of the position of the vessel or aircraft with respect to a desired course or location. Pilotage or Piloting is the most common method of air navigation. In this method, the pilot keeps on course by following a series of landmarks on the ground.



Fig 2.3 Pilotage

2.1.3.4 Celestial Navigation

Computing position or attitude by estimating angles between local vertical or horizon and the line to known celestial objects (Stars). Celestial Navigation is the science of finding the way by using the sun, stars, moon, and planets.

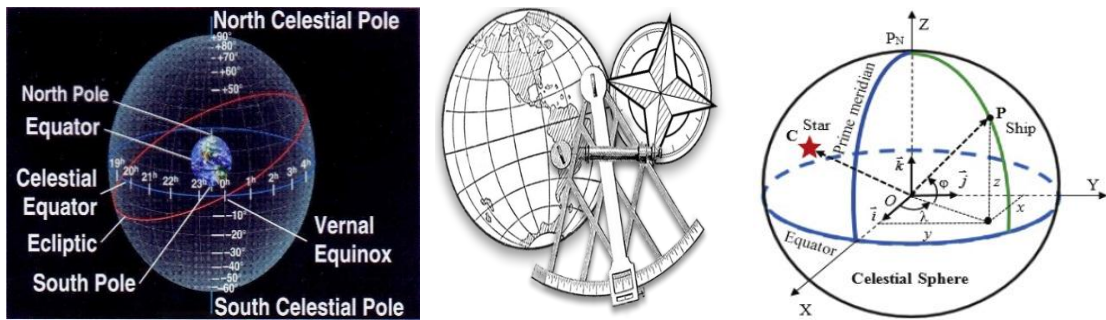


Fig 2.4 Celestial navigation

2.1.3.5 Radio and Satellite navigation

Radio navigation relies on transmitting-beacons with known locations on earth. For example LORAN/ Pseudolites use beacons on ground. Whereas, satellite navigation relies on transmitting-beacons or satellites above earth. For example GPS, GLONASS, BEIDOU and GALILEO are transmitting-beacons above earth.

Since 1930, radio navigation has been used. Some basic navigation techniques are as follows

- marker beacons
- direction finding
- Bearing elevation
- passive ranging
- two-way ranging
- hyperbolic ranging
- Doppler positioning



Fig 2.5 Radio & Satellite Navigation

2.1.3.6 Inertial Navigation

Inertial Navigation relies on inertial sensors. It comprises three rate sensors named as gyroscopes and three acceleration sensors named as accelerometers. Actually accelerometers measure the specific force in terms of gravity. They can calculate position and heading with respect to some reference system. Therefore, they provide navigation solution of the body where they are installed. The navigation solution is obtained by integrating the angular rates and acceleration provided by the inertial sensors. Therefore, navigation system can provide the roll, pitch & yaw in angular rates which are named as attitude of body. They also provide acceleration, by integrating acceleration we can find velocity and from the integration of velocity we can find the position of the body. The inertial navigation systems are self-sufficient as they do not rely on external source for navigation solution calculations.

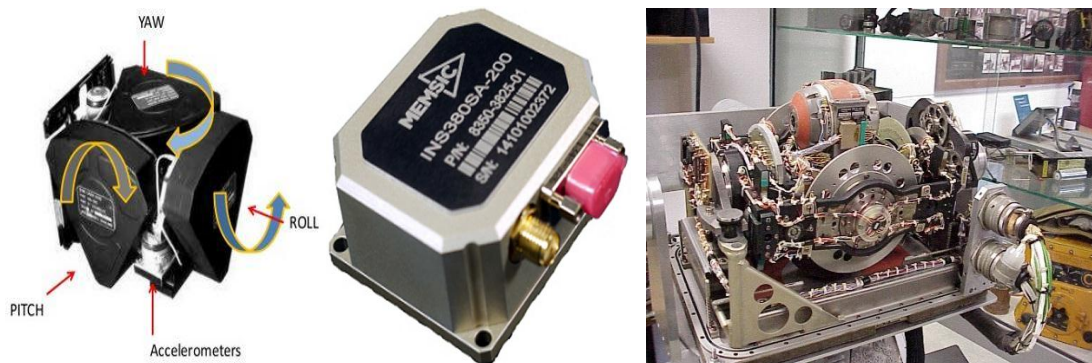


Fig.2.6 Inertial Navigation System

2.2 Coordinate Frames

To express the position and orientation of an object, body or navigation system w.r.t. some reference frame a system of coordinates is used called as coordinate frames. It can also be defined as a coordinate frame is a certain number of reference frames required for computing & processing the Navigation Information. Any reference frame is a right-handed set of three mutually orthogonal axes with defined origin. Important part of navigation system consists of finding the relationship between different frames. There are two ways that a coordinate frames may be expressed as

- An origin is provided by them
- They consists of a set of axes.
- The motion of objects may be described by using set of axes.

The position and orientation of an object is also defined by the coordinate frames [1].

Here some important coordinate frames are given below which are used for navigation. They are also mutually transformed.

- ◆ ECI frame or Earth Centered Inertial Frame
- ◆ ECEF frame or Earth Centered Earth Fixed Frame
- ◆ Local Level Frame or Local Navigation Frame
- ◆ Wander Frame
- ◆ Computational Frame
- ◆ Body Frame
- ◆ Orbital Coordinate System

2.2.1 ECI Frame or Earth Centered Inertial Frame

A non-rotating frame w.r.t fixed stars is known as inertial frame. Center of earth is origin of inertial frame. Axes of inertial frames are defined w.r.t fixed stars. Therefore, ECI frame is stationary in space or have no acceleration due to moving at constant velocity. The important features of ECI frame are described below

- i. The center of the Earth is its origin.
- ii. Along the axis of the Earth's rotation is z-axis.
- iii. In the equatorial plane pointing towards the vernal equinox is x-axis
- iv. Right handed system completes y-axis.

2.2.2 ECEF Frame or Earth Centered Earth Fixed Frame

ECEF is alike to the inertial frame. Same origin & z-axis is shared by this frame. It rotates along with earth. The ECEF is represented by symbol e. Earth's epicenter which is center of mass is origin. It can be explained as follow.

- i. Center of mass of earth is its origin.
- ii. Its z-axis is along the conventional terrestrial pole (CTP).
- iii. Through the intersection of the equatorial plane the x-axis passes and also through the reference meridian.
- iv. In the equatorial plane, the right-hand coordinate system is completed by y-axis

2.2.3 Navigation or Local Level Frame

The frame which has origin at navigation system's center of mass is called local level frame. It is also called navigation frame. Its other names are geodetic frame or geographic frame. It is represented by symbol 'n'. It is most common frame used for

navigation. It represent the navigation system's velocity and attitude on or near the surface of earth. Some major features of local level frame are as follows.

- i. The origin of local level or navigation frame corresponds with the sensor frame's center of mass.
- ii. Toward true north, y-axis of local level frame points.
- iii. Towards east, x-axis of local level frame points.
- iv. By pointing upward, z-axis which is vertical to reference ellipsoid and it also completes coordinate systems according to right hand rule.

2.2.4 Wander Azimuth Frame

Since singularities may occur at poles during computations for navigation applications. To avoid singularities at poles during navigation wander azimuth frame is used. The only difficulty with this frame is that it is very hard to visualize. It is most common and popular frame which is used in inertial navigation applications. It is also called w frame. It is locally level like local navigation frame. It is revolved through wander angle about local vertical [2].

2.2.5 System Computational Frame

A frame used to implement the equations of motion in any reference frame is known as computational frame. It may be any frame from the above mentioned coordinates frames [2].

2.2.6 System frame or Body Frame

Body frame is actually the sensor frame. As we know that navigational frame remains fixed overtime. But body frame changes with change in sensor orientation. For

navigation applications, most of time sensor's sensitive axes are made to coincide with moving platform axes. Usually these axis are body frame. It can be defined as

- i. With the vehicle's center of gravity in body frame, its origin usually coincides.
- ii. In body frame towards the forward direction, the y-axis points. It is also roll axis.
Since the roll angle is defined using the right-hand rule around this axis.
- iii. In the transverse direction of body frame, the x-axis points towards. It is pitch axis.
- iv. In the vertical direction in body frame, the z-axis points. This completes right hand rule for coordinates system.
- v. Z-axis is yaw axis. Since, yaw angle using the right-hand rule around this axis corresponds to the rotations.

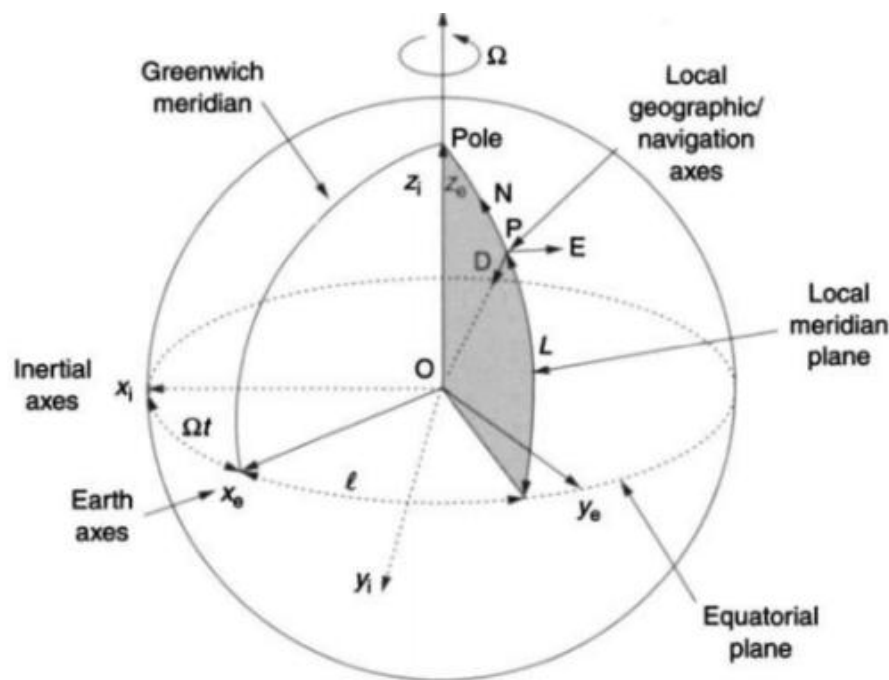


Fig 2.7 Reference Frames

2.2.7 Orbitals Coordinates System

The orbital coordinates system is used to locate a satellite in inertial space with Keplerian elements. This can be expressed in following way

- i. In orbital coordinate system, at the focus of an elliptical orbit its origin is situated. It coincides with the sphere's mass center.
- ii. In orbital coordinate system, in the direction of the descending node its Y-axis points. It is in the direction of minor axis of the satellite's orbit.
- iii. In orbital system of coordinates, towards the perigee its x-axis points. It is in the direction of major axis of the satellite's orbit.
- iv. In orbital coordinate system, to the orbital plane z-axis is orthogonal.

2.3 Coordinates Frames Transformations

The technique for transmuting a vector from one coordinates system of frame to another coordinate frame is called coordinate frame transformations. It involves different techniques for coordinate transformations like, direction cosines, Euler (rotation) angles and Quaternions etc. The rotation matrix used for frame transmutation is called transformation matrix. It also well-known as direction cosine matrix. To transform kinematics between coordinate frames is important and essential property for attitude representation related to inertial navigation. To express the attitude of one frame to another frame some equations are used. Acceleration, velocity, position and angular rates in Cartesian coordinates are transformed into inertial frame and local navigation frames.

For transmuting a vector information from one coordinates frame into other coordinates, there are a number of techniques are available. To define a body attitude with respect to a coordinate frame of reference various mathematical representations can be used.

Representation of Attitude:

Three attitude representations can be presented here, namely:

- a. Direction Cosine

b. Euler Angles

c. Quaternions

a. Directional Cosine of a Vector

A vector or line in three-dimensional geometry makes cosine angles with X, Y & Z-axis respectively. The angles cosines part of that vector forms with the coordinate axes are represented through direction cosines. It is represented by a 3x3 cosine matrix. In cosine matrix the column represents a 'unit' vector in body's frame which is projected along frame of the reference as follows.

$$C_b^n = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{33} & C_{33} \end{bmatrix}$$

Where C_b^n a 3x3 matrix is represents the direction cosine used for transformation to navigation frame from body frame.

b. Euler Angles

The orientation of an object in 3D using a combination of three rotations about different three axis can be represented by using the way called Euler's angles. For describing the sensor's orientation, we use multiple coordinate frames like inertial frame, body frame and vehicle frame. Since it includes three different rotations about different three axis, there its basic necessity is to transform one coordinate system to another coordinate system. Three ordered right hand rotations are used for transformation of frames. These rotations are named as roll, pitch and yaw. Mathematically, we can express these rotations as

$$\begin{aligned} \text{About Z-axis rotation is } \Psi, \quad C_1 &= \begin{bmatrix} \cos(\Psi) & \sin(\Psi) & 0 \\ -\sin(\Psi) & \cos(\Psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \text{About Y-axis rotation is } \theta, \quad C_2 &= \begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \end{aligned}$$

About X-axis rotation is $\emptyset$,

$$C_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\emptyset) & \sin(\emptyset) \\ 0 & -\sin(\emptyset) & \cos(\emptyset) \end{bmatrix}$$

By multiplying coordinate transformations matrix between two references frames, the body attitude can be obtained w.r.t. a new reference frame. Now, the transformation to body's frame from navigation's frame can be obtained by following equation

$$C_b^n = C_3 C_2 C_1$$

c. Quaternions

To translate any rotation in three dimensional coordinate system, a four elements vector representation is used named as quaternion. In quaternion one real element and three complex elements are used. For a single rotation about a vector a four element vector demonstration is used. This single vector defines the reference frame. The transmutation from one frame of coordinates to other reference frames of coordinate, quaternions are used. We can write quaternions in several ways

$$q = q_0 + q_1(i) + q_2(j) + q_3(k)$$

Or

$$q = (q_0, q_1, q_2, q_3)$$

Or

$$q = q_0 + \mathbf{q}$$

2.4 Globally Available Satellite Navigation System (G.N.S.S)

A navigation system based on satellites which is used globally for users for estimation of their position, velocity & time and some time for position, velocity and navigation measurements is acknowledged as globally available satellite navigation system or simply G.N.S.S. These satellites are revolving around the earth in different orbits above earth about 20000 Km to 23000 Km. the satellites complete their revolution period

twice a day. The satellites of GNSS are placed in medium earth orbits (MEO). The satellites broadcast signals in several frequencies ranges and bands. Currently, four satellites systems are globally fully operational. These are GPS, GLONASS, BEIDOU and GALILEO. The detail of four globally working & largely utilized satellites navigation systems are as following.

G.P.S which is abbreviation of Global Positioning System. It is the US satellite navigation system.

GL.O.NA.S.S which is abbreviation of Global Orbiting Navigation Satellite System. It is the Russia's satellite navigation system.

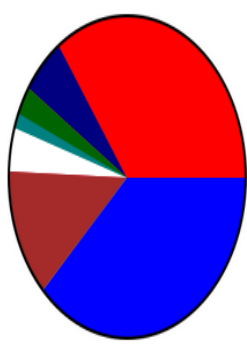
BEIDOU is the China's Satellite Navigation System

GALILEO is the European Union's satellite navigation system.

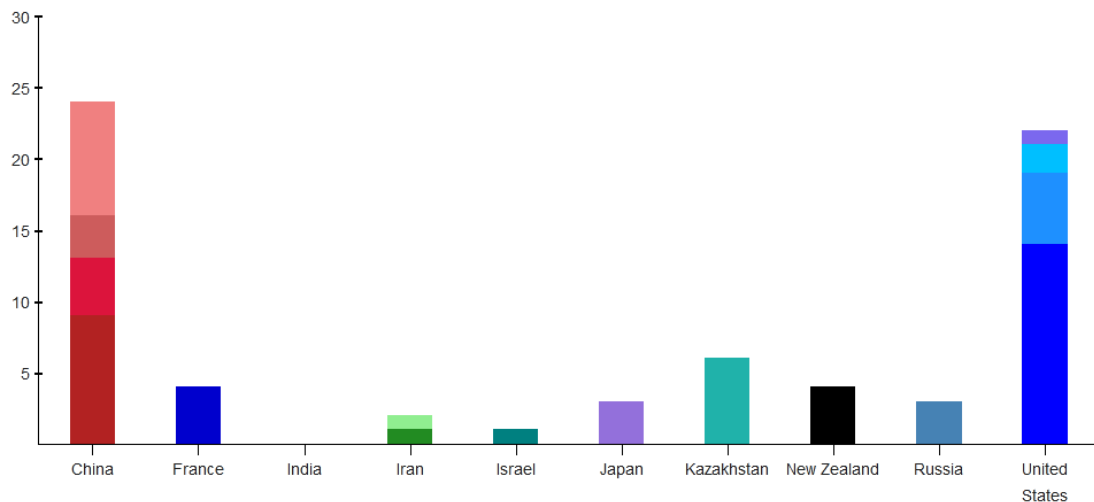
GPS, GLONASS & BEIDOU are fully functional, whereas GALILEO will be fully function after 2020. Two satellite navigation systems are operational as regional based which are NAVIC and QZSS. Whereas, NAVIC is Indian navigation satellite system which is advance version of regional navigation system and is going to become a global navigation system. Also, QZSS is Japan navigation satellite system working as regional navigation system

The major parts of GNSS are, first satellites its which is called space segment, second are ground control stations or master stations which are called control segment and third is the user which uses these satellites to find location. The location is determined from latitude, longitude and altitude. The ground control stations and user estimates position through trilateration methodology. For this purpose minimum four satellites are required.

Table 2.1 Comparison of different Global Navigation Satellite Systems



Country	Launches	Successes	Failures	Partial failures	Remarks
 China	24	21	3	0	
 Europe	4	4	0	0	
 India	0	0	0	0	
 Iran	2	1	1	0	
 Israel	1	1	0	0	
 Japan	3	3	0	0	
 Russia	9	9	0	0	Includes Soyuz launches from Kourou
 United States	26	24	2	0	Includes Electron launches from Mahia
World	69	63	6	0	



GNSS Applications

Now a days, GNSS is used for a variety of purposes. Initial these were designed for military applications. GNSS is being used in modern technology for many civil applications like.

- Aviation.
- Military precision-guided munitions.
- Emergency response and disaster warning.
- Maritime.
- Land transportation.

- Surveying and Mapping.
- Environment Monitoring.
- Agricultural Precision.
- Management of Natural resources.
- Climate change.
- Ionospheric studies.
- Wireless Networking.
- Photographic Geocoding.
- Communications of Mobile Satellite.
- Precise time reference.

2.4.1 Architecture or Segments of GNSS

There are three basic segments of GNS as follows.

1. Satellite or space section
2. Ground Control section
3. End User section

2.4.1.1 Satellite or Space Section

Satellite space segment is consist of satellite constellations which are inclined at various angles orbiting around the earth. Global navigation satellites are orbiting around the earth at medium earth orbit with different orbital periods. They broadcast velocity, position, time & navigation data which are used by satellite users. They estimates their position & orientation by using these information.

There are four fully functional and globally operated satellite systems which are revolving around the globe in medium earth orbits. These satellites are used for navigation purposes. In order to solve the distance formula equation for four variables

i.e. x , y , z and t . Therefore, minimum four satellites are necessary to calculate the user position, as demonstrated in below Fig. In the GNSS constellation where each satellite broadcasts a specific signal that recognizes its satellite and provides information about its time, orbit, and status.

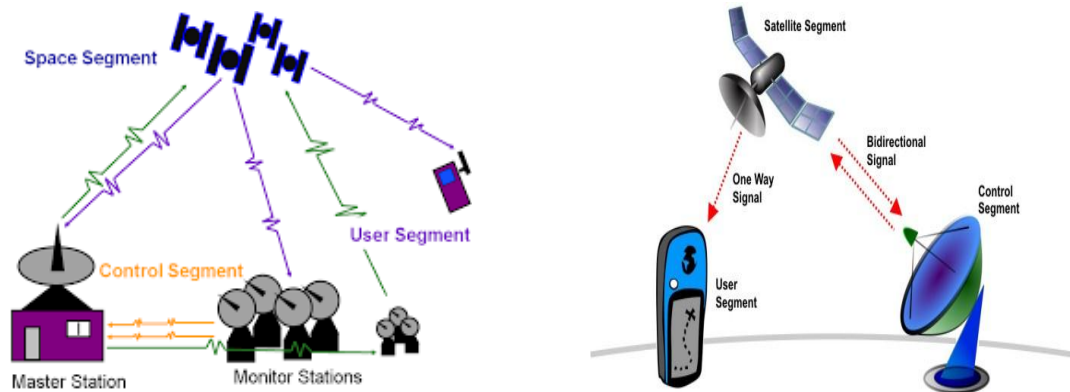


Fig 2.8 GNSS space segment

2.4.1.2 Satellite Control Section

GNSS satellite are constantly monitored by ground control satellite section. This control satellite segment is consists of network of monitor stations, control stations & data uploading stations. The monitoring stations are controlled by master station. The monitoring stations monitors & estimates satellites orbital errors, clock biases and ephemeris errors. These stations are located in different areas on the ground. The clock integrity and satellite signals are observed by monitoring stations and also information is guided to the master stations. The orbital and time corrections to the satellites are uploaded and compensated in satellite data by data uploading stations.

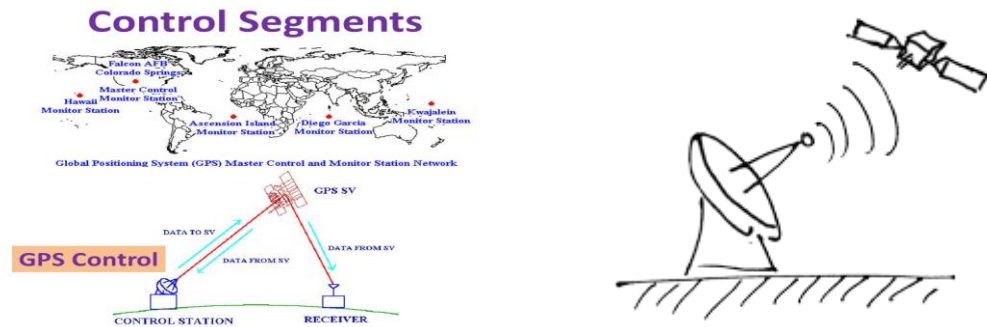


Fig 2.9 GNSS control segment

2.4.1.3 GNSS User Section

Global navigation satellites signal are used to determine the user location or position. So GNSS user is an important segment of navigation satellite system. Generally, GNSS user section comprises of GNSS receiver. GNSS signals are processed in GNSS receiver for absolute positioning. The user section comprises of GNSS receivers which use information provided by the GNSS signals data to estimate their location i.e. lat, long, height & time etc.

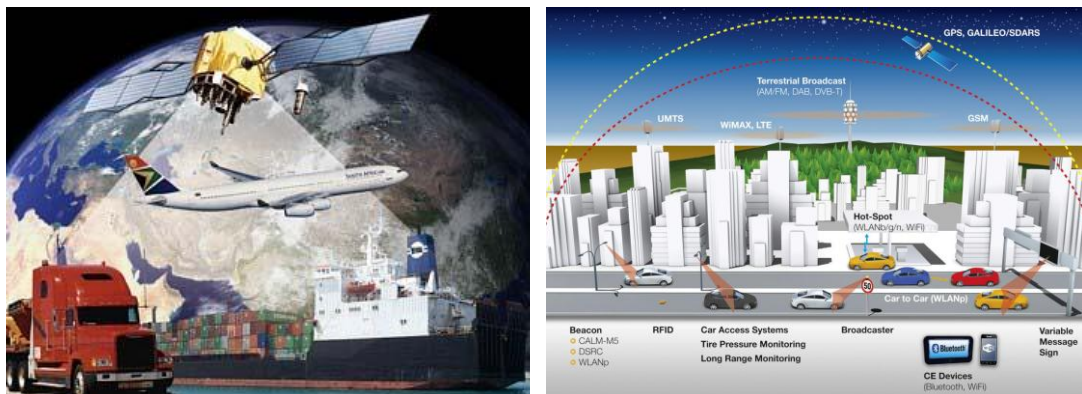


Fig 2.10 GNSS user segment

2.4.2 GNSS Orbits

The orbits where global navigation satellites are revolving around the earth is called GNSS orbits. Global navigation satellites are revolving in different orbits above the

earth's atmosphere. Generally, four fully functional navigation satellite systems are working globally. The altitude of GPS and GLONASS satellites orbits is about 20,000 km from earth's surface. The altitude of BEIDOU is 21,500 km and for GALILEO satellite is 23,000 km above the globe. The GNSS orbits are named as medium earth orbits or MEO. These orbits are highly stable, predictable and nearly circular. The pressure of solar radiations and earth's gravitational force effects the GNSS orbits. These effects are normally corrected. During correction the satellite status altered to out of service. So the user does not use the GNSS signals when its status is out of service.

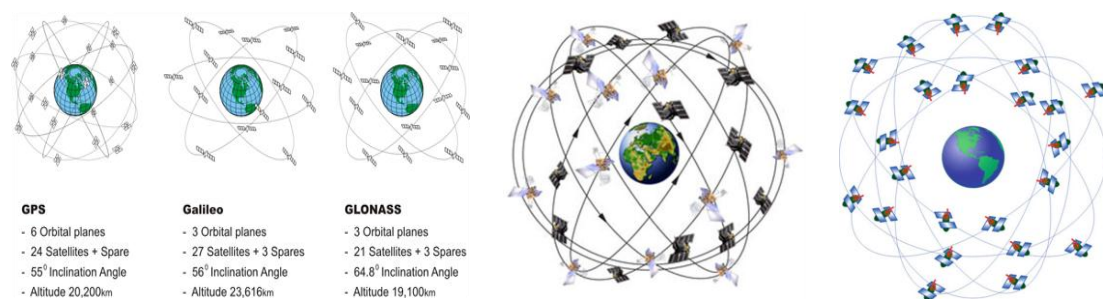


Fig 2.11 GNSS orbits

2.4.3 GNSS Signals

GNSS satellites use a complex signals scheme. The GNSS signals frequencies are higher than radio FM frequencies. These are lower than microwave frequencies. The range of GNSS signals frequencies is around 1.5 GHz to 1.5 billion Hz. When these signals are received at the ground, by this time they are very, very weak.

GPS satellites produce a central L-band frequency of 10.23 MHz using very stable clocks. Satellites then multiply this frequency by 154 and 120 to generate two carrier frequencies at $L1 = 1575.42 \text{ MHz}$ and $L2 = 1227.60 \text{ MHz}$. GPS signals consist of a carrier signal with frequency $L1$ or $L2$, a unique code assigned to each satellite, and a data message conveying information about satellite position, velocity, and clock bias.

Table 2.2 GNSS frequencies and signals

GNSS Frequencies and Signals		
System	Signal	Frequency (MHz)
GPS	L1 C/A	1575.42
	L2 C	1227.6
	L2 P	1227.6
	L5	1176.45
GLONASS	L1 C/A	1598.0625-1609.3125
	L2 C	1242.9375-1251.6875
	L2 P	1242.9375-1251.6875
	L3 OC	1202.025
Galileo	E1	1575.42
	E5a	1176.45
	E5b	1207.14
	E5 Alt BOC	1191.795
	E6	1278.75
BeiDou	B1I	1561.098
	B2I	1207.14
	B3	1268.52
	B1C	1575.42
	B2a	1176.45
NAVIC	L5	1176.45
SBAS	L1	1575.42
	L5	1176.45
QZSS	L1 C/A	1575.42
	L1 C	1176.45
	L1S	1575.42
	L2C	1227.6
	L5	1176.45
	L6	1278.75

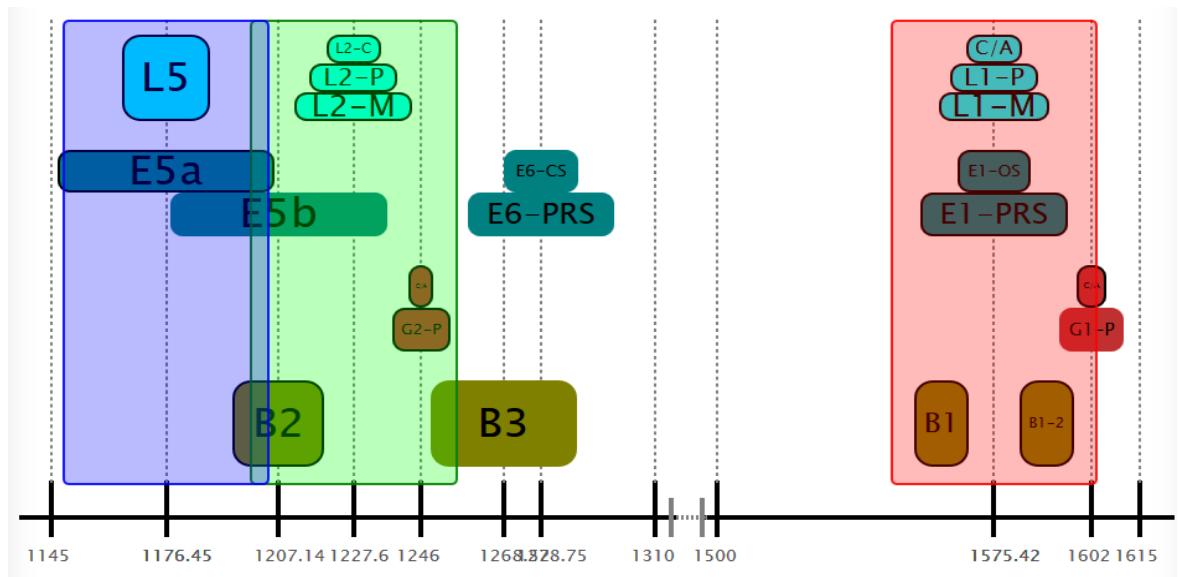


Fig 2.12 GNSS signals

The main signal components are described as follows:

- *Carrier:*

Sinusoidal signal consisting radio frequency at a given frequency.

- *Ranging code:*

Sequences of 0s and 1s (zeroes and ones), which allow the receiver to determine the travel time of radio signal from satellite to receiver. They are called Pseudo-Random Noise (PRN) sequences or PRN codes.

- *Navigation data:*

A binary-coded message providing information of clock bias parameters, satellite health status, almanac (with a reduced accuracy ephemeris data set), the satellite ephemeris (Keplerian elements or satellite position and velocity) and other complementary information.

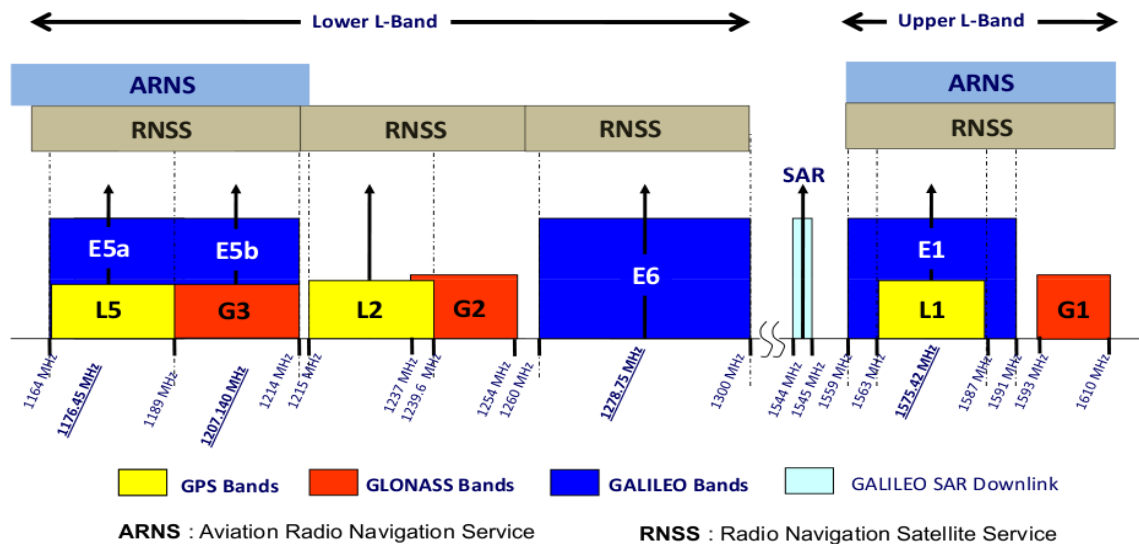


Fig 2.13 GNSS signals components

2.4.4 GNSS Error Resources

The measurements obtained by using the GNSS signals are not perfect. There are some kind of errors experience by GNSS receiver due to which measurements become noisy. These errors can be removed by known their source. Some typical error sources are given as follows.

- Satellite clock error
- Error of receiver's clock
- Receiver's noise error
- Multipath errors
- Ionospheric delay
- Tropospheric delay
- Satellite orbital errors
- Dilution of precision
- Ephemeris data error
- Selective availability

Contributing Source	Error Range
Satellite clocks	± 2 m
Orbit errors	± 2.5 m
Ionospheric delays	± 5 m
Tropospheric delays	± 0.5 m
Receiver noise	± 0.3 m
Multipath	± 1 m

Fig 2.14 Some GNSS error sources

2.4.4.1 Satellite Clock Error

Highly accurate and precise atomic clock are installed and used in global navigation satellites. These clock have an accuracy of nanoseconds. The GNSS signal received by the receiver have time information. This time refers to the emission time of signal from satellite. The distance between user and satellite is calculated by using this time. If there is some error in satellite clock, then accurate measurements cannot be achieved by using this time. 10 nanoseconds error in satellite time can produce 3 meters error in pseudo range measurements. The time of atomic clocks at each satellite is monitored continuously and corrected by the ground control stations or master stations.

2.4.4.2 Receiver Clock Error

The clock installed on GNSS receiver are not too accurate. The price of receiver increased with increasing the accuracy of clock. To minimize the clock error differential GNSS technique is best for clock error minimization. The clock bias error can also be compensated for the best estimation.

2.4.4.3 Receiver Noise

GNSS receivers are based electronic components and have some offsets. These offsets are to known in order to minimize the error. These error are thermal noise, biases or offsets and components non-linearity.

2.4.4.4 Multipath Error

One of the major source of GNSS errors is multipath error. Multipath errors are due to GNSS signal reflected by buildings, tunnels, trees or bridges etc. When these reflected signals are received by the receiver they have to travel more distance after their emission from GNSS. They arrived with some slight delay as compared to other direct signals. Some error in position estimation can be occurred due such delay.

2.4.4.5 Ionosphere Delay

The atmospheric layer about 60 to 100 km. above the earth's surface which contains ionized gases is called ionosphere. The solar activity effects this layer due to ionization level. As the ionization level is changed by the solar activity, due to which ionospheric layers refractive index is changed. The GNSS signals experience more transit time to reach the GNSS receiver. The satellite elevation angle also adds an additional error. Approximately, ionospheric delay can cause position error of ± 5 meters.

2.4.4.6 Tropospheric Delay

The atmospheric layer about 8 to 40 km above the surface of earth which contains dry gases& water vapors is called troposphere. It is electrically neutral and also non dispersive for GNSS signals. It can cause a decrease in speed of GNSS signals due to which an error in position estimation can occur. Approximately, ± 0.5 to ± 1 meters can occur due to the tropospheric delay.

2.4.4.7 Errors of Satellite's Orbit

Orbital errors of a satellite are the errors between its actual position in space and measured position estimated by using ephemeris data. These error are predicted and estimated by the control stations by using the satellite previous position and earth's gravity. For broadcast to the users these error are uploaded to the satellites as ephemeris data. Satellite orbital errors can cause ± 2 to ± 3 meters error in user position.

2.4.4.8 Dilution of Precision (DOP)

The available satellites geometry plays a major role to calculate the location of the user. To calculate the location of the user, DOP error may be caused by the relative positions in three dimensional space of the satellites used. Well disseminated satellites results in good values, whereas bad disseminated satellites results in poor estimations. Dilution of precision is also known as geometrical dilution of precision.

2.4.4.9 Ephemeris Data Error

Ephemeris data is used to calculate the satellites orbital position. Sometimes the estimated orbital position of a satellite is different by the actual orbital position. Thus the difference between expected orbital position & actual orbital position of a GNSS satellite by using the ephemeris data is also called as ephemeris data error. For satellite pseudo range estimation, the GNSS receivers uses the ephemeris data. Therefore, error in ephemeris data can cause an error in user position.

2.4.4.10 Selective Availability

Selective availability errors are artificial errors. These errors are added intentionally to the satellites signals to degrade the position and velocity of users. The errors are modeled but it requires lot of calculations.

2.5 Inertial Navigation System. (INS)

Inertial Navigation System (INS) is a navigational system comprises on inertial sensors like gyroscopes and accelerometers. It calculates position and heading w.r.t some reference. It is very stable and accurate. It has small size and light weight. It updates the data rapidly. It gives updates after every 20 milliseconds. It provides navigation data like position, speed, attitude and heading of the host or carrier. Thus for military and civilian navigation, the INS plays an important in their navigation applications. It does not depend on any external source that's why it is autonomous navigation system. It does not emits energy to external space. This made it applicable in airspace, underground and sea. To calculate orientation, position and velocity of a moving vehicle, we use motion estimation sensors like accelerometers and angular rate sensor like gyroscope. These are also used on ships, aircrafts, submarines, guided missiles and space crafts of gyroscopes are used for the navigation applications.

Inertial navigation systems are self-sufficient navigation systems which does not rely on any external signals. The measurements are provided by inertial sensors i.e. accelerometers and gyroscopes. Inertial navigation system is comprise of 3×gyroscopes and 3×accelerometers which are orthogonally installed. To track the position & velocity of moving object the accelerometers readings are utilized. Similarly, to track the orientation of moving objects, the reading of gyroscopes are used for navigation solution. For this purpose the initial conditions are required. Once if INS is initialized, then it uses its internal sensors to estimate the velocity, position and orientation of moving system. Any change in geographic position e.g. east or north, a change in velocity of moving object e.g. speed or direction and a change in orientation of moving system can be detected and recorded by the inertial navigation system.

Inertial navigation systems based on mems sensors are developed and used now a days. These are very cheap, small in size, having very low power consumption and are light weight. Their accuracy is not too good as compared to the tactical grad inertial navigation systems. They can give good performance with added navigation systems like satellite global navigation system or augmented navigation systems.

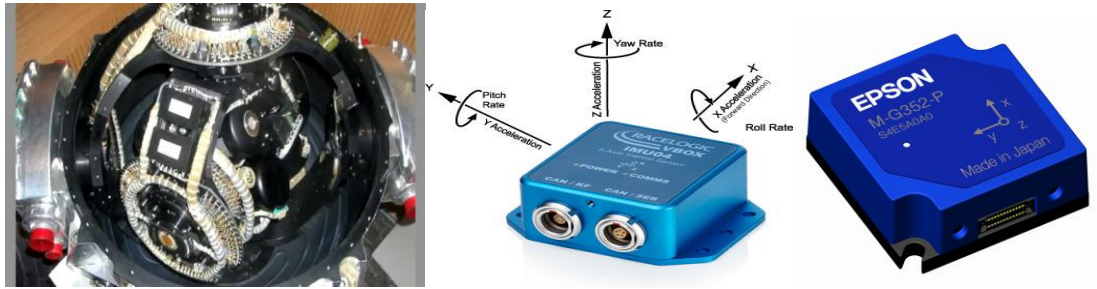


Fig 2.15 Inertial Navigation System

IMU i.e. inertial measurement unit is the heart of inertial navigation system. On the basis of their installation in navigation systems, two basic types of navigation systems are as follows.

- ◆ Gimballed or Platform based Inertial Navigation System
- ◆ Strapdown Inertial Navigation Systems

2.5.1 Gimballed / Platform Inertial Navigation Systems

In gimballed or platform system the inertial navigation system is installed at fixed position with respect to the body frame. Therefore, if system on which a platform navigation system is installed experiences any sort of angular rates or orientation then the platform navigation system remains at its initial fixed position. Thus the position of platform navigation system does not change with host system orientation. The roll, pitch and yaw of the host system is sensed by the navigation system and navigation system undergoes same through same orientation to keep its self to initial fixed position.

The host system orientation is sensed by platform navigation system and further send to the controls for further working.

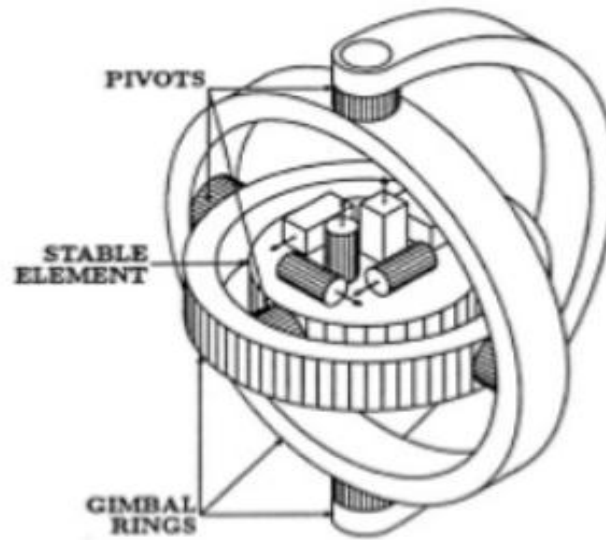


Fig 2.16 Gimbal INS

2.5.1.1 Advantages

- Less computation time is requires.

2.5.1.2 Disadvantages

- Expensive
- Complex in nature
- Maintenance is require
- Due to mechanical parts suffers from wear and tear
- Over time output drifts

2.5.2 Strapdown Inertial Navigation Systems

The navigation system which is tightly mounted parallel to the body axes of the vehicle is called strapdown inertial navigation system. In a strapdown inertial navigation system $3 \times$ accelerometers and $3 \times$ gyroscopes are installed in 3D plane of the vehicle. The accelerometers directly sense the acceleration of vehicle in terms of specific force due to earth's gravity. The gyroscopes sense the angular rates experienced by the vehicle and also due to earth's spinning normally known as earth's rate. By the integration of acceleration, we can find velocity of the vehicle. Similarly by the integration of the velocity we get the position. Similarly, by the integration of angular rates we get the attitude in terms of roll and pitch. Thus current velocity, position and attitude information of the vehicle are estimated over desired trajectory by using inertial navigation system.

Modern inertial navigation systems used on orbiting satellites, booster rockets and commercial jetliners use highly sophisticated inertial navigation systems. Hence, these inertial navigation systems are formed by integrating very accurate sensors like ring laser or fibre optics gyroscopes. Tiny error in the measuring sensors can introduce large errors in navigation solution provided by guidance system. The errors contributed by the sensors are not constant and are not bounded. The error can grow with time.

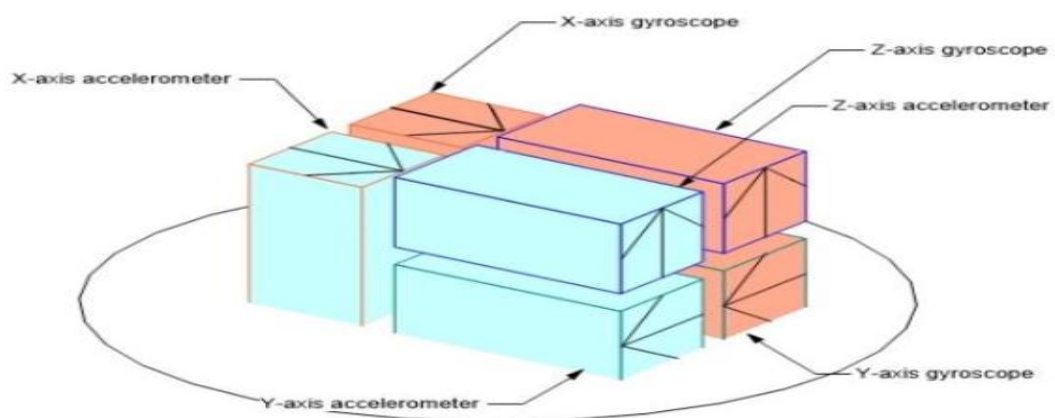


Fig 2.17 Layout of a strapdown INS, accelerometers are blue and gyroscopes are red

2.5.2.1 Advantages

- Cheap in Cost
- Small size
- Light in weight
- No mechanical part
- Maintenance is not required
- Reliable over time

2.5.2.2 Disadvantages

- Greater computation time is required
- Highly precise and accurate Gyroscopes are used

2.5.3 Mechanisation of INS

Inertial navigation system is comprises on accelerometers and gyroscopes sensors with data processing unit. Accelerometers outs acceleration whereas, gyroscopes outs angular rates. Acceleration and angular rates are transformed using transformation methods. To get the velocity, position and attitude of vehicle, the acceleration and angular rates are integrated.

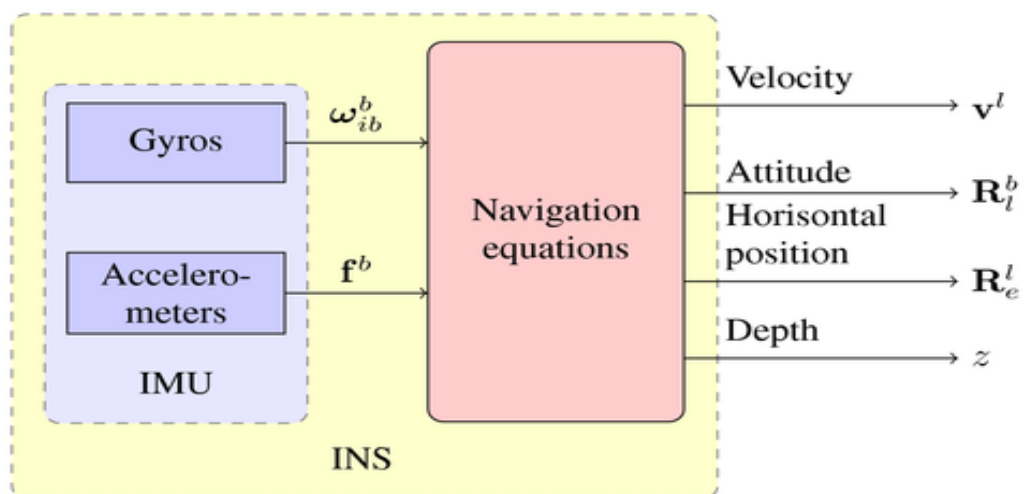


Fig 2.18 Basic mechanisation of INS (courtesy by google images)

2.6 MEMS based Inertial Navigation Systems

Inertial navigation system comprises on micro electro-mechanical systems sensors like MEMS based accelerometers & MEMS based gyroscopes are known as MEMS INS. Through mems technology mechanical elements, mems sensors, actuators & electronics components are integrated on silicon substrate to form a mems based inertial navigation system. This type of navigation system are miniature in size, have low power consumptions and are very cheap. Key challenge for navigation systems is making them smaller, less expensive and more robust while preserving high-end performances grade. They are used now a days in a variety of applications due to their reliability and robustness. Today's, each smart phone mobile and gaming devices have built-in mems based sensors like accelerometers & gyroscopes. For development of key technologies which are used in transportation, telecommunications and health care are going to be used mems technology. There is a large market for development of mems based automotive, aerospace and marine applications.

Mems sensors are quickly gaining market e.g. for crash air-bag deployment systems in automobiles, mems accelerometer has displaced conventional accelerometer which is bulky in weight. Mems based inertial navigation systems are much smaller. They are more functional and lighter. They are now becoming more reliable and robust. Their cost is very less than conventional navigation systems. They are mainly used for short timed navigation applications as their positioning and attitudinal errors increase steadily with time. Their performance is degraded with time for longer time applications as their errors are not bounded. Research is going on to increase their performance for longer time navigation solutions.

Mems based inertial navigation systems are used in integrated navigation applications where their errors are compensated by a reliable navigation system like GNSS. Through

GNSS velocity and position updates, the errors of mems INS are kept in bound and are continuously compensated during the entire flight.

Some examples of MEMS based INS are MEMSIC INS (NAV440CA-200, NAV440CA-400), SBG systems (Ellipse series, Ekinox series, Apogee series)



Fig 2.19 MEMS based INS

2.7 AHRS (Attitude & Heading Reference System)

The device which is used to estimate the orientation or attitude and heading of a vehicle or system is known as attitude and heading reference system or in short AHRS. An AHRS is a navigation system or device that is comprises of multi-axes accelerometers, gyroscopes and magnetometers which provides estimation of an objects orientation in 3-D space. Measurements of pitch, roll, and yaw are typical data outputs of AHRS.

In other words, AHRS is a 3-axis IMU which is combined with a 3-axis magnetic sensor. Roll and pitch angular rates are sensed by the IMU using accelerometers & gyroscopes readings. While the heading measurements are done by the magnetometer readings. The on-board processor installed in this system, creates a virtual 3-axis sensor. This is capable of measuring attitude and heading of an object which is moving in 3D space during navigation applications.

Composition of AHRS:

An AHRS is basically comprises on following sensors

- i. Accelerometers
- ii. Gyroscopes
- iii. Magnetometers
- iv. 32-bit processor or DSP

AHRS Uses:

AHRS are used for different platform navigation applications as follows.

- They provide flight data recordings of platform system's orientation which are used as time function.

- For platform stabilization they are also used. To maintain platform stability their angular rate outputs are utilized into a control loop directly.
- As an attitude control system AHRS also used. They are used in different navigation applications like aircraft, helicopter, blimps, quad-rotors & underwater robotic vehicles.
- During land navigation applications AHRS is also used to provide the heading estimations during global navigation system unavailability.
- With desired attitude angles to directly command a platform with estimation of orientation angles. The angular rate outputs at high bandwidth is used.
- To make singularity free attitude control by using quaternion output for many applications.



Fig. 2.20 Attitude and heading reference system (AHRS) *courtesy google images*

2.8 MAGNETOMETERS

The sensor used to estimate the earth magnetic field or magnetism is known as magnetometer. It measures the direction of magnetic field, strength of magnetic field and also change in magnetic field. Compass is the simplest example of magnetometer. These days, magnetometers are used for variety of purposes and for different applications. Magnetometers are also used as heading sensor in integrated navigation applications.

2.8.1 Principle and Classification of Magnetometers

On the basis of working principle and their physical properties magnetometers are categorized accordingly.

Magnetometers can be further classified according their physical effects as

Induction Magnetometers;

Induction magnetometers are magnetometers made according to the working principle of electromagnetic induction made according to Faraday's law.

2.8.1.1 Magnetic Magnetometers;

Magnetic magnetometers are magnetometers made according to the working principle that a Lorentz force can be generated by the current in magnetic field.

2.8.1.2 Magneto-resistive Magnetometer;

Magneto resistive magnetometer are magnetometers made according to the working principle that magnetic field changes the resistivity of the conductor.

2.8.1.3 Magneto-optical Magnetometers:

Magneto-optical magnetometers are magnetometers which are made according to the working principle of the Faraday magneto-optical effect. The example is optical pump magnetometer.

In regard to some other characteristics they can also be classified as follows.

2.8.1.4 Vector Magnetometer:

Vector magnetometers are devices that estimate the vector components of a field magnet. To study the magnetic field of Earth, these magnetometers are used. In terms of declination and the inclination they express the vector components of the field. Declination is the angle between the horizontal component of magnetic north and the field vector. Whereas, inclination is the angle between the horizontal surface and the field vector

2.8.1.5 Scalar Magnetometer:

Scalar magnetometers are also known as total field magnetometers. They measure the vector magnetic field magnitude.

2.8.2 Uses of Magnetometers:

- Archaeology
- Auroras
- Aerospace
- Biology
- Cesium vapor magnetometer
- Coal exploration
- Directional drilling
- Defense
- Military
- Mineral exploration
- Mobile phones
- Oil exploration
- Spacecraft Magnetic surveys

2.8.3 Magnetometer as heading Sensor

Magnetometer is also used for heading measurements. The heading can be best calculated by 3-axis magnetometer in a magnetic disturbance free environment. Consider, with an arbitrary orientation a magnetometer reading h taken will be given by:

$$h = R_x(\phi)R_y(\theta)R_z(\varphi)h_0 \quad \dots\dots\dots (2.1)$$

Where

h_0 is the local Earth magnetic field

$R_x(\phi)$, $R_y(\theta)$, $R_z(\varphi)$ are matrices of orientation around reference frames

Magnetometers are utilized to determine the heading during navigation applications. The intensity of magnetic field due to the earth is 0.5 to 0.6 gauss. Its component are parallel to the earth surface and always pointing towards magnetic north. Earth's magnetic north and earth's true north are not coincide. There is an angle amongst the earth's magnetic north and earth's true north. This angle is named as declination angle or inclination angle depends on the direction of measurements. The heading of any vehicle w.r.t. the magnetic north can be estimated by measuring the earth's magnetic field components.

For estimation of heading w.r.t. geographic or true north the declination angle is added for west direction values. It is subtracted for east direction values obtained by using magnetometer. An angle between axis of rotation which is geographic north or true north and magnetic north is called declination angle. By using updated earth's magnetic field model, the declination angle at any place on earth can be easily determined. By

using earth's magnetic field components and declination angle the heading can be estimated by using this equation

$$\psi = \arctan \left(\frac{h_y}{h_x} \right) \pm Dec\ angle \dots\dots\dots (2.2)$$

3. MAGNETOMETER CALIBRATION

3.1 Magnetometer Calibration Methods and Magnetometer Performance Analysis

There are different methods for magnetometer parameters calculation through calibration and performance analysis.

- Ellipsoidal Fitting Method
- Med wick Method
- Magnetometer calibration using inertial sensors
- Magnetometer 3D Calibration With Minimum Space Coverage

3.2 Magnetometer 3D Calibration with Minimum Space coverage when heading information is available;

This is fully automatic technique of magnetometer calibration, where user is not involved in calibration process. Also the user has not to perform any special instructions or any certain process during the calibration of magnetometer. The user has not to move the portable device in figure eight during calibration process. Moreover, as compared to other calibration techniques, a little space coverage is required during this calibration technique.

Updated version of world's magnetic field model is required to calculate the accurate reference earth magnetic field components at the region where calibration is being performed. The following information is required from the earth's magnetic field model

- (i) The earth's magnetic field vector components value
- (ii) The angle between geomagnetic north & geographic north usually known as declination angle.

The position of the device on globe in form of latitude, longitude and altitude is required to estimate the earth's magnetic field information. To estimate the position in terms of latitude, longitude and altitude the global navigation satellite system is used during magnetometer calibration. An external heading information source is required during proposed calibration technique to provide different readings of device heading. Different pitch and roll angles of the device are required to perform magnetometer proposed calibration technique. To estimate the roll and pitch angles following sensors readings may be used.

- (i) Accelerometers readings
- (ii) Gyroscopes readings
- (iii) Integrated navigation solution

To correct all different error sources that affect magnetometers by using calculated different calibration parameters, the calibration techniques is capable. For reliable heading estimation the calibrated magnetometer readings can be used. At the same time, simpler calibration equations will be used and less space coverage during calibration is needed.

From proposed technique of calibration of magnetometer, following parameters can be estimated precisely.

- Biases
- Scale factors
- Misalignments
- Soft Iron effects
- Hard Iron effects

The proposed technique of calibration of magnetometer depending on existing external heading source of information and calibration is performed automatically in fast way with small space. To estimate the calibration parameters in device frame following inputs are required.

- (i) *magnetometer raw reading*
- (ii) *External heading information*
- (iii) *magnetic 3D Reference field components*
- (iv) *Pitch and roll angles*

By using two levelled horizontal magnetometer readings the heading can be estimated as follows.

$$\psi = -\arctan(h_y/h_x) \pm D \quad \dots\dots\dots (3.1)$$

The magnetometer output can be expressed by following equations. The reference vector of magnetic field in device frame can be defined as given below.

$$h'_x = b_x + \gamma_x h_x \quad \dots\dots\dots (3.2)$$

$$h'_y = b_y + \gamma_y h_y \quad \dots\dots\dots (3.3)$$

$$h'_z = b_z + \gamma_z h_z \quad \dots\dots\dots (3.4)$$

Where,

h_x, h_y & h_z are magnetic field vectors obtained from earth magnetic model, transformed in body frame.

h'_x, h'_y & h'_z are raw/un-calibrated magnetometer output

b_x, b_y & b_z are biases readings in the device frame X, Y and Z axes respectively,

γ_x, γ_y & γ_z are readings scale factor

From equations (3.2), (3.3) and (3.4) we can proceed as follows

$$\begin{bmatrix} h'_x & h'_y & h'_z \end{bmatrix} = \begin{bmatrix} 1 & h_x & h_y & h_z \end{bmatrix} \begin{bmatrix} b_x & b_y & b_z \\ \gamma_x & 0 & 0 \\ 0 & \gamma_y & 0 \\ 0 & 0 & \gamma_z \end{bmatrix} \dots\dots\dots (3.5)$$

The misalignments of magnetometer are considered as zero means there are negligible misalignments. Equation (3.5) can be transformed as follows

$$H'_{1 \times 3} = H_{1 \times 4} X_{4 \times 3}$$

$$H_{4 \times 1}^T H'_{1 \times 3} = H_{4 \times 1}^T H_{1 \times 4} X_{4 \times 3}$$

$$(H_{4 \times 1}^T H_{1 \times 4})^{-1} H_{4 \times 1}^T H'_{1 \times 3} = X_{4 \times 3}$$

$$(H^T H)_{4 \times 4}^{-1} H_{4 \times 1}^T H'_{1 \times 3} = X_{4 \times 3} \dots\dots\dots (3.7)$$

From equation (3.7), the unknown biases and scale factors for magnetometer raw readings can be estimated and further uses for estimation of magnetometer output.

Magnetic field raw readings are stored before the magnetometer calibration. The required external heading which is used during magnetometer calibration can be estimated by using global navigation system or any reliable external heading source. Sufficient data of roll and pitch sectors is required for online calibration of the magnetometer. The external heading will only be used during magnetometer calibration. After magnetometer calibration using proposed technique, its heading independently can be used.

The device is assumed to be aligned to the platform on which the magnetometer is installed and having zero heading misalignment during calibration. The X-axis is towards the east direction, the Y-axis is towards the north direction and the Z-axis is towards the up direction for both platform and device frames.

To transform from local level frame to the device frame of the earth magnetic field components, following equation is devised.

$$M^b = C_n^b M^m \dots\dots\dots (3.8)$$

With the corresponding pitch and roll values the heading readings are used for transforming to device frame from local level ENU frame.

Different sets of roll & pitch values are required to estimate different sets of heading and to solve the aforementioned equations.

3.3 Error Compensation Parameters of Magnetometer

Magnetometers is a sensor that is used to detect the Earth's magnetic field. The measured readings by this sensor have some imperfections. Several different sources of error effect these sensors. These sources can be characterized as follows

- Biases / Null shift errors
- Scale factors of Magnetometer readings
- Misalignment errors of axis
- Hard iron effects of magnetometer
- Soft iron effects of magnetometer

3.3.1 Null Shift Error or Constant Bias:

When a magnetometer measures a magnetic field value even if the magnetic field along this axis is zero, then this may be called a null shift, off set or zero bias. Normally this error remains constant, hence the offset may be named as constant bias.

3.3.2 Scale Factors:

Scale Factor is ratio between input and output. In generally a scale factor is a number which multiplies or scales the readings or some quantity. In the line slope equation

$$y = Cx \quad \quad C \text{ is the normally scale factor for } x.$$

3.3.3 Misalignment:

The Magnetometers axis are mounted with the body axis in perfect alignment. If magnetometer axis are not completely aligned With the body axis, then misalignments error will be happened.

3.3.4 Hard Iron Effects:

The error observed due to some unwanted magnetic field and effect the magnetometer readings are called hard iron effects or errors. These errors are produced by materials that effect the earth's magnetic field and exhibit a constant additive effect.

3.3.5 Soft Iron Effects:

Soft iron distortions or effects are due to materials that effect or influence the earth's magnetic field. These materials does not itself generate magnetic field. Therefore no additive field is experienced by the magnetometer in hard iron effects.

4. MEMS INS THEORY AND ALGORITHM

MEMS based inertial navigation system consists of accelerometers and gyroscopes, have benefits of not depending on external signals, so there is no possibility of signal jamming or loss like GNSS. However, accelerometers and gyroscopes have some imperfections, which are bias and scale factor errors. A number of errors of inertial sensors are deterministic errors. They can be determined in lab through some specific lab calibrations. The remaining errors of inertial sensors are not systematic and they need to be modelled stochastically. These errors contribute in velocity and position errors that increase with time and are unbounded. This behaviour is largely due to

- The dead reckoning nature of navigation system,
- The impurity of the inertial measurements by systematic and random errors
- Imperfections in data processing such as gravity modelling error, linearization error and other numerical processing errors.

The inertial sensors implemented in this work are MEMS inertial sensors which are light weight, small size and low cost with respect to high-end inertial. These features make these MEMS sensors an attractive option for applications such as pedestrian or vehicular navigation. However, MEMS sensors are characterized by poorer performance too. They cannot be used in autonomous mode for prolonged periods. But they are well suited to integrated navigation systems

4.1 Inertial Sensors Errors:

The performance of inertial sensors are affected by a variety of errors which varies with time, environment and dynamics of the moving object on which these sensors are installed. These errors can be categorized as sensor fixed biases, scale factors, axis misalignment, & random noises. Some errors of inertial sensors are constant (i.e. does not change with

time) and appear first time the sensor is turned on and can be estimated through inertial sensors calibration techniques. The measurement model of inertial sensors is given below to show how these errors appear in the sensor output. The gyro and accelerometer output models are expressed by equations

$$\tilde{\omega} = B_g + S_g \omega + \omega M_g + \eta_g \quad \dots\dots\dots (4.1)$$

$$\tilde{f} = B_a + S_a f + f M_a + \eta_a \quad \dots\dots\dots (4.2)$$

Where

ω and $\tilde{\omega}$ are true and measured angular rate

f and $\tilde{f}$ are true and measured specific force

b_g and b_a are biases of gyroscopes and accelerometers respectively

M_g and M_a are misalignments of gyroscopes and accelerometers respectively

η_g & η_a are gyroscopes and accelerometers random noises respectively

4.1.1 Bias

Bias is normally defined as an output when there is no input. In case of inertial sensors some amount of data obtained when there is no input imply to the sensor. The input for gyroscope sensor is angular rate. The gyroscope senses the angular rate either provided by an external source or the earth spin rate. Similarly, the input for accelerometer is acceleration either provided externally or due to earth gravity. When inertial sensors give some useful data without any input source then it is called bias. The gyroscope biases are expressed in unit of degree per hour ($^{\circ}/hr$) or radian per second (rad/sec). Biases are two types, one is deterministic and other is random. The deterministic bias or offset can be determined by calibration.

4.1.2 Scale Factors

Scale factor is a known output at some known input. The change in output of inertial sensor divided by the change in input is also defined as scale factor. Generally scale factors are calculated by the method of least squares as the slope of the straight line to input output data. Scale factor errors are deterministic errors and can be estimated by inertial sensors calibration. Scale factor stability is very important for inertial sensors and must not be varied beyond prescribed limit.

4.1.3 Misalignment

Misalignment are inertial sensor mounting errors. The non-orthogonality of the axes of inertial sensors defining the INS with respect to the body frame. Misalignment of inertial sensors also estimated by the inertial system calibration can be compensated or modelled in the INS error equation.

4.1.4 Random Noise

Random Noise is some time called as sometimes random walk noise. It is generally non-systematic noise. Therefore cannot be removed from the data using deterministic models. It can only be modelled by stochastic procedure. A stochastic process is random time process.

4.2 Attitude Parameterization

The attitude for inertial navigation can be represented by a number of ways. The most commonly implemented approaches are given as follow.

- DCM (Direction cosine matrix)
- Euler angles
- Quaternion
- Rotation vector

4.3 World Geodetic System 1984 (WGS84) Model

World Geodetic system model WGS84 is a 3-dimensional coordinate reference frame for establishing latitude, longitude and heights for navigation, positioning. It is based on a consistent set of constants and model parameters that describe the Earth's size, shape, and gravity and geomagnetic fields.

In this model, following parameters are introduced;

$$R_M = a(1 - e^2)/(1 - e^2 \sin(\varphi)^2)^{3/2} \dots\dots\dots (4.3)$$

$$R_N = a/(1 - e^2 \sin(\varphi)^2)^{1/2} \dots\dots\dots (4.4)$$

Where,

R_M The meridian radius of curvature

R_N The prime vertical radius of curvature.

φ The latitude of the local point.

a The length of the semi major axis (~6378 137.0 m)

e The major eccentricity of the ellipsoid (~0.0818191908426)

ω_{ie} Earth Rate (15.041067 °/hour)

4.4 Algorithm MEMS Inertial Navigation System

The gyroscope and accelerometers of MEMS inertial measurement unit give raw data which is further processed for navigation purposes. The accelerometers output the acceleration due to gravity. Whereas, gyroscopes output angular rates. These rates may be due to earth rotation are externally applied to gyroscope. These raw outputs of MEMS are transformed in physical parameters by using transformations. The navigation equations are used to get the desired output. Position and velocity are obtained by sensors outputs in the form of latitude, longitude & height and attitude in the form of pitch, yaw and roll.

4.5 Algorithm

MEMS Inertial Navigation system requires a specific set of coordinate system in which all the computations are performed. In this research, we have used strapdown MEMS INS, in which the sensors are mounted in vehicle coordinate system such that the x-axis pointed towards east, the y-axis pointed towards north and the z axis pointed towards upward. Earth is assumed to be ellipsoidal in shape.

The typical flow chart of implementation of strapdown MEMS INS is given below:

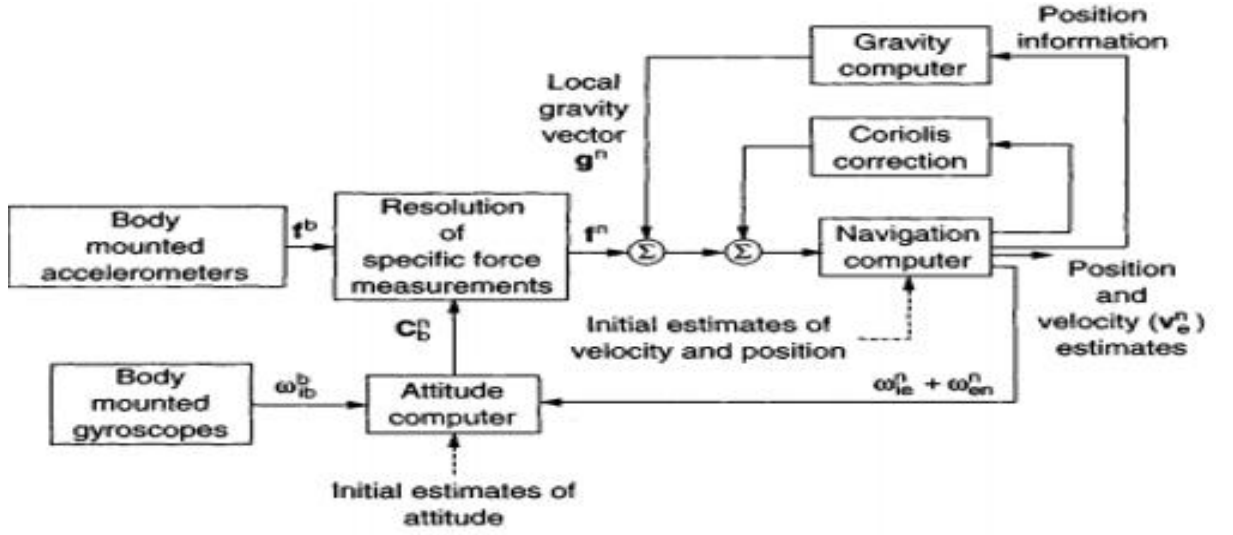


Fig 4.1 Scheme of Algorithm of MEMS INS

4.5.1 Transformation to East-North-Upward (ENU Navigation) Frame from Earth-Centered-Earth-Fixed Frame

The transformation to East-North-Upward (E-N-U) frame from Earth-Centered-Earth-Fixed Frame is achieved by performing sequential angular rotations which may starting from ECI frame.

$$X_i Y_i Z_i \xrightarrow[\text{about } Z \text{ axis}]{\omega_{ie}^e t} X_e Y_e Z_e \xrightarrow[\text{about } Z_e \text{ axis}]{\lambda + 90^\circ} \xrightarrow[\text{about } X' \text{ axis}]{90^\circ - \phi} X_g Y_g Z_g (ENU)$$

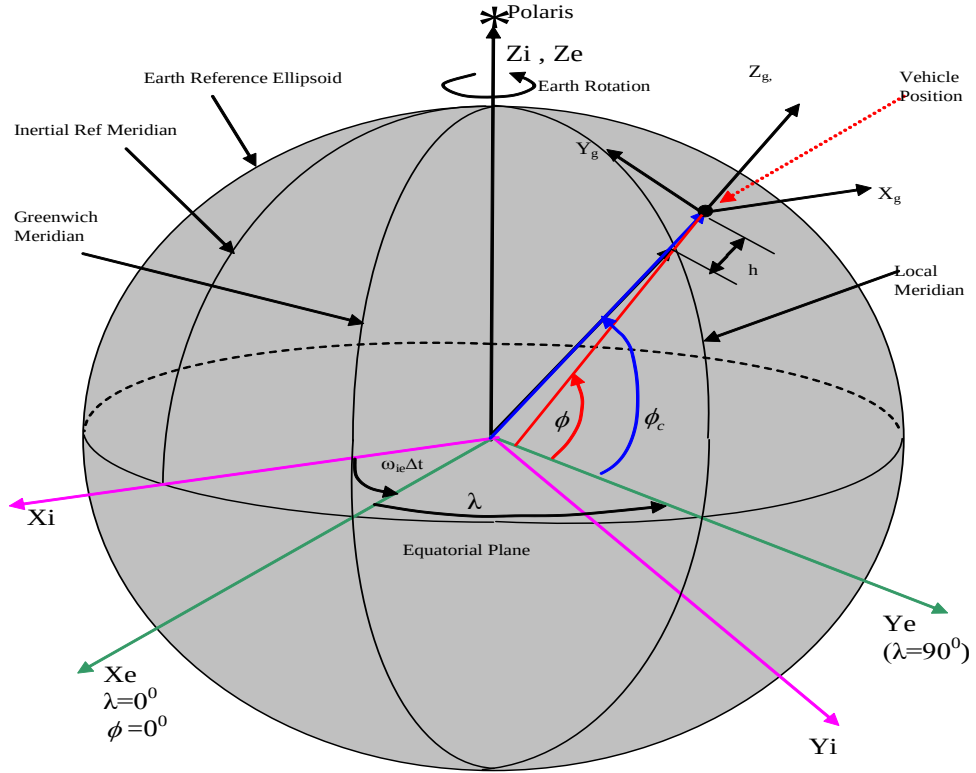


Fig. 4.2 ENU Scheme Transformation

From Sequence of rotation the following transformation matrix are obtained as:

$$C_{ecf}^{enu} = \begin{bmatrix} -\sin\lambda & \cos\lambda & 0 \\ -\sin\phi\cos\lambda & -\sin\phi\sin\lambda & \cos\phi \\ \cos\phi\cos\lambda & \cos\phi\sin\lambda & \sin\phi \end{bmatrix} \dots\dots\dots (4.5)$$

Where C_{ecf}^{enu} matrix is position matrix for the geodetic or navigation frame mechanization.

The navigation system positions i.e. latitude ϕ and longitude λ are estimated from the C_{ecf}^{enu} matrix solution updated as given below:

$$\phi = \phi_m = \sin^{-1}(C_{ecf}^{enu}_{3,3})$$

$$\lambda = \lambda_m = \tan^{-1}\left(\frac{C_{ecf\ 3,2}^{enu}}{C_{ecf\ 3,1}^{enu}}\right)$$

$$\lambda = \begin{cases} \lambda_m & \text{if } C_{ecf\ 3,1}^{enu} > 0 \\ \lambda_m + 180^\circ & \text{if } C_{ecf\ 3,1}^{enu} < 0 \text{ and } \lambda_m < 0 \\ \lambda_m - 180^\circ & \text{if } C_{ecf\ 3,1}^{enu} > 0 \text{ and } \lambda_m > 0 \end{cases}$$

To update the position matrix, aforementioned matrix is used.

4.5.2 Updating of Position Matrix

To update the position matrix using Euler's method the differential equation are solved as given below

$$\dot{C}_{ecf}^{enu} = \Omega_{eg}^g C_{ecf}^{enu} \dots\dots\dots (4.6)$$

Where

$$\Omega_{eg}^g = \begin{bmatrix} 0 & -\omega_{egz}^g & \omega_{egy}^g \\ \omega_{egz}^g & 0 & -\omega_{egx}^g \\ -\omega_{egy}^g & \omega_{egx}^g & 0 \end{bmatrix}$$

4.5.3 in ENU Frame Earth Spin Rate

To transform the earth's spin rate from earth's fixed frame into ENU navigation frame, following matrix is used.

$$\omega_{ie}^n = \begin{bmatrix} 0 \\ \omega_{ie} \cos \emptyset \\ \omega_{ie} \sin \emptyset \end{bmatrix} \dots\dots\dots (4.7)$$

Where

ω_{ie} is the Earth's spin rate.

4.5.4 Measurement of Transport Rate

The measurement of transport rate can be achieved by the following equation

$$\omega_{eg}^g = \left[-\frac{V_N}{R_N+h} \frac{V_E}{R_E+h} \frac{V_E}{R_N+h} \tan\phi \right]^T \dots\dots\dots (4.7)$$

4.5.5 Updated Velocity Components used for Position Updates

Information as Latitude, longitude and altitude rates are expressed as follows

$$\dot{\phi} = \frac{V_N}{R_N+h} \dots\dots\dots (4.8)$$

$$\dot{\lambda} = \frac{V_E}{(R_E+h)\cos\phi} \dots\dots\dots (4.9)$$

$$\dot{h} = V_U \dots\dots\dots (4.10)$$

Where V_E, V_N & V_U are east, north and up components of velocity respectively in body frame.

By using the rates the updates of lat. long. and altitude can be obtained by solving theses equations.

$$\phi(t + \Delta t) = \phi(t) + \int_t^{t+\Delta t} \dot{\phi} dt$$

$$\lambda(t + \Delta t) = \lambda(t) + \int_t^{t+\Delta t} \dot{\lambda} dt$$

$$h(t + \Delta t) = h(t) + \int_t^{t+\Delta t} \dot{h} dt$$

..... (4.11)

4.5.6 Updating the Gravity Vector

By using following equation, the Gravity is updated as follows.

$$g = g_0 \frac{(1+0.001932851*\sin^2 \phi)(1-2h/Re)}{\sqrt{1-0.006694379*\sin^2 \phi}}$$

Where $g_0 = 9.7803267714$

4.5.7 Computation of Velocity

To calculate the acceleration vector the below equation can be used

$$\dot{V}_e^g = f^g - (\omega_{eg}^g + 2\omega_{ie}^g) * V_e^g - g^g \dots\dots\dots (4.12)$$

$$\text{Where } f^g = f_{ib}^g = C_b^g f_{ib}^b$$

f_{ib}^g Output in body frame obtained from triad of accelerometer

$$\begin{bmatrix} \dot{V}_{ex}^g \\ \dot{V}_{ey}^g \\ \dot{V}_{ez}^g \end{bmatrix} = \begin{bmatrix} f_x^g + 0 \\ f_y^g + 0 \\ f_z^g + g \end{bmatrix} - \begin{bmatrix} 0 & 2\omega_{iez}^g + \omega_{egz}^g & -(2\omega_{iey}^g + \omega_{egy}^g) \\ -(2\omega_{iez}^g + \omega_{egz}^g) & 0 & 2\omega_{iex}^g + \omega_{egx}^g \\ 2\omega_{iey}^g + \omega_{egy}^g & -(2\omega_{iex}^g + \omega_{egx}^g) & 0 \end{bmatrix} \begin{bmatrix} V_{ex}^g \\ V_{ey}^g \\ V_{ez}^g \end{bmatrix} \dots\dots\dots (4.13)$$

By integration of above equation, in reference frame we can obtain velocity vector as

$$V_e^g(t_{n+1}) = V_e^g(t_n) + \int_{t_n}^{t_{n+1}} \dot{V}_e^g dt$$

4.5.8 Navigation frame (geographic) Transformation into Strapdown Frame:

Angular rotations are performed for transformation into strapdown frame from navigation frame as follows

$$X_g Y_g Z_g \xrightarrow[\text{about } Z_g \text{ axis}]{-\psi} X'_g Y'_g Z'_g \xrightarrow[\text{about } X'_g \text{ axis}]{\theta} X''_g Y''_g Z''_g \xrightarrow[\text{about } Y''_g \text{ axis}]{\gamma} X_b Y_b Z_b \text{ Following}$$

the sequence of rotations to obtain the transformation matrix as

$$C_n^b = \begin{bmatrix} \cos\gamma\cos\psi + \sin\gamma\sin\theta\sin\psi & -\cos\gamma\sin\psi + \sin\gamma\sin\theta\cos\psi & -\sin\gamma\cos\theta \\ \cos\theta\sin\psi & \cos\theta\cos\psi & \sin\theta \\ \cos\gamma\cos\psi - \cos\gamma\sin\theta\sin\psi & -\sin\gamma\sin\psi - \cos\gamma\sin\theta\cos\psi & \cos\gamma\cos\theta \end{bmatrix} \dots\dots\dots (4.14)$$

4.5.9 Attitude Direction Cosine Matrix (DCM) C_n^b updates

During the small span of time from time interval t to $(t + \Delta t)$ subsequent the sequence of rotations to update the Direction Cosine Matrix following transformation is used

$$X_b Y_b Z_b(t) \xrightarrow[\text{about } X_b]{\Delta\theta_x} \xrightarrow[\text{about } Y_b]{\Delta\theta_y} \xrightarrow[\text{about } Z_b]{\Delta\theta_z} X_b Y_b Z_b(t + \Delta t)$$

At time interval $(t + \Delta t)$ the corresponding transformation matrix for Direction Cosine Matrix is given as follows

$$\dot{C}_n^b(t) = \begin{bmatrix} 0 & \omega_{nbz}^b & -\omega_{nby}^b \\ -\omega_{nbz}^b & 0 & \omega_{nbx}^b \\ \omega_{nby}^b & -\omega_{nbx}^b & 0 \end{bmatrix} C_n^b(t)$$

$$\dot{C}_n^b(t) = -\Omega_{nb}^b C_n^b(t) \dots\dots\dots (4.15)$$

Where Ω_{nb}^b is the skew symmetric matrix corresponding to ω_{nb}^b .

4.5.10 Computation of ω_{nb}^b

ω_{nb}^b can be calculated as follows

$$\omega_{ib}^b = \omega_{in}^b + \omega_{nb}^b$$

$$\omega_{nb}^b = \omega_{ib}^b - \omega_{in}^b$$

$$\omega_{nb}^b = \omega_{ib}^b + C_n^b \omega_{in}^n$$

ω_{ib}^b is the rates obtained from gyroscopes and ω_{in}^n is obtained as follow:

$$\omega_{in}^n = \omega_{ie}^n + \omega_{en}^n$$

$$\omega_{in}^n = \begin{bmatrix} -\frac{V_y^n}{R_{M+h}} \\ \omega_{ie} \cos \phi + \frac{V_x^n}{R_{N+h}} \\ \omega_{ie} \sin \phi + \frac{V_x^n}{R_{M+h}} \tan \phi \end{bmatrix} \dots\dots\dots (4.16)$$

4.5.11 Using Quaternion scheme to update the Attitude DCM C_n^b

For Quaternion estimation the dynamic equation is given as follows

$$\dot{Q} = \frac{1}{2} M(Q) \omega_{nb}^b$$

$$\omega_{nb}^b = [0 \quad \omega_{nbx}^b \quad \omega_{nby}^b \quad \omega_{nbz}^b]^T$$

$$Q = [q_0 \quad q_1 \quad q_2 \quad q_3]^T$$

DCM (C_n^b .) Strapdown matrix is estimated from the updating of quaternion as follows

$$C_n^b = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{33} & C_{33} \end{bmatrix}$$

4.5.12 Updating Quaternion

By following process Quaternion will be updated

$$\Delta \theta = \begin{bmatrix} \Delta \theta_x \\ \Delta \theta_y \\ \Delta \theta_z \end{bmatrix}$$

$$N(\Delta \theta) = \sqrt{(\Delta \theta_x)^2 + (\Delta \theta_y)^2 + (\Delta \theta_z)^2}$$

$$\begin{bmatrix} q_0(t + \Delta t) \\ q_1(t + \Delta t) \\ q_2(t + \Delta t) \\ q_3(t + \Delta t) \end{bmatrix} = \left\{ \cos\left(\frac{N(\Delta \theta)}{2}\right) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \left\{ \frac{\sin\left(\frac{N(\Delta \theta)}{2}\right)}{N(\Delta \theta)} \right\} M \right\} \begin{bmatrix} q_0(t) \\ q_1(t) \\ q_2(t) \\ q_3(t) \end{bmatrix}$$

$$C_n^b = \begin{bmatrix} \cos\gamma\cos\psi + \sin\gamma\sin\theta\sin\psi & -\cos\gamma\sin\psi + \sin\gamma\sin\theta\cos\psi & -\sin\gamma\cos\theta \\ \cos\theta\sin\psi & \cos\theta\cos\psi & \sin\theta \\ \cos\gamma\cos\psi - \cos\gamma\sin\theta\sin\psi & -\sin\gamma\sin\psi - \cos\gamma\sin\theta\cos\psi & \cos\gamma\cos\theta \end{bmatrix} \dots\dots\dots (4.17)$$

4.5.13 Attitude Computation

Attitude is computed form the updated C_n^b , as follows

$$\psi_g = \tan^{-1}\left(\frac{C_{21}}{C_{22}}\right) \dots\dots\dots (4.18)$$

$$\gamma_g = \tan^{-1}\left(\frac{C_{13}}{C_{33}}\right) \dots\dots\dots (4.19)$$

$$\theta_g = \sin(C_{23}) \dots\dots\dots (4.20)$$

For practical/actual values, the attitude angles adjustment can be done by following way

4.5.14 Heading

If $C_{22} > 0$ and $\psi_g > 0$ then $\psi = \psi_g$
If $C_{22} > 0$ and $\psi_g < 0$ then $\psi = \psi_g + 2\pi$
If $C_{22} < 0$ then $\psi = \psi_g + \pi$

4.5.15 Roll

If $C_{33} > 0$ then $\gamma = \gamma_g$
If $C_{33} < 0$ and $\gamma_g < 0$ then $\gamma = \gamma_g + \pi$
If $C_{33} < 0$ and $\gamma_g > 0$ then $\gamma = \gamma_g + \pi$

4.5.16 Pitch

$$\theta = \theta_s$$

5. THE KALMAN FILTER

5.1 State Space Model

A Kalman filter algorithm is a recursive predictive process which utilizes information of system state dynamics and the measurements sequentially to estimate the state vector which minimize the error covariance. It is a very powerful filter in the sense that it will estimate past, present and even future states of integrated solution. Generally, Kalman filter process assumes that the system is linear and white noise can disturb it. Measurements are usually linear correlated to the system states but white noise can corrupt it.

Navigation system's attitude measurements will be done in discrete time. The samples from the accelerometer and the gyroscope sensors are discrete. In discrete time the linear state space model for many applications is given as follows.

$$x_{k+1} = F_k \cdot x_k + w_k \quad (5.1)$$

$$z_k = H_k \cdot x_k + v_k \quad (5.2)$$

Where,

k specifies the time instance

x_k is n -dimensional vector demonstrating state variable

F_k is n by n matrix

z_k is m -dimensional vector representing the measurement of the system

H_k is m by n matrix

w_k & v_k are white noise uncorrelated with covariance matrices.

Furthermore, for process and measurement noises there is no time correlation. They are uncorrelated with each other i.e.

$$E[w_k w_j^T] = Q_k \delta_{kj} \quad (5.3)$$

$$E[v_k v_j^T] = R_k \delta_{kj} \quad (5.4)$$

$$E[w_k v_j^T] = 0 \quad (5.5)$$

Where,

δ_{kj} is Kronecker delta function.

Where, $\delta_{kj} = 1$ if $k=j$ and $\delta_{kj} = 0$ if $k \neq j$.

The Kalman filter work in two steps:

1. Using system dynamic model and previously estimated state, the next state and its covariance is predicted termed as *prediction*. Mathematically, it can be written as:

$$\hat{X}_k^- = \Phi_{k,k-1} * \hat{X}_{k-1}^+ \quad (5.6)$$

The covariance of predicted state is computed as:

$$P_k^- = \Phi_{k,k-1} * P_{k-1}^+ * \Phi_{k,k-1}^T + Q_{k-1} \quad (5.7)$$

The equation (5.7) propagates in time, theoretically the current covariance which adds uncertainty to the prediction.

2. When the measurements at next epoch are available, the predicted state and its covariance are merged with these measurements.

The innovation sequence can be given as:

$$V_k = Z_k - H_k \hat{X}_k \quad (5.8)$$

Where,

V_k is innovation sequence. Which is the difference of predicted & measured observation. This represent new information added to the system. Its covariance is given by equation

$$C_k = H_k P_k^- H_k^T + R_k \quad (5.9)$$

The Kalman gain is an increment factor which determines that how much information from the innovation sequence is acceptable which will be added to the systems. If the measurements exceedingly are noisy, then matrix for Kalman gain reduces the effect to produce the minimum error covariance.

The matrix for Kalman gain is given by the equation

$$K_k = P_k^- H_k^T \left(H_k P_k^- H_k^T + R_k \right)^{-1} \quad (5.10)$$

After the computation of gain, the state and its covariance are updated using new measurement information as:

$$\hat{X}_k = \hat{X}_{k,k-1} + K_k * (Z_k - H_k * \hat{X}_{k,k-1}) \quad (5.11)$$

$$P_k^+ = (1 - K_k H_k) * P_k^- * (1 - K_k H_k)^T + K_k * R_k * H_k^T \quad (5.12)$$

The ‘^’ represent the estimated quantity, the superscript ‘-’ represent priori estimation & ‘+’ posterior estimation (after measurement update). This procedure is repeated for every epoch with previous epoch state as initial value. That’s why it is called a recursive filter.

5.2 The Extended Kalman filter (EKF)

By replacing the linear terms $F_k X_k$ & $H_k X_k$ by the potentially nonlinear functions $f[X_k, k]$ & $h[X_k, k]$ respectively in equations (5.1) & (5.2), the generalized form can be attained. The relationship between X_k & X_{k+1} is linear and relationship between X_k & Z_k is no longer established for this standard Kalman filtering technique could not be used. Therefore, the below equation for EKF will be derived for a navigation system as:

$$X_{k+1} = f[X_k, k] + w_k \quad (5.13)$$

$$Z_k = h[X_k, k] + v_k \quad (5.14)$$

Where white noise sequences are w_k and v_k which again mutually uncorrelated with Q_k and R_k covariance matrices respectively. For the EKF following equations can be expressed as;

Prediction:

$$\hat{x}_{k|k-1} = f[\hat{x}_{k-1|k-1}, K] \quad (5.15)$$

$$P_{k|k-1} = F[k, \hat{x}_{k-1|k-1}] * P_{k|k-1} * F[k, \hat{x}_{k-1|k-1}]^T + Q_k \quad (5.16)$$

Update:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \cdot [Z_k - h[\hat{x}_{k|k-1}, k]] \quad (5.17)$$

$$P_{k|k} = P_{k|k-1} - K_k H \cdot [k, \hat{x}_{k|k-1}] \cdot P_{k|k-1} \quad (5.18)$$

While,

$$K_k = P_{k|k-1} H^T * [k, \hat{x}_{k|k-1}] * \{ H[k, \hat{x}_{k|k-1}] P_{k|k-1} H^T + R_k \}^{-1} \quad (5.19)$$

Equation (5.20) is gain for Kalman filter and

$$H[k, \hat{x}_{k|k-1}] = \frac{\partial h[x,k]}{\partial x} \Big|_{x=\hat{x}_{k|k-1}} \quad (5.20)$$

$$F[k, \hat{x}_{k|k-1}] = \frac{\partial f[x,k]}{\partial x} \Big|_{x=\hat{x}_{k|k-1}} \quad (5.21)$$

Where,

$$\hat{x}_{k-1|k-1} \quad \& \quad P_{k-1|k-1}$$

Above are the state variable and from the previous time instance is covariance matrix.

And

$$\hat{x}_{k|k-1} \quad \& \quad P_{k|k-1}$$

are the state variable estimates.

Now, from the prediction step the covariance matrix can be established as.

$$\hat{x}_{k|k} \quad \& \quad P_{k|k}$$

On to next time instance will be delivered.

The state variable $\hat{x}$ is the estimation of an n -dimensional vector and also is the measurement of the navigation system [22].

6. MEMS INS/GNSS & MAGNETOMETER INTEGRATION

6.1 MEMS INS / GNSS Integration Techniques/Strategies

Usually MEMS INS and GNSS are integrated by using Kalman filter. The INS position & velocity errors are corrected by using position and velocity updates of GNSS. In order to combine MEMS-INS & GNSS readings several methodologies are used to integrate the both systems. Some most common techniques are described as;

- Uncoupled Integration scheme
- Loosely Coupled Integration scheme
- Tightly Coupled Integration scheme
- Deep or Ultra-Tight Integration scheme

The brief description of above mentioned techniques are given in following.

6.1.1. Uncoupled Integration Scheme

In uncoupled integration scheme, the systems are independent. Each system will provide navigation solution independently. Therefore, two or more distinct navigation solutions are available. Since GNSS is all weather available navigation system and generally considered as more accurate. The errors of INS are grows with time and needs to be corrected. Therefore, GNSS solution is utilized to correct the MEMS INS errors which are accumulated during navigation. When GNSS is not available the navigation solution is entirely provided by MEMS inertial navigation system. The standalone INS solution is not reliable for long endurance navigation applications as the errors are accumulated. Due to this drawback, the uncoupled integration scheme is not used for navigation applications.

6.1.2. Loosely Coupled (LC) Integration Approach

The loosely coupled integration scheme is most commonly used integration scheme. In this type of integration scheme MEMS INS & GNSS are integrated in loose coupling. The GNSS derived position & velocity are merged and updates the position & velocity of INS by using Kalman filter. In next step to improve the navigation solution for navigation the error states are feedback to INS equations to moderate the errors. Loosely coupled integration scheme block diagram is shown in below Fig. 6.1

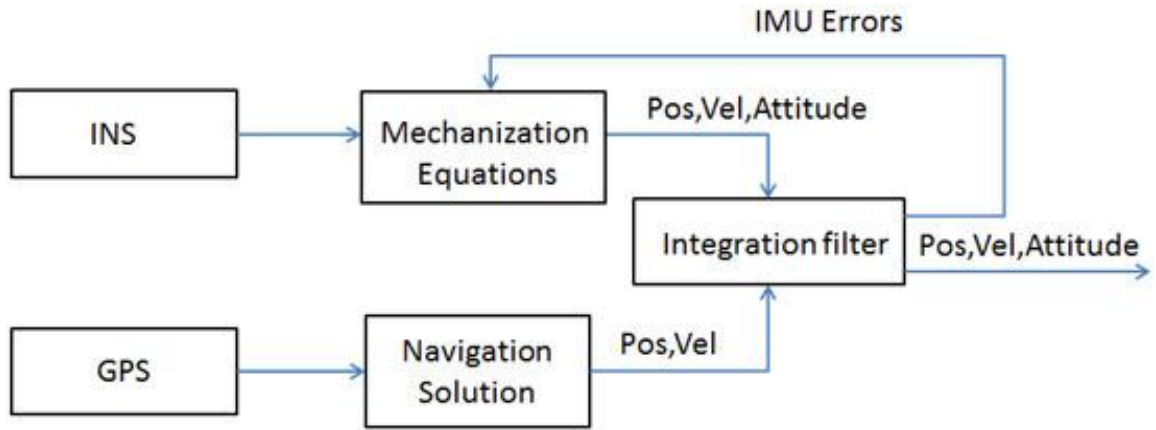


Fig.6.1 Loosely coupled integration scheme (in Visio)

Mechanization equations of MEMS INS are utilized to transform the acceleration & angular rates of inertial measurements to navigation frame from body frame. The distinctive error state in ECEF frame which are estimated by the MEMS INS & GNSS (especially GPS) integration using Kalman filter are as follow

$$\hat{x}_k = [\varphi_x, \varphi_y, \varphi_z, \delta V_x^e, \delta V_y^n, \delta V_z^u, \delta \phi, \delta \lambda, \delta h, \varepsilon_{gbx}, \varepsilon_{gby}, \varepsilon_{gbz}, \nabla_{Abx}, \nabla_{Aby}, \nabla_{Abz}]^T \quad (6.1)$$

Discrete time measurement model for Kalman filter is stated in below equation

$$z = H_k \hat{x}_k + v_k \quad \dots\dots\dots (6.2)$$

Where

H_k matrix is the connexion between the measurements and the error states.

z_k at the k -th time instant

v_k is a covariance matrix equal to R_k & the white noise associated with the measurements

In ECEF coordinates for loosely coupled integration scheme, z_k can be expressed as

$$z = \begin{bmatrix} X_{INS} - X_{GPS} \\ Y_{INS} - Y_{GPS} \\ Z_{INS} - Z_{GPS} \end{bmatrix} \dots\dots\dots (6.3)$$

For error state, the discrete Kalman filter dynamic model can be written as

$$\delta x_k = \Phi_{k-1} \delta x_{k-1} + \mu_{k-1} \dots\dots\dots (6.4)$$

Whereas,

Φ is the transition matrix

μ indicates in the discrete time system the noise propagation.

6.1.3. Tightly-Coupled (TC) Integration Approach

In tightly-coupled integration scheme all the measurements are directly used to estimate the final results. It is dual integration approach. One is processing of the measurements & the other one is sensor fusion.

A centralized Kalman filter is used in tightly coupled algorithm. It integrates the assessed pseudo-ranges (ρ_{GPS}) and Doppler-shift ($freq_Doppler_GPS$), from the GNSS receiver. It also assessed the information of velocity, position and attitude of inertial sensors. Tightly coupled scheme block diagram is given in fig. below.

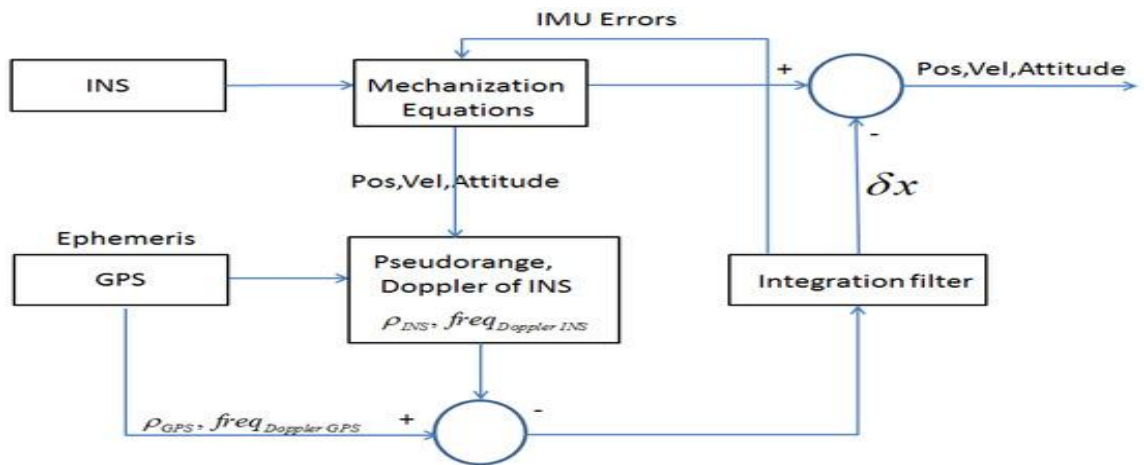


Fig.6.2 Tightly coupled integration scheme (in Visio)

The error states estimated by using Kalman-filter for the integration of MEMS INS & GNSS, considering an *ECEF* frame for tightly coupled integrated navigation scheme may be given as.

$$\hat{x}_k = [\varphi_x, \varphi_y, \varphi_z, \delta V_x^e, \delta V_y^n, \delta V_z^u, \delta \phi, \delta \lambda, \delta h, \varepsilon_{gbx}, \varepsilon_{gby}, \varepsilon_{gbz}, \nabla_{Abx}, \nabla_{Aby}, \nabla_{Abz}, \delta t, \delta t']^T. (6.5)$$

Where, $\delta t, \delta t'$ are the clock drifts error also known as biases which affect the receiver of GNSS.

Also for tightly-coupled structural design, the centralized Kalman-filter is designed by using a measurements model and a dynamic error states model.

6.1.4 Integration Implementation Approach

For the MESM based INS & GNSS integration system loosely and tightly coupling integration schemes are implemented. In mechanization derived navigation information the Kalman filter estimates the errors. By linearizing the mechanization equations MENS INS error model is obtained. Where second order terms are neglected. Mechanization error grows rapidly without feedback scheme & they can make the neglected terms important. Therefore can produce large errors into the integrated system. If high end inertial sensors are installed in inertial measurement device then they are can introduce small errors and integration scheme is valid for such functions. However in a short time interval, a low cost MEMS INS propagates large errors.

Magnetometer of AHRS system is used for availability & reliability during partially or fully GNSS signal outages or data loss and to enhance the system performance during signal obstruction in urban surroundings due to bridges, tunnels and reflections of signal. To mitigate the MEMS-INS errors during GNSS signal outages, Magnetometer is integrated by using Kalman filter data fusion. A Kalman filter which uses

magnetometer heading during GNSS signal outages is applied in the loosely and tightly coupled. To evaluate the performance of integrated navigation system van tests or road tests in urban area have been conducted. A feedback loop is used in a closed loop integrated navigation scheme. Using the error estimation obtained from the Kalman filter. Due to this raw sensor outputs are corrected and also other mechanization parameters. Thus maintaining the small error assumptions, mechanization propagates small errors. After every filter updates the error states in this case must be reset to zero.

6.2 System Model

6.2.1 Dynamic Error Model

Navigation equations describe the dynamics of the navigation states and are the non-linear system. The information from INS and GPS is normally blended using Kalman Filter in order to estimate the navigation states error. To formulate EKF, the establishment of linear dynamic model to estimate navigation error states is necessary.

Attitude Errors:

Attitude error in ENU frame in most extended linearized form is given as,

$$\begin{bmatrix} \delta \dot{\rho} \\ \delta \dot{r} \\ \delta \dot{A} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{M+h} & 0 \\ -\frac{1}{N+h} & 0 & 0 \\ \frac{-\tan \varphi}{(N+h)} & 0 & 0 \end{bmatrix} * \begin{bmatrix} \delta v^e \\ \delta v^n \\ \delta v^u \end{bmatrix} + R_b^l * \begin{bmatrix} \delta \omega_x \\ \delta \omega_y \\ \delta \omega_z \end{bmatrix} \\ + \begin{bmatrix} 0 & 0 & -\frac{V^n}{(M+h)^2} \\ \omega^e \sin \varphi & 0 & \frac{V^e}{(N+h)^2} \\ -\frac{\omega^e \cos \varphi - V^e \sec^2 \varphi}{N+h} & 0 & \frac{V^e \tan \varphi}{(N+h)^2} \end{bmatrix} * \begin{bmatrix} \delta \phi \\ \delta \lambda \\ \delta h \end{bmatrix}$$

$$+ \begin{bmatrix} 0 & \frac{V^e * \tan \varphi}{N+h} + \omega^e \sin \varphi & -\frac{V^e * \tan \varphi}{N+h} - \omega^e * \sin \varphi \\ -\frac{V^e * \tan \varphi}{N+h} - \omega^e * \sin \varphi & 0 & -V^n / (M+h) \\ \frac{V^e}{N+h} + \omega^e * \cos \varphi & V^n / (M+h) & 0 \end{bmatrix} * \begin{bmatrix} \delta \rho \\ \delta r \\ \delta A \end{bmatrix} \dots\dots\dots (6.6)$$

Linearization Equation for Position in ENU Frame

In the E-N-U-frame the position equation is given as

$$\delta \dot{r}^l = \begin{bmatrix} \delta \dot{\varphi} \\ \delta \dot{\lambda} \\ \delta \dot{h} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{M+h} & 0 \\ -\frac{1}{(N+h) * \cos \varphi} & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} \delta V^e \\ \delta V^n \\ \delta V^u \end{bmatrix} + \begin{bmatrix} 0 & 0 & -\frac{V^n}{(M+h)} \\ \frac{V^e * \tan \varphi}{(N+h) * \cos \varphi} & 0 & \frac{-V^e}{(N+h)^2 * \cos \varphi} \\ 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} \delta \varphi \\ \delta \lambda \\ \delta h \end{bmatrix} \dots\dots\dots (6.7)$$

Velocity Linearization Equation in ENU Frame

In the E-N-U-frame, Velocity equation is given as

$$\begin{bmatrix} \delta \dot{V}^e \\ \delta \dot{V}^n \\ \delta \dot{V}^u \end{bmatrix} = \begin{bmatrix} 0 & f_u & -f_n \\ -f_u & 0 & f_e \\ f_n & f_e & 0 \end{bmatrix} \begin{bmatrix} \delta \rho \\ \delta r \\ \delta A \end{bmatrix} + R_b^l \begin{bmatrix} \delta f_x \\ \delta f_y \\ \delta f_z \end{bmatrix} + \begin{bmatrix} 2\omega^e * \sin \varphi * V^u + 2\omega^e * \cos \varphi * V^n + \frac{V^n * V^e}{N+h * \cos^2 \varphi} & 0 & 0 \\ -2\omega^e * \cos \varphi * V^e - \frac{V^n * V^e}{N+h * \cos^2 \varphi} & 0 & 0 \\ -2\omega^e * \sin \varphi * V^e & 0 & \frac{2\gamma}{R} \end{bmatrix} \begin{bmatrix} \delta \varphi \\ \delta \lambda \\ \delta h \end{bmatrix} +$$

$$\begin{bmatrix} -\frac{V^e}{N+h} + \frac{\tan \varphi}{N+h} & 2\omega^e \sin \varphi + \frac{V^e \tan \varphi}{N+h} & -2\omega^e \cos \varphi - \frac{V^e}{N+h} \\ -\frac{V^e \tan \varphi}{N+h} - 2\omega^e \sin \varphi & -V^n/(M+h) & -V^n/(M+h) \\ 2\omega^e \cos \varphi + \frac{2V^e}{N+h} & 2V^n/(M+h) & 0 \end{bmatrix} * \begin{bmatrix} \delta V^e \\ \delta V^n \\ \delta V^u \end{bmatrix} \dots\dots\dots (6.8)$$

Table 6.1

Angular Rate	Derivative	Assumption
$\omega_{ie}^e = \begin{pmatrix} 0 \\ \omega^e \cos \varphi \\ \omega^e \sin \varphi \end{pmatrix}$	$\delta \omega_{ie}^e = \begin{pmatrix} 0 \\ -\delta \varphi \sin \varphi \omega^e \\ \delta \varphi \omega^e \cos \varphi \end{pmatrix}$	ω^e is constant and is given without any errors.
$\omega_{et}^e = \begin{pmatrix} -\frac{V^n}{M+h} \\ \frac{V^e}{N+h} \\ \frac{V^e \tan \varphi}{N+h} \end{pmatrix}$	$\delta \omega_{et}^e = \begin{pmatrix} -\frac{\delta V^n}{M+h} + \frac{\delta h V^n}{(M+h)^2} \\ \frac{\delta V^e}{N+h} - \frac{\delta h V^e}{(N+h)^2} \\ \frac{\delta V^e \tan \varphi}{N+h} + \frac{V^e \delta \varphi \sec^2 \varphi}{N+h} - \frac{\delta h V^e \tan \varphi}{(N+h)^2} \end{pmatrix}$	Assume M and N are error free because the terms with $(M+h)^2$ and $(N+h)^2$ are very small.

6.2.2 INS-GNSS Measurement Model

To formulate a Kalman filter to update the navigation system a linear dynamic model that describes the errors is needed. The equations used for navigation are non-linear. These equations are used to demonstrate the navigation states dynamics. The information from INS and GPS is normally blended using Kalman filter to approximate the error of navigation states. To formulate Kalman filter, the establishment of linear dynamic model to estimate navigation error states is necessary.

Kalman Filter Model:

The Kalman filter equations for the state space model is given as:

$$\dot{x}_t = F(t) * x(t) + G(t) * w(t)$$

$$z_t = H x(t) + v(t)$$

The prediction & update are two main steps for the Kalman filter algorithm.

Prediction:

Estimation of Predicted state: $\hat{x}_k^- = \Phi_k * \hat{x}_{k-1}$

Estimation of Predicted covariance: $P_k^- = \Phi_k * P_{k-1} * \Phi_k^t + Q_{k-1}$

Update:

Kalman gain $K_k = P_k^- * H^t * (HP_k^- H^t + R)^{-1}$

Updated state estimate $\hat{x}_k^+ = \hat{x}_k^- + K_k * (z_k - H\hat{x}_k^-)$

Updated covariance estimate $P_k = (I - K_k H) * P_k^- * [I - K_k H]^T + K_k R K_k^T$

The error state vector $\hat{x}_k$ for an ENU frame are given below.

$$\hat{x}_k = [\varphi_x, \varphi_y, \varphi_z, \delta V_x^e, \delta V_y^n, \delta V_z^u, \delta\phi, \delta\lambda, \delta h, \varepsilon_{gbx}, \varepsilon_{gby}, \varepsilon_{gbz}, \nabla_{Abx}, \nabla_{Aby}, \nabla_{Abz}]^T \text{ ..(6.9)}$$

Where:

$\varphi_x, \varphi_y, \varphi_z$ Errors in pitch, roll and yaw.

$\delta V_x^e, \delta V_y^n, \delta V_z^u$ Errors in constant velocity for an ENU frame

$\delta\phi, \delta\lambda, \delta h$ Errors in latitude, longitude and altitude

$\varepsilon_{gbx}, \varepsilon_{gby}, \varepsilon_{gbz}$ Gyroscopes constant drifts

$\nabla_{Abx}, \nabla_{Aby}, \nabla_{Abz}$ Accelerometers constant biases

In above equation, the navigation error states are the velocity errors, position errors & attitude errors $\delta\phi, \delta\lambda, \delta h$. The other symbols are related to the sensor error models.

Whereas, gyroscopes and accelerometers scale factors can also be added to the above equation.

It is a common practice to model the residual stochastic error part as a first-order Gauss Markov process, after the removal of the deterministic error parts of the inertial sensor biases.

The process co-variance matrix P is given as:

$$P = \text{diag}[\sigma_{\varphi_x}^2, \sigma_{\varphi_y}^2, \sigma_{\varphi_z}^2, \sigma_{\delta V_x^e}^2, \sigma_{\delta V_y^n}^2, \sigma_{\delta V_z^u}^2, \sigma_{\delta \phi}^2, \sigma_{\delta \lambda}^2, \sigma_{\delta h}^2, \sigma_{\varepsilon_{gbx}}^2, \sigma_{\varepsilon_{gby}}^2, \sigma_{\varepsilon_{gbz}}^2, \sigma_{\nabla_{Abx}}^2, \sigma_{\nabla_{Aby}}^2, \sigma_{\nabla_{Abz}}^2] \dots (6.10)$$

$w(t)$ is process noise vector and it contain the random errors of gyroscopes and accelerometers.

$$w(t) = [\varepsilon_{grx}, \varepsilon_{gry}, \varepsilon_{grz}, \nabla_{arx}, \nabla_{ary}, \nabla_{arz}]^T \dots (6.11.)$$

Where, Q is the process noise co-variance matrix consisting of covariance of gyroscopes and accelerometers.

$$Q = \text{diag}([\sigma_{gx}^2, \sigma_{gy}^2, \sigma_{gz}^2, \sigma_{ax}^2, \sigma_{ay}^2, \sigma_{az}^2, 1, 1, 1, 0, 0, 0, 0, 0, 0]) \dots (6.12)$$

Measurement:

$$z = H * \hat{x}_k + v \dots (6.13)$$

Where, measurement vector is z & it can be written as;

$$z = \begin{bmatrix} v_{INS} - v_{GPS} \\ \varphi_{INS} - \varphi_{GPS} \\ \lambda_{INS} - \lambda_{GPS} \\ h_{INS} - h_{GPS} \end{bmatrix}$$

Or

$$z = \begin{bmatrix} v_{INS} - v_{GPS} \\ (R_M + h) * (\varphi_{INS} - \varphi_{GPS}) \\ (R_N + h) * \cos(\lambda_{INS} - \lambda_{GPS}) \\ h_{INS} - h_{GPS} \end{bmatrix} \dots (6.14)$$

Where, v is measurement noise vector consisting of velocity and position noise.

$$v = [v_{Vx}, v_{Vy}, v_{Vz}, v_{\phi}, v_{\lambda}, v_h]^T \quad \dots\dots\dots (6.15)$$

Also, R is the measurement noise covariance matrix and is given as:

$$R = diag[\sigma_{VGx}^2, \sigma_{VGy}^2, \sigma_{VGz}^2, \sigma_{\phi_G}^2, \sigma_{\lambda_G}^2, \sigma_{h_G}^2] \quad \dots\dots\dots (6.16)$$

The measurement matrix H is given by:

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & (R_M + h) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & (R_N + h)Cos\phi & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \dots\dots\dots (6.17)$$

From the position & velocity covariance matrix obtained by GNSS (GPS) is fed to Kalman filter as measurement noise. The measurement matrix shows that the velocities and position are given in off diagonal terms. They are properly accounted and ensures the correlation between the GPS and present in the covariance matrix. GPS provide the measurements updates to the INS-filter. Hence, the MEMS INS will not work properly if GNSS/GPS is not available or outages occur.

6.3 Magnetometer Heading Aiding

During integrated navigation applications, the GNSS may suffer from its signal outages due to obstruction of signals, jamming of its signals, multipath, spoofing and interferences. For critical navigation applications, stand-alone GNSS cannot assist for the best purpose. To obtain position, attitude and velocity solutions of a vehicle, MEMS INS measurements are mechanized during GNSS signal blockage. The navigation solution obtained by INS during GNSS obstruction is mostly depends on inertial sensors quality of measurements. By using top grade inertial sensors the improved navigation solution can be obtained by the INS. Nowadays, best grade MEMS INS are manufactured and utilized in navigation applications. The MEMS based inertial navigation system used in this research has very good performance and with the added navigation it gives best performance. Besides these characteristics, modern inertial navigation system constructed from micro-electro-mechanical system technology cannot give best performance with any added navigation system. Thus, they are integrated with global navigation system or any other reliable navigation system, for their best performance and best navigation.

The large errors in the measurements cannot be compensated by using 15 state Kalman filter. These errors contribute large errors in attitude estimation. The errors in velocity grows due to growing errors in heading assessment. Since, the projection of forward velocity component in lateral direction contribute due to heading. While velocity errors in the horizontal direction by projection of very strong gravity signal are contributed by roll and pitch. Roll & pitch angles can be measured by using accelerometer's readings. Whereas, inertial navigation system can estimate the correct heading. By the integration of velocity position can be estimated. Therefore, large errors in velocity measurements

ultimately leads to large errors in position measurements, when GNSS signal are not available in integrated navigation of MEMS-INS & GNSS.

Magnetometer is utilized for heading estimation from true magnetic north using magnetic field model. In many low cost MEMS INS & GNSS integrated navigation applications, the magnetometer may be the one of the reliable way to have a good heading calculations and its heading can be utilized during GNSS signals outages. Using magnetometers heading observations in MEMS INS-GNSS integrated navigation, provides reasonable accuracy during GNSS signals outages.

To incorporate the Magnetometer's heading observation into the MEMS INS & GNSS integrated navigation filter measurement model, the equation involving the heading errors is required with the system error states [35]. From two levelled horizontal magnetometer's signals, its heading information can be obtained from the following equation.

$$\psi = - \arctan (h_y/h_x) \dots\dots\dots (6.18)$$

Where

h_y & h_x represent the magnetic field vector of the two levelled measurements in a coordinate system attached to the device comprising of the magnetometer (or compass body).

ψ is the angle with the true north made by the device.

6.4 Magnetometer Online Calibration

To use the accurate heading of magnetometer during integrated navigation solution, its calibrated readings and compensated parameters are required. These parameters are based on magnetometer's biases, scale factors, misalignments etc. The existing method of its calibration are mainly based on magnetometer's calibration in lab using different calibrations techniques. From these calibration methods, one of the most accurate and precise calibration method is 12 position calibration method. In this method magnetometer axes are place at different 12 positions and data is recorded. From this recorded data its parameters are calculated. From calculated parameters the calibrated and compensated magnetometer readings are obtained which is used to find the accurate heading. For this process reference magnetic field is also required to obtain the precise and calibrated output of the magnetometer. To obtain the reference magnetic field we use the updated version of world's magnetic model 2020-2025. We use ENU frame for magnetometer parameters calculation through calibration.

We calibrated magnetometer in lab using 12 positions calibration method. The parameters obtained are given in the table below.

Table.6.2 Magnetometer Calibration Parameters

Axis/Parameters	Mx	My	Mz
Bias	0.0029164	0.0014059	-0.0020832
Scale Factor	0.91139	0.90828	0.91916
Misalignments	-0.0030401	0.0050938	-0.00032331
	-6.8934e-06	0.001792	0.0011975

The main challenge of today's novel research is the online calibration of magnetometer. Since the magnetometer online heading can be used for the best navigation solution. That's why online calibration of magnetometer is very useful to estimate the accurate

heading. To perform online calibration of magnetometer a field test on a van was devised in which an AHRS comprising on magnetometer, a high grade reference INS (RINS i.e. R for reference) and a GNSS receiver was installed on a van used for field testing. Different pitch-roll sectors were defined. The value of pitch-roll sectors lies in +25 deg to -25 deg. The raw output of magnetometer was divided into sectors according to the pitch-roll sectors. Since magnetometer equations for raw output are as follows.

$$h'_x = b_x + \gamma_x h_x \quad \dots\dots\dots (6.19)$$

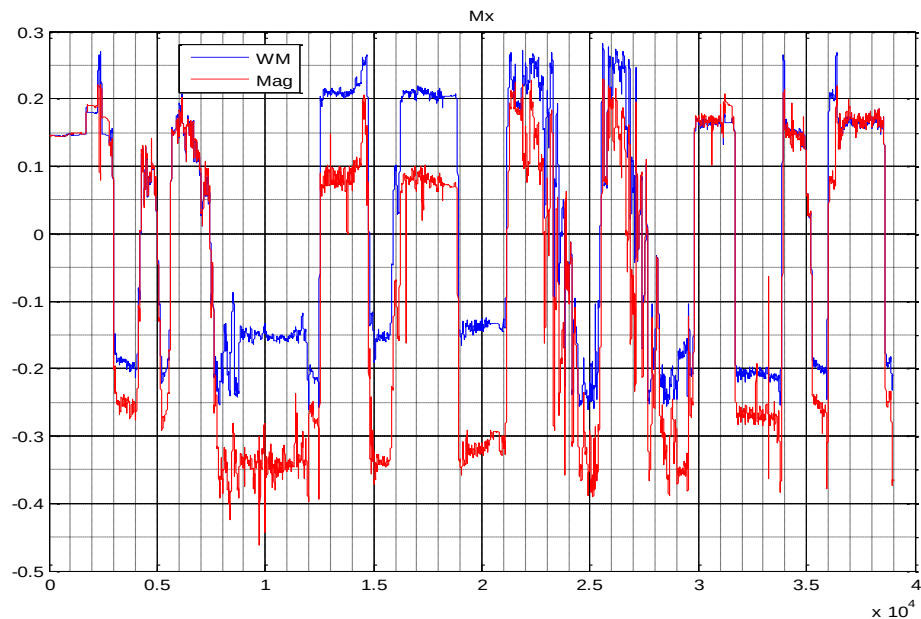
$$h'_y = b_y + \gamma_y h_y \quad \dots\dots\dots (6.20)$$

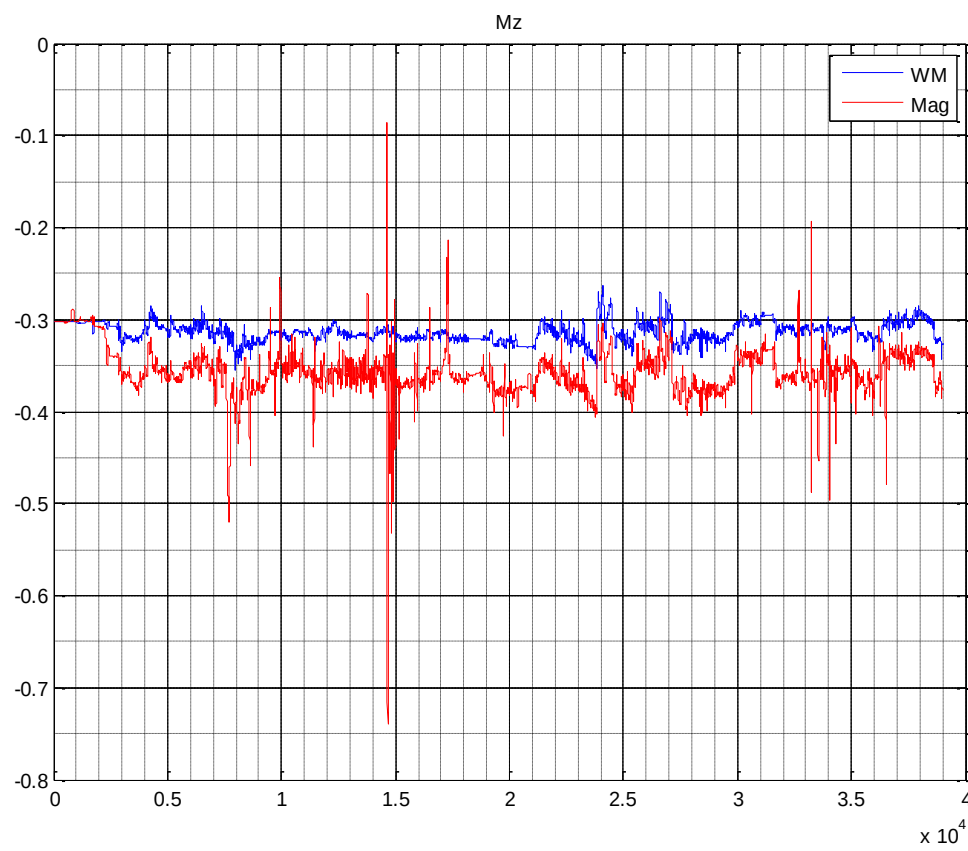
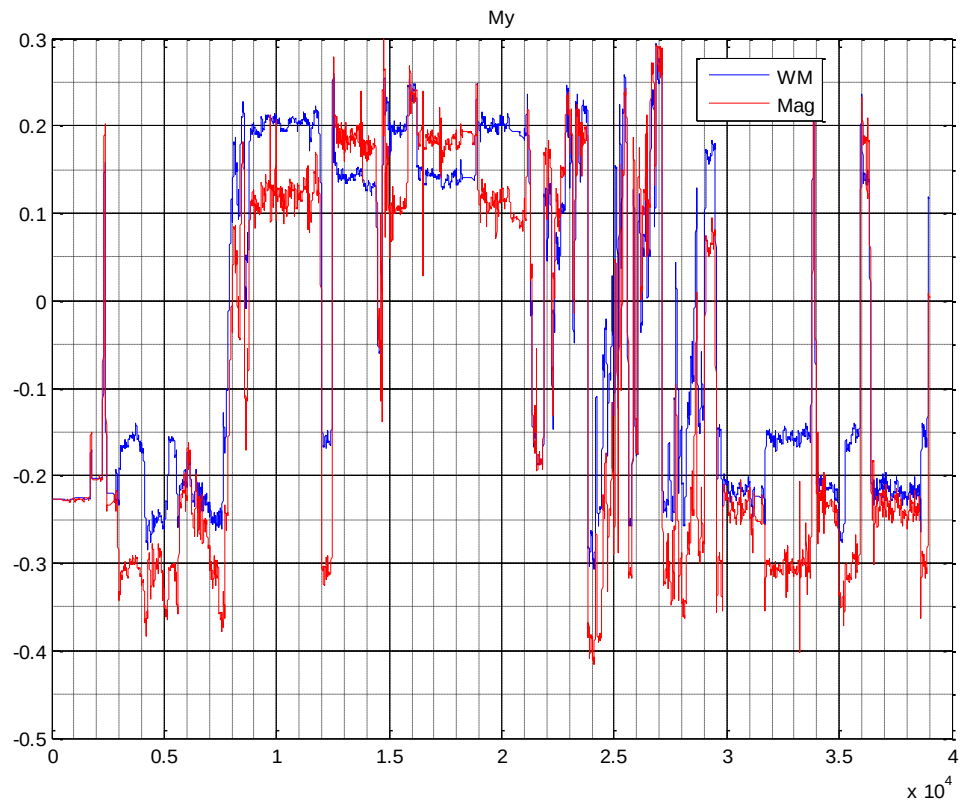
$$h'_z = b_z + \gamma_z h_z \quad \dots\dots\dots (6.21)$$

Where, h'_x , h'_y & h'_z are raw/un-calibrated magnetometer output

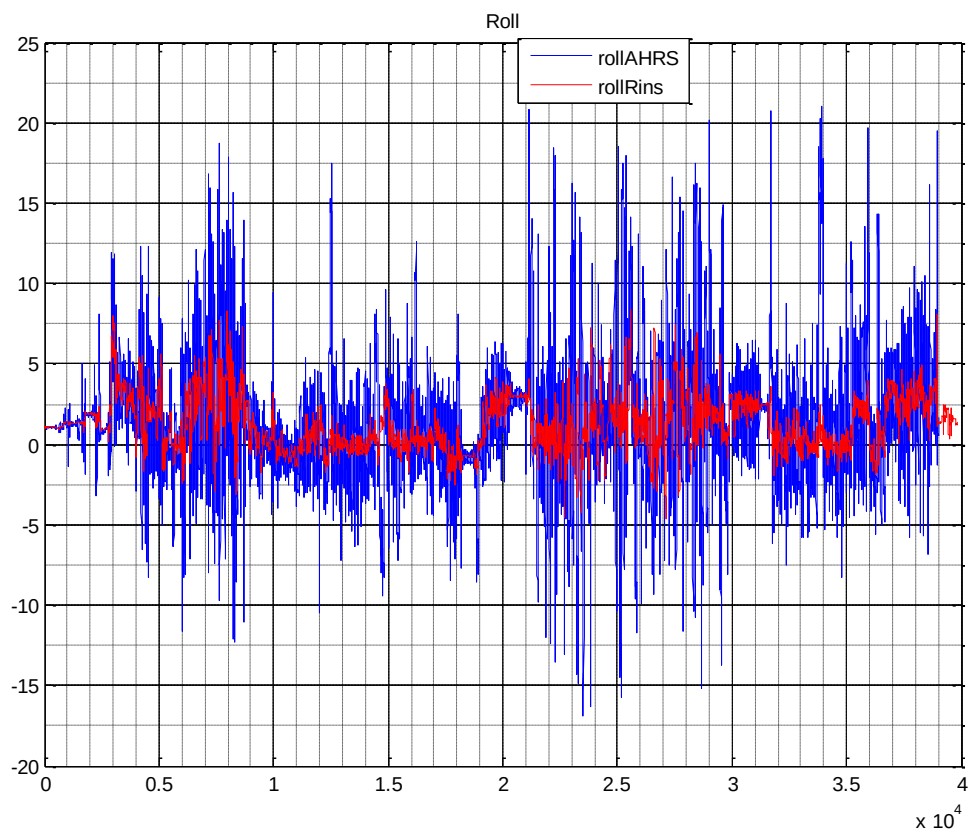
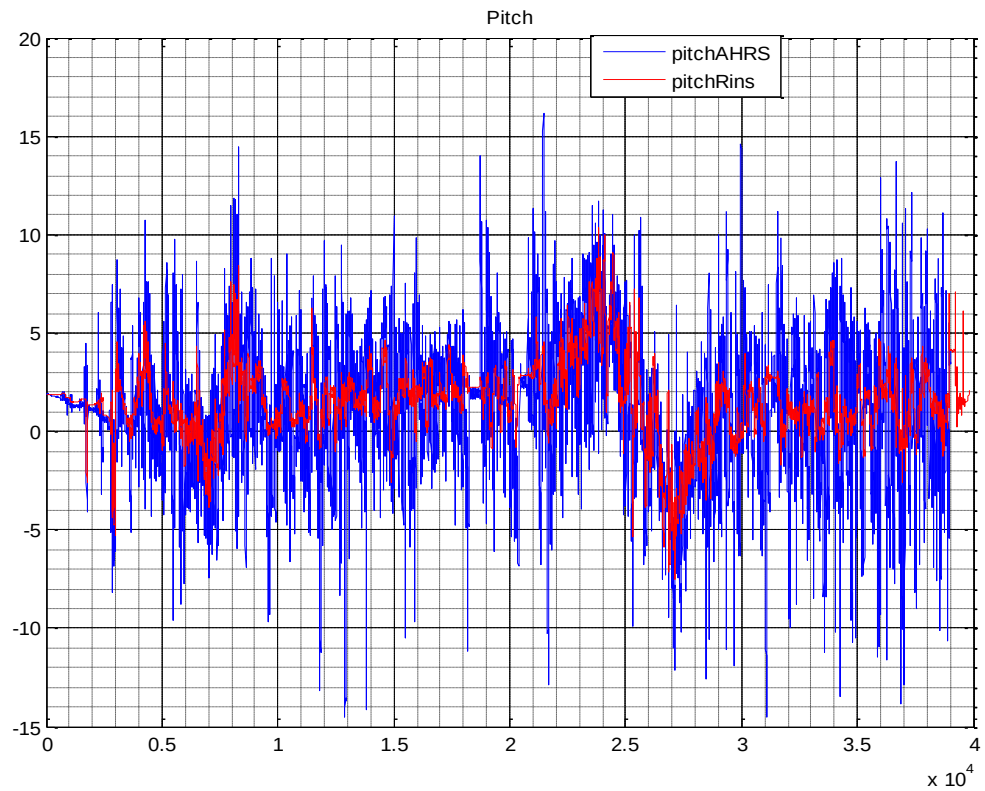
From aforementioned equations, it is clear that there are minimum three unknowns. To solve these equations minimum three different raw outputs of magnetometers are required to calculate the calibrated parameters of magnetometer for each pitch-roll sectors. Magnetometers related different plots are shown in the following figures.

1. Plots of Magnetometer Mx, My & Mz components Vs World Magnetic Field Components

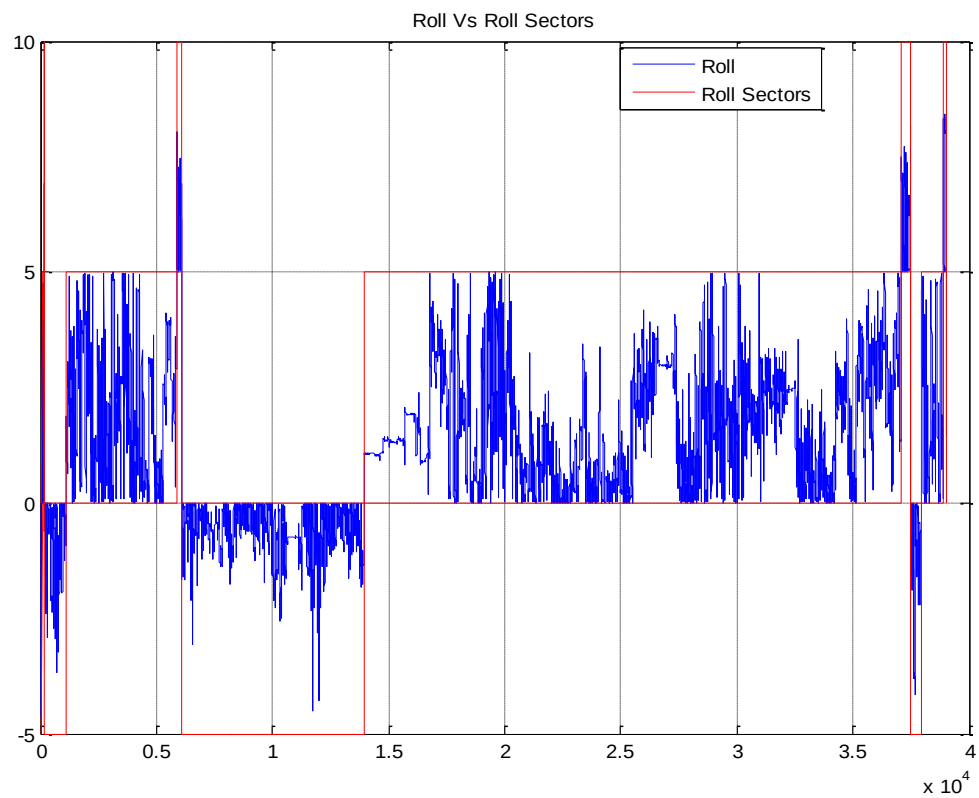
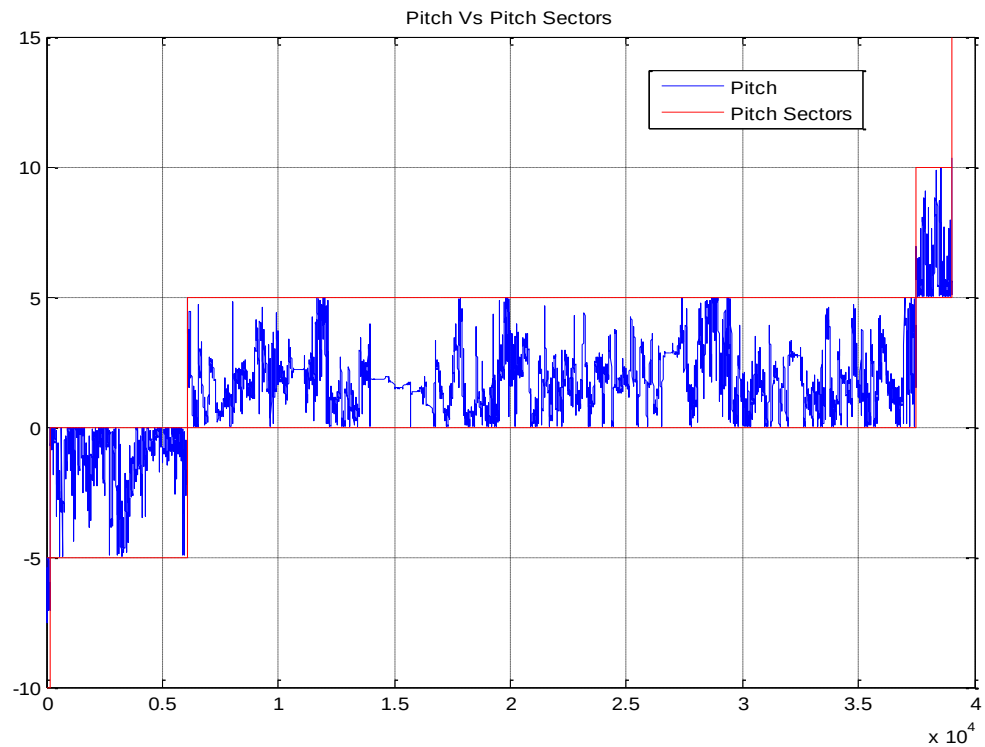




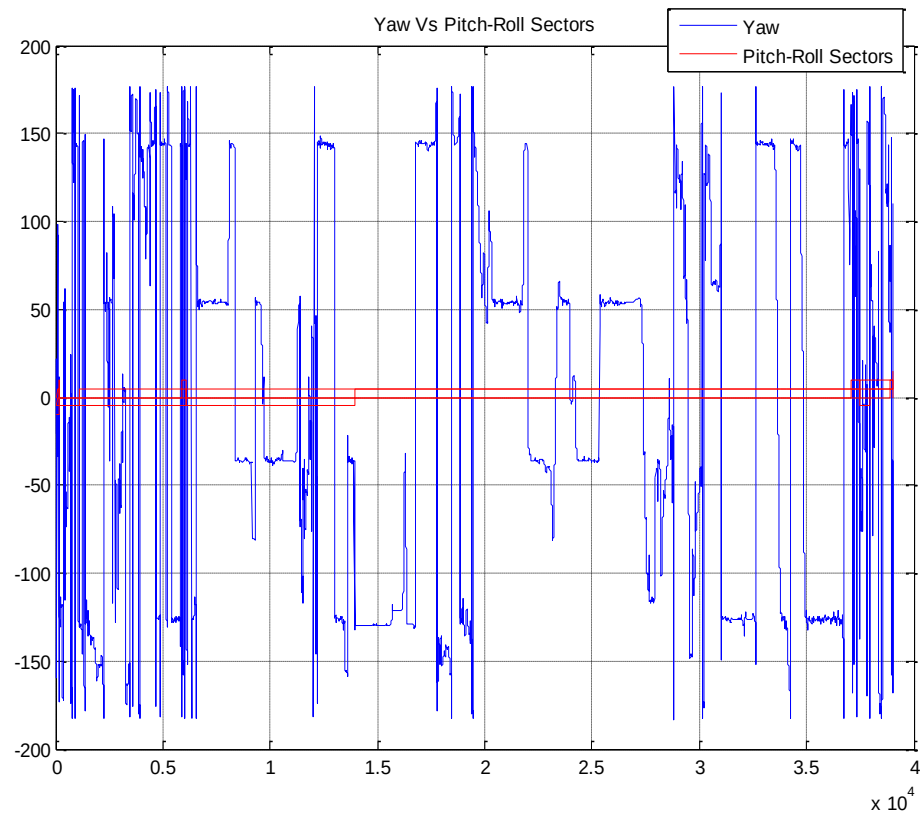
2. AHRS & Reference INS pitch & roll plots



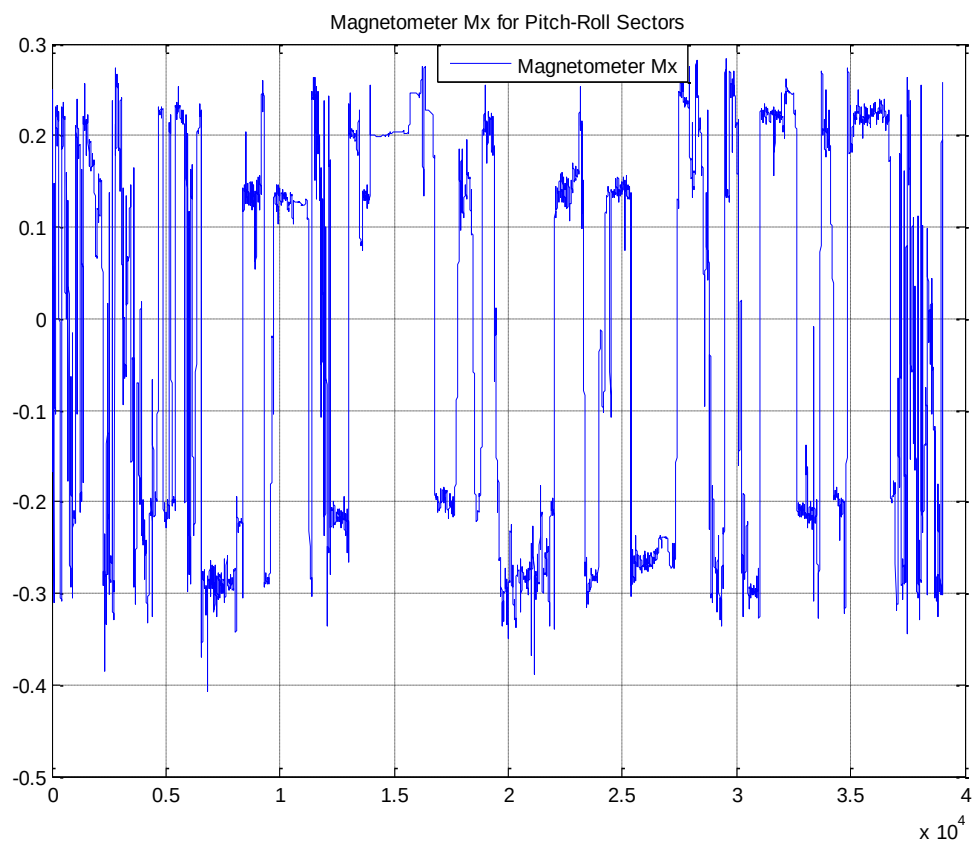
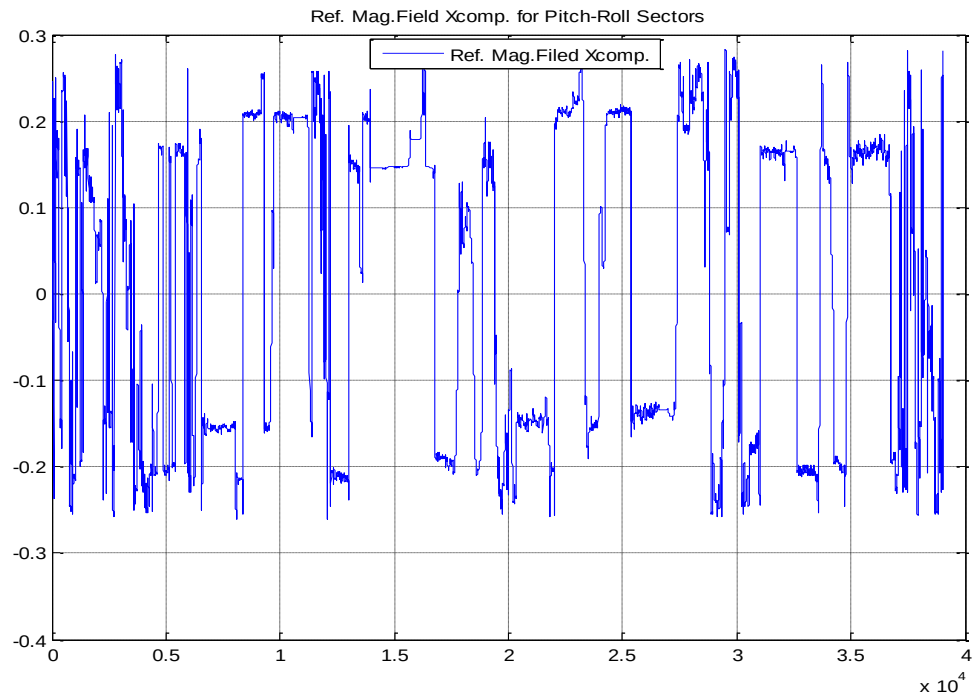
3. Plots of AHRS Pitch & roll for different Pitch-Roll Sectors during Magnetometer online calibration

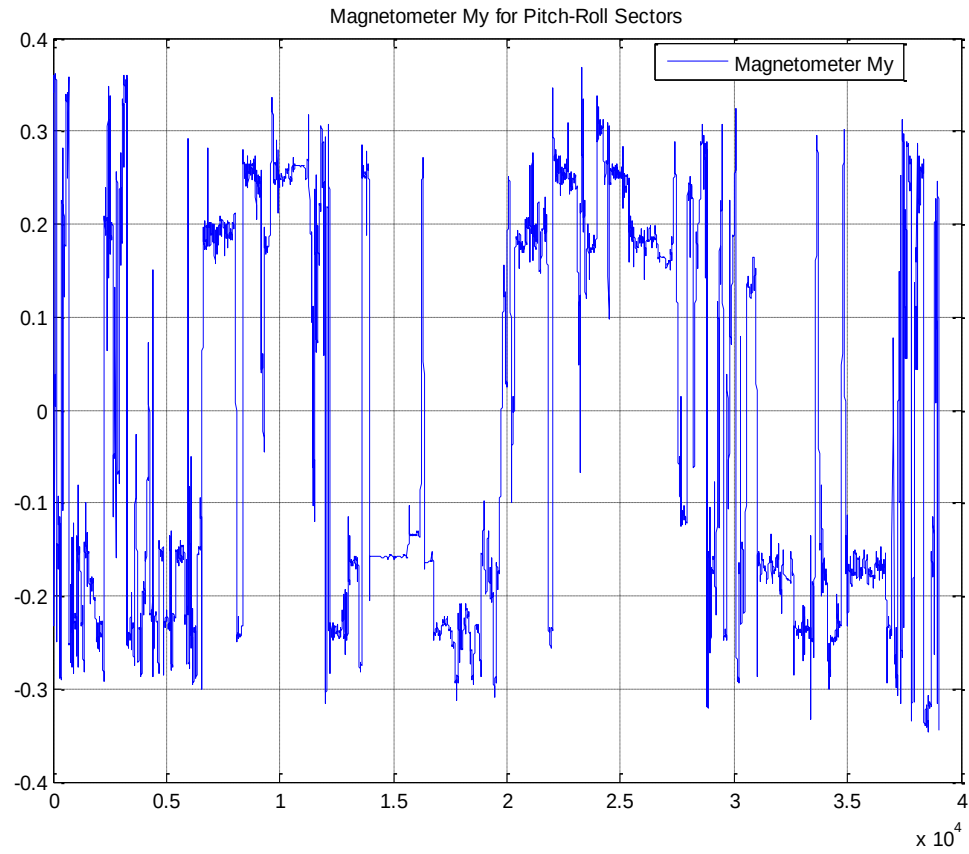
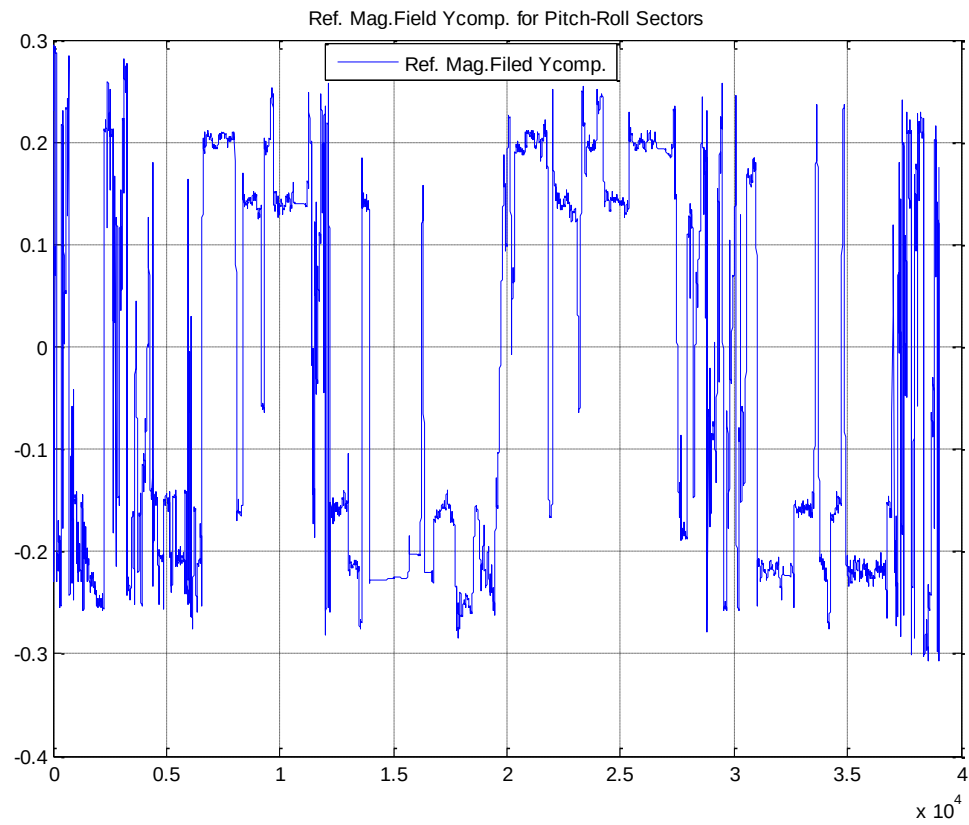


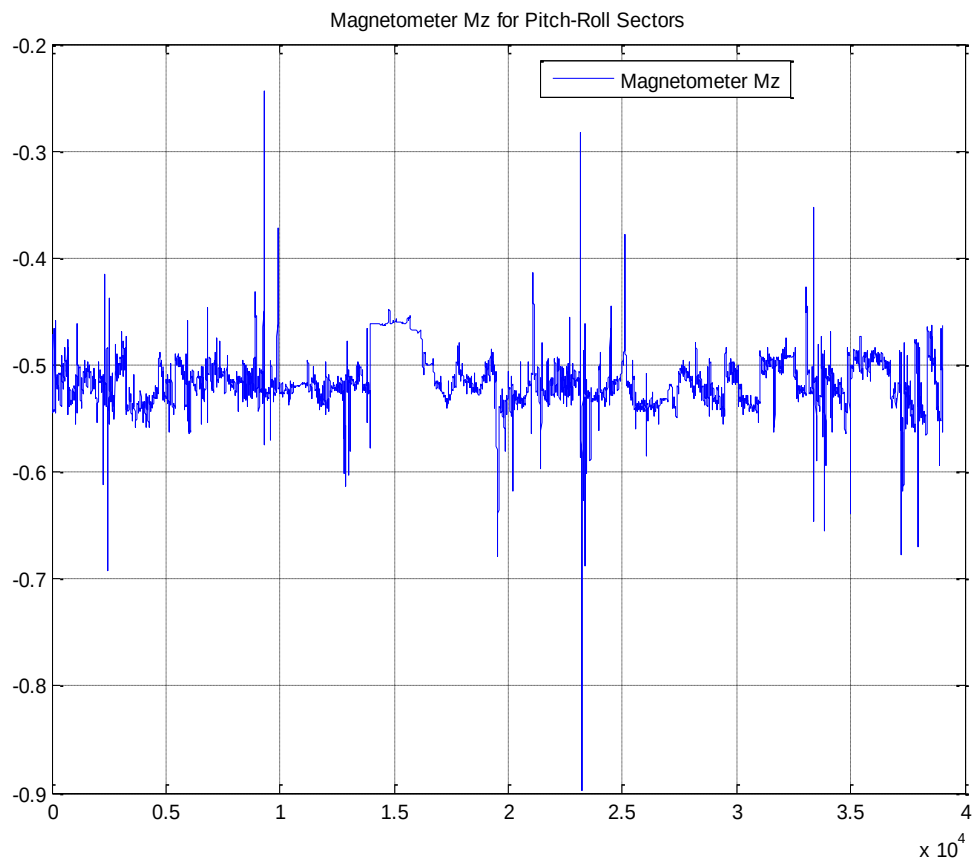
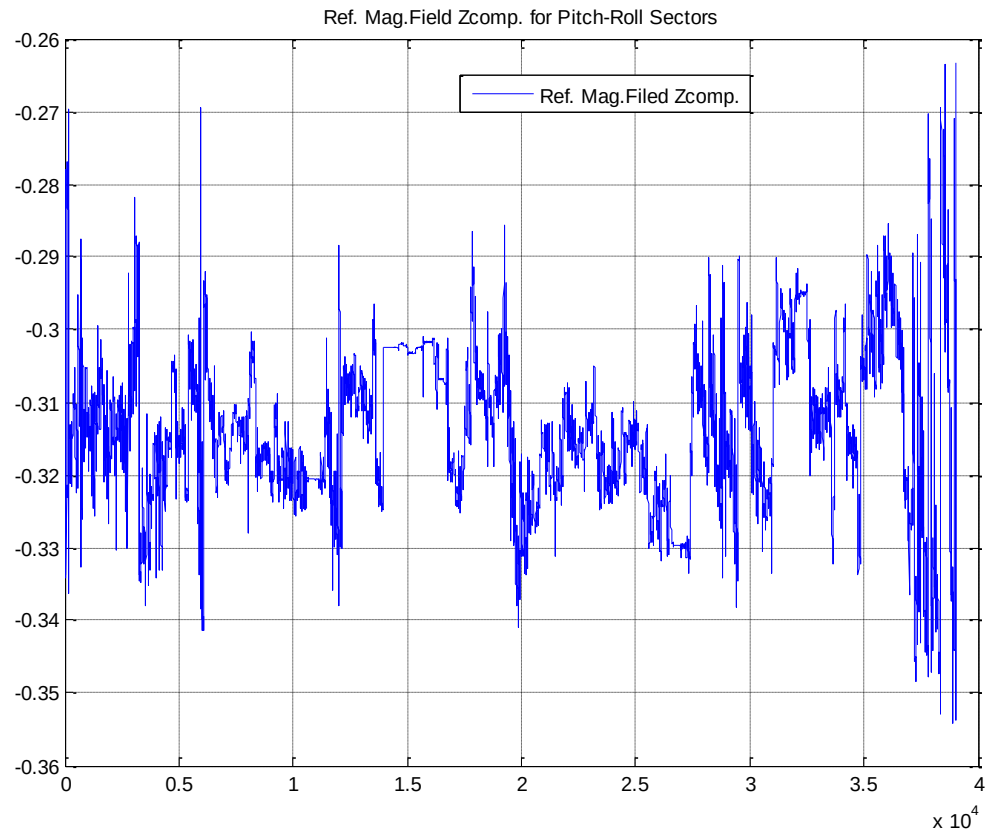
4. Plots of Magnetometer Heading for different Pitch-Roll Sectors during Magnetometer online calibration



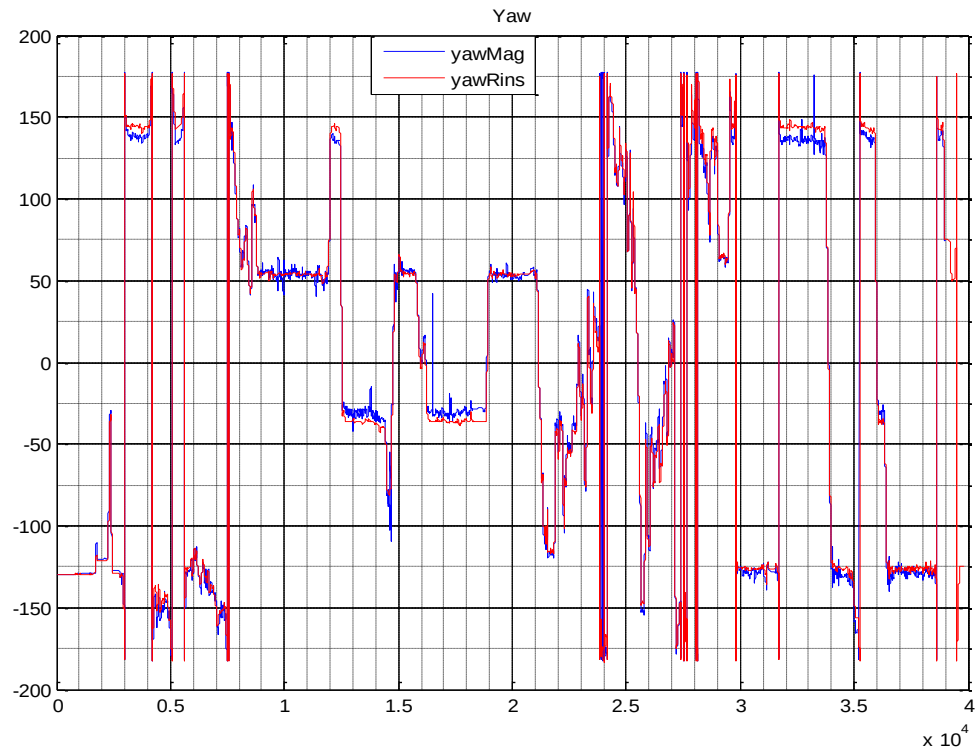
5. Reference magnetic Field X, Y & Z components & Magnetometer Mx, My & Mz magnetic components for Different Pitch-Roll Sectors



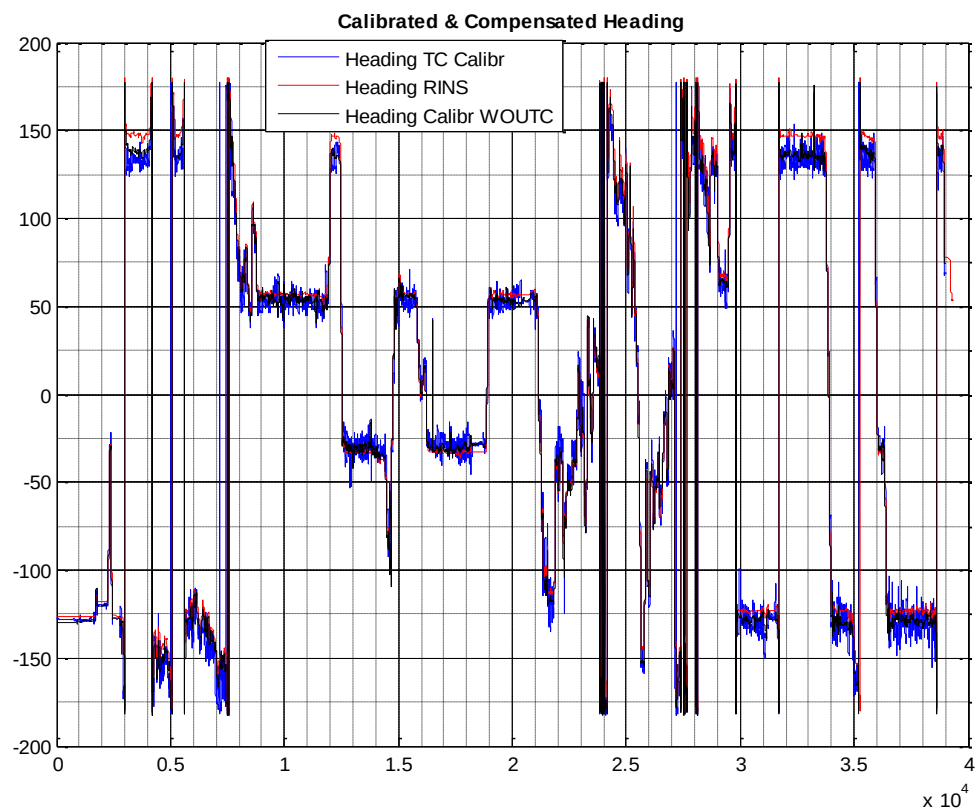




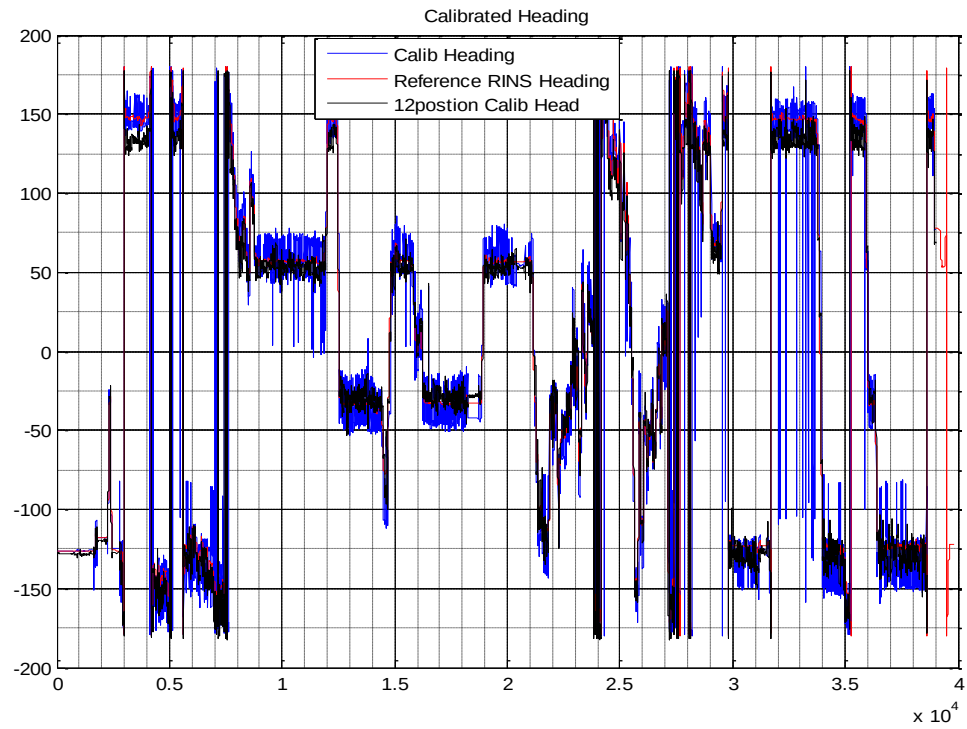
6. Plots of Magnetometer Calibrated Heading and Reference Heading by INS



7. Plots of Magnetometer Calibrated & Compensated heading, calibrated without compensated heading & reference heading by INS



8. Plots of Magnetometer online Calibrated Heading, Heading Estimated by 12 Position Test & Reference Heading of INS



7. RESULTS AND DISCUSSION

Experimental Results and Performance Analysis

In this chapter, the performance analysis algorithms for low cost MEMS INS & GNSS simple integrated navigation developed & evaluated. Also for MEMS INS, GNSS & Magnetometer integrated navigation the algorithm for performance evaluation is designed. For the performance evaluation, the integrated navigation algorithm implemented in MATLAB-based program has been used to generate the integrated navigation solutions by processing the data collected from the MEMS INS, GNSS & Magnetometer in field tests. Details of system design and test description for data collection and processing procedures are presented. Navigation performance analysis with and without GNSS updates is evaluated. The heading updates from Magnetometer are used during GNSS outages.

7.1 Specification of MEMS INS & GNSS

7.1.1 MEMS INS Specification

MEMS INS is comprises on MEMS based Gyroscopes and Accelerometers. These MEMS sensors are installed in 3D arrangement in this way that they are perpendicular to each other i.e. they make an angle of 90 degree with each other. The specification of MEMS INS used here is given in below table.

Table: 7.1 MEMS INS Specification

MEMS Gyroscope Specification		MEMS Accelerometers Specification	
Bias Stability	0.5 deg/hr	Bias Stability	1 mili g
Angular Rate	± 300 deg/sec	Acceleration	± 30 g
Data output	Digital	Data output	Digital

7.1.2 GNSS Specification

The specifications of GNSS receiver is given in below table.

Table: 7.2 GNSS Specification

GNSS Specification	
Position Accuracy	5m
Velocity Accuracy	0.2 m/sec

7.1.3 Reference Navigation System

A reference navigation system of tactical grade is used for the comparison of MEMS INS & GNSS performance. A tactical grade reference navigation system which is consisting of GNSS receiver and a tactical grade INS. The attitude of MEMS INS is compared with the attitude of reference navigation system.

7.2 MEMS-INS, GNSS & AHRS-Magnetometer Field Testing Description

To analyse the performance of integrated navigation solution of the MEMS INS, GNSS & Magnetometer, a field test is carried out in the test vehicle at Kashmir highway Islamabad. The MEMS INS, AHRS Magnetometer & reference navigation system were installed on testing vehicle in the centre. The GNSS antenna was also installed in the centre of the vehicle on its roof. The raw data from the MEMS INS, AHRS Magnetometer & GNSS was collected and stored for navigation solution processing. The navigation data was also recorded and processed from reference navigation system. For self-alignment process the initial 5 minute data was recorded in stationary state. Then field test using testing van was carried out for about 65 minutes. Approximately

a total distance of 40 Km was travelled during the van test / field test. The trajectory used during van test of integrated navigation systems is shown in Figure

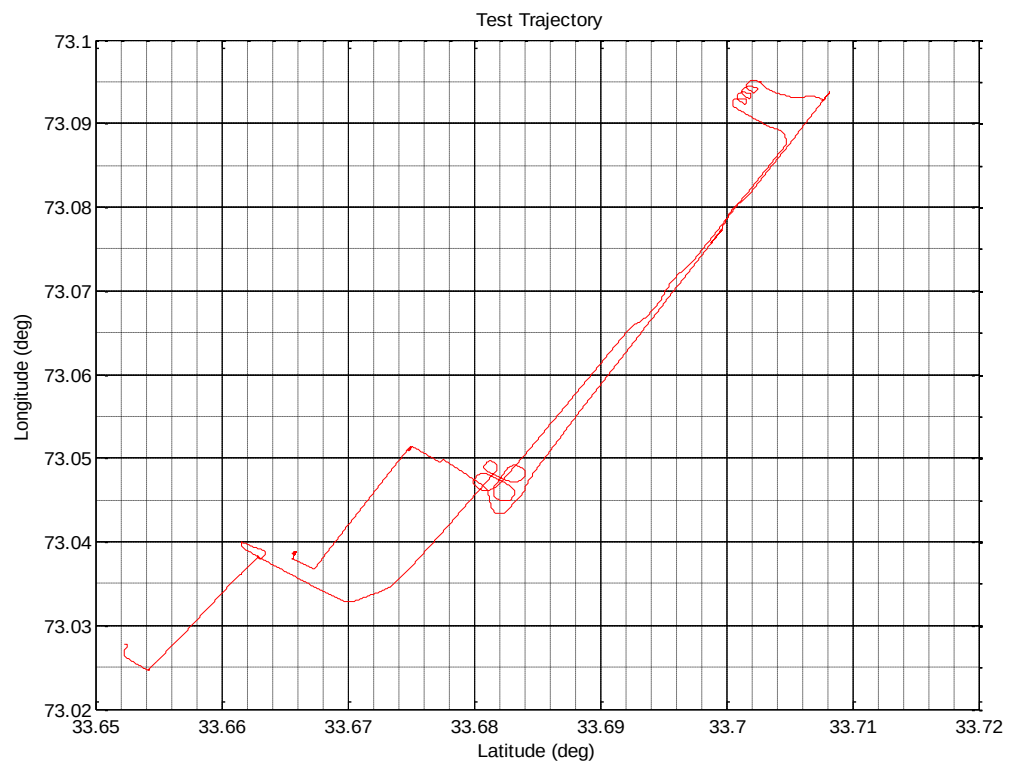


Figure 7.1 **Reference test trajectory for INS/GNSS/Magnetometer integration**

7.2.1 The GNSS Performance over test trajectory

The GNSS latitude, longitude and height performance at test trajectory is in the figure

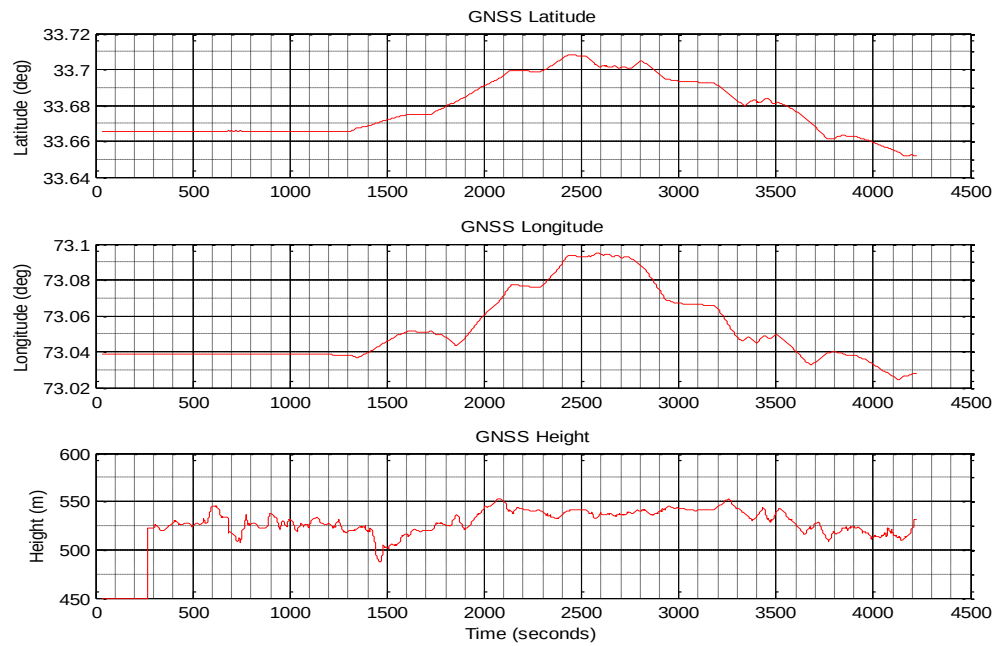


Figure 7.2 GNSS latitude, longitude & height over test trajectory

The GNSS east, north & up velocity components over the trajectory are shown in figure

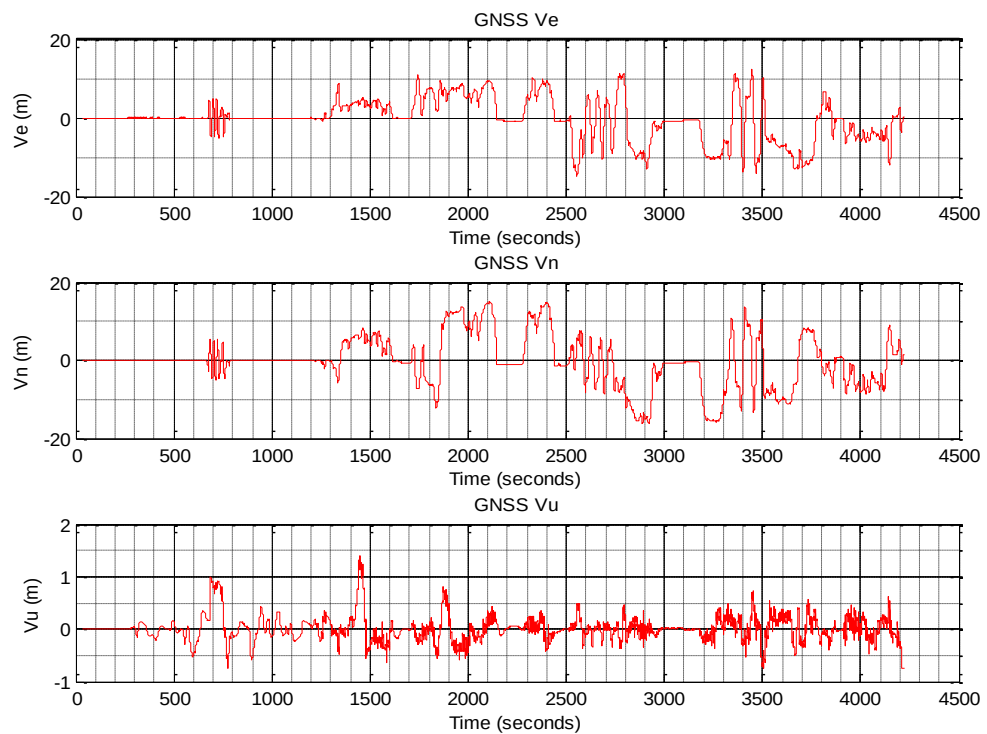


Figure 7.3 GNSS east, north & up velocity components over test trajectory

7.2.2 MEMS INS Performance Comparison with Tactical Grade Reference INS

The pitch, roll and yaw of MEMS INS and tactical grade reference INS is given in figures below. The MEMS INS has drifts in pitch, roll and yaw angles measured over the entire test trajectory.

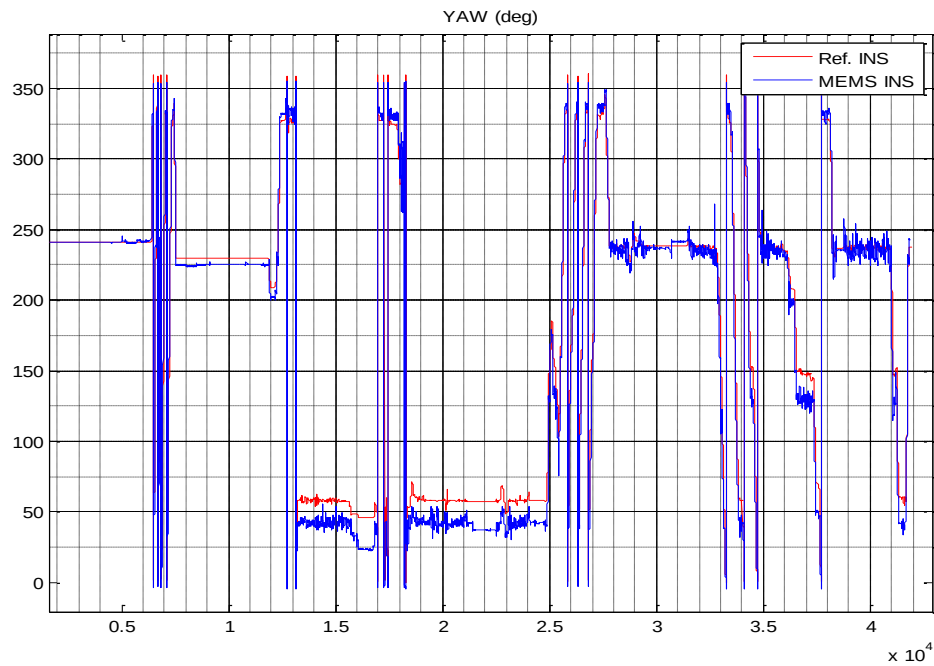


Figure 7.4 Comparison of MEMS INS & Ref. INS Yaw

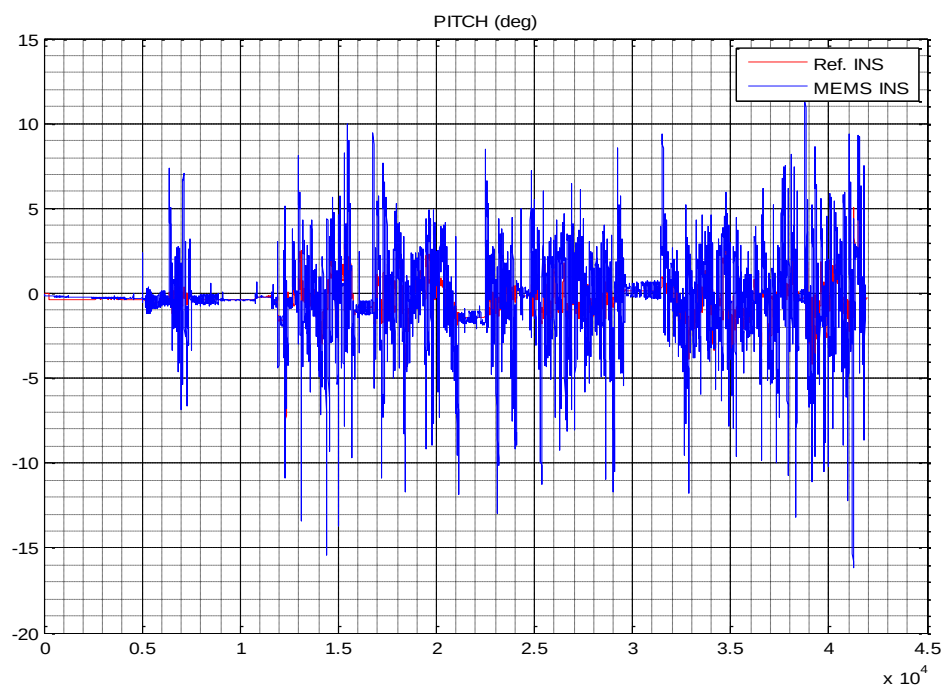


Figure 7.5 Comparison of MEMS INS & Ref. INS Pitch

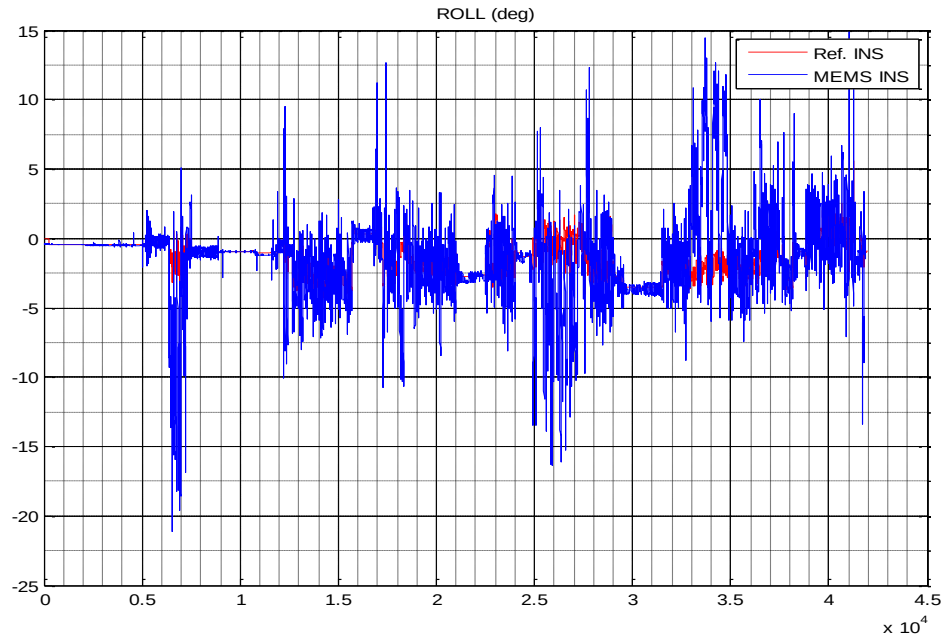


Figure 7.6 Comparison of MEMS INS & Ref. INS Roll

7.2.3 Magnetometer heading comparison with GNSS Heading over Test Trajectory

Magnetometer heading comparison with the GNSS heading over the test trajectory is shown in figure below

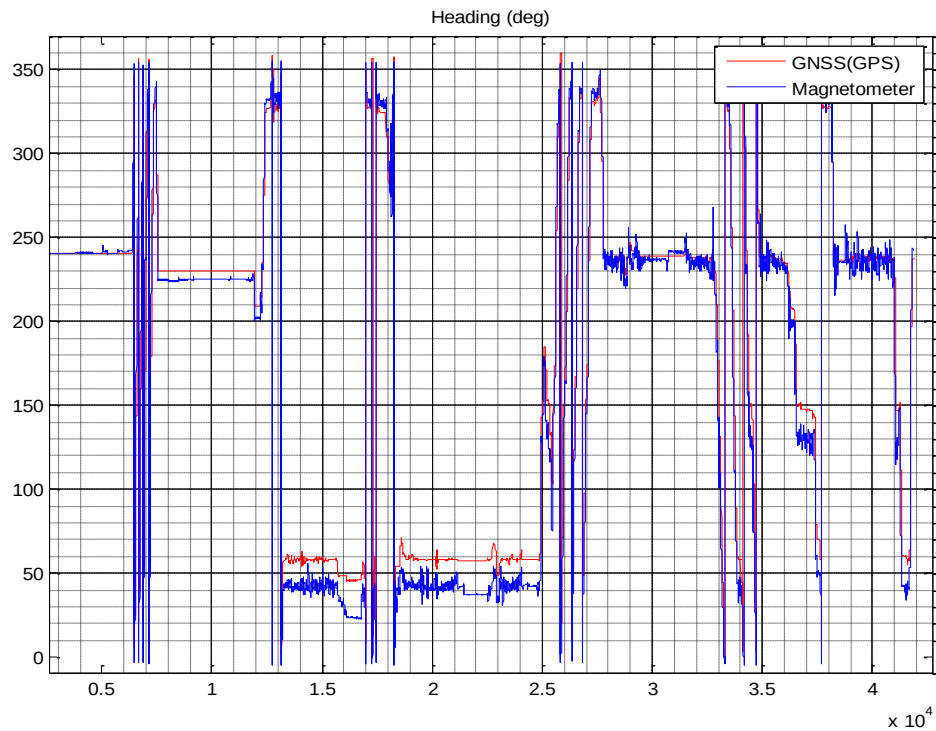


Figure 7.7 Magnetometer & GNSS heading comparison over trajectory

7.2.4 MEMS INS and GNSS Simple Integrated Performance Comparison with MEMS INS

MEMS INS attitude comparison with integrated attitude. The integrated results are obtained by updating GNSS position & velocities. The figures below indicates that the integrated solution is more precise and have less errors as compared to the MEMS INS standalone navigation solution. Also at start the drifts in MEMS INS solution are low. With the passage of time the drifts become very large. Hence the MEMS INS is low accuracy navigation system and can't be used for long time due to unbounded drifts.

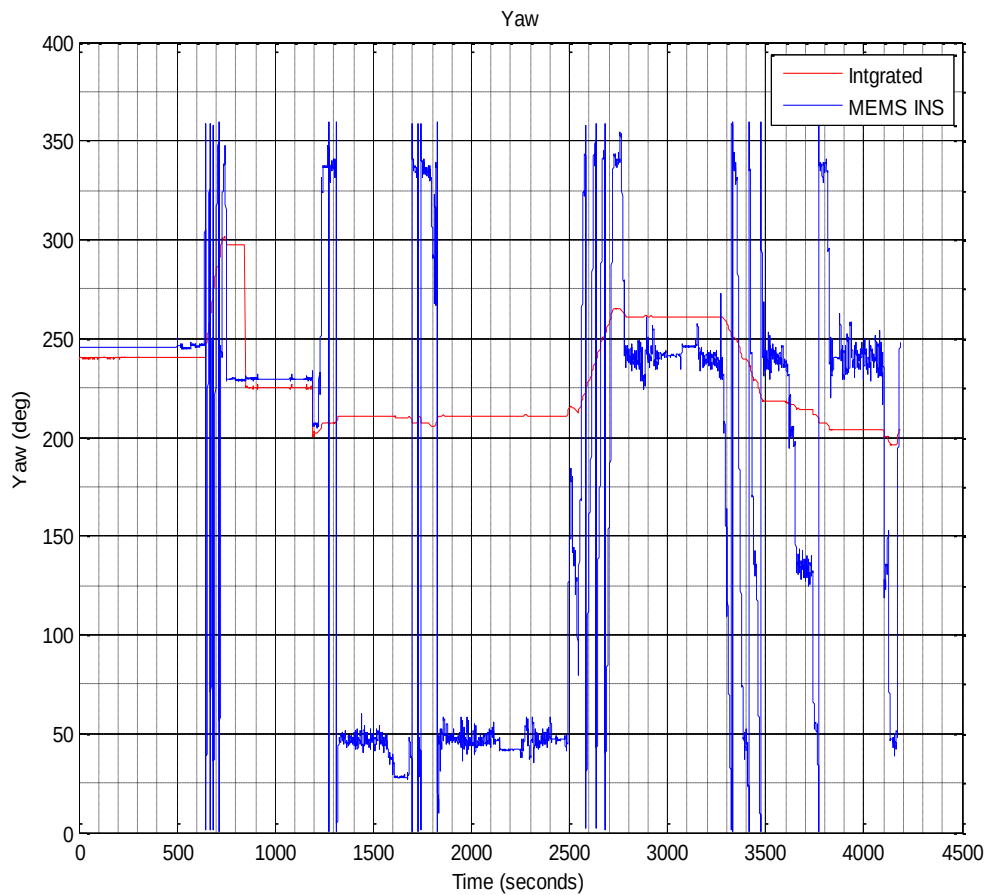


Figure 7.8 Comparison of yaw of MEMS INS and simple integrated

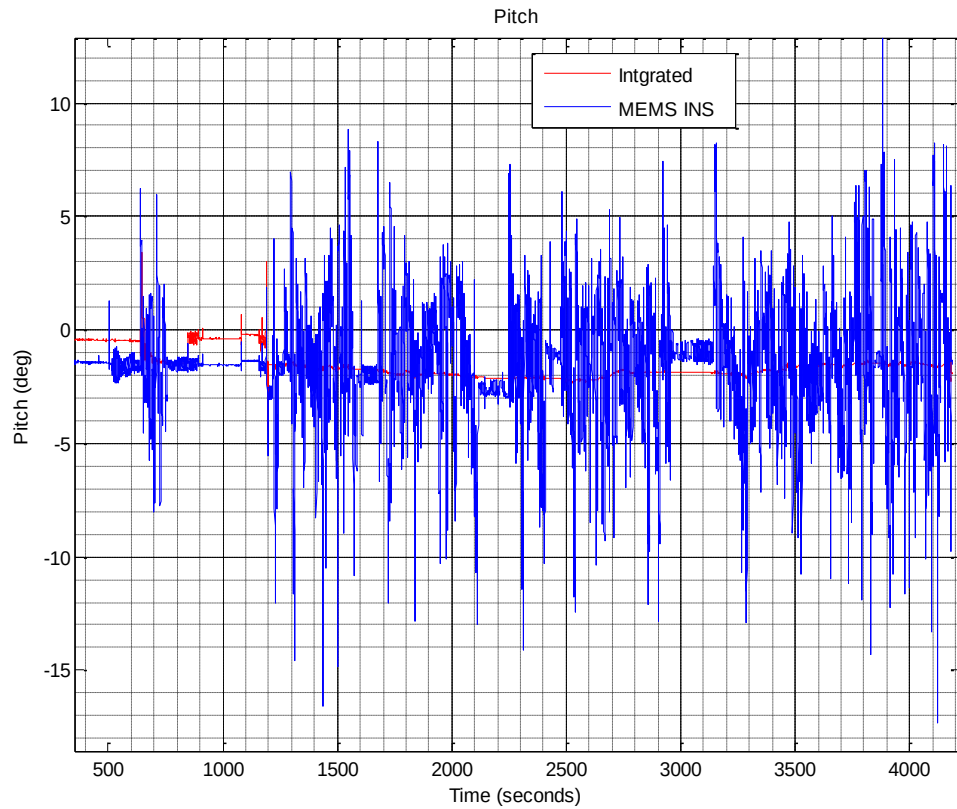


Figure 7.9 Comparison of pitch of MEMS INS and simple integrated

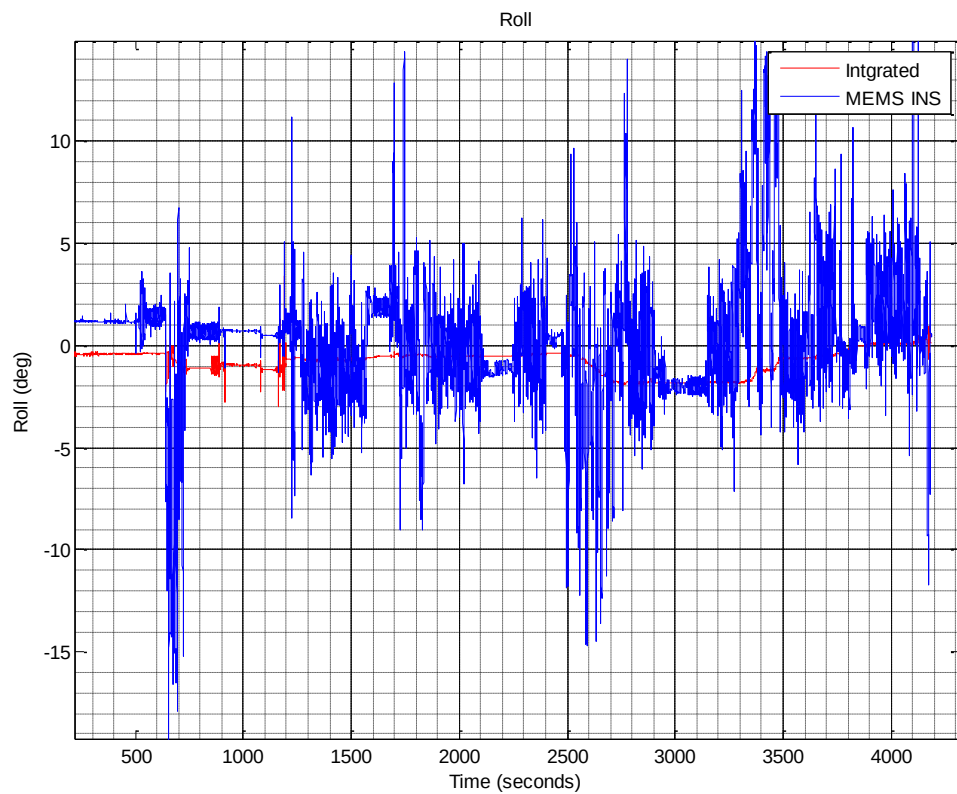


Figure 7.10 Comparison of roll of MEMS INS and simple integrated

7.2.5 Comparison of GNSS Outages and Corrected/Integrated Solution

The latitude and longitude will be zero when GNSS signal are obstructed. The corrected solution when GNSS signal are available is also given for comparison.

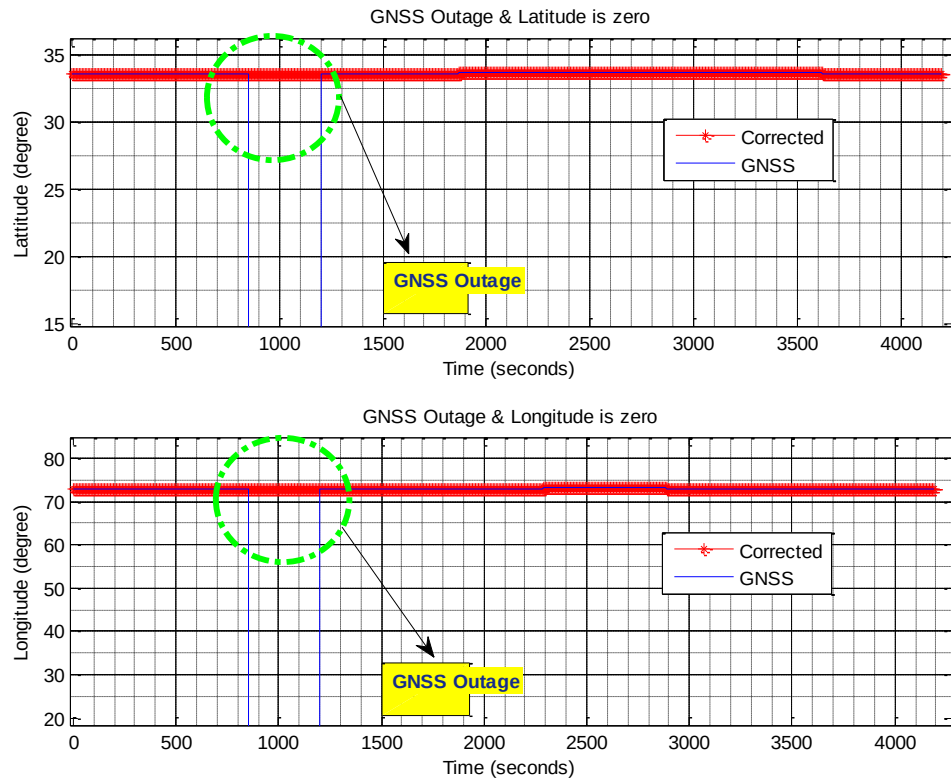


Figure 7.11 Comparison of Latitude & Longitude Error of Corrected and GNSS outages where Latitude & Longitude is zero

Simple integrated solution and GNS based solution is shown in the figure below. Clearly, it is seen that when GNSS signals are outage the GNSS solution deviated and simple integrated solution is based on MEMS INS only which have also drifts

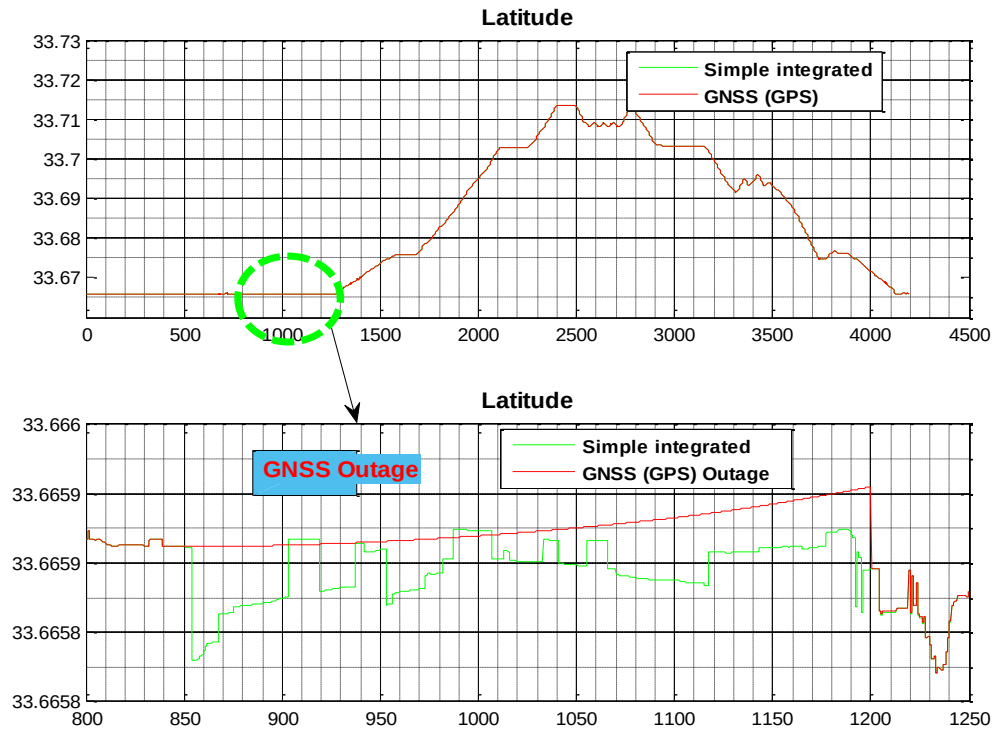


Figure 7.12 Comparison of Longitude error of Simple Integrated/ GNSS outage/ GNSS based solutions

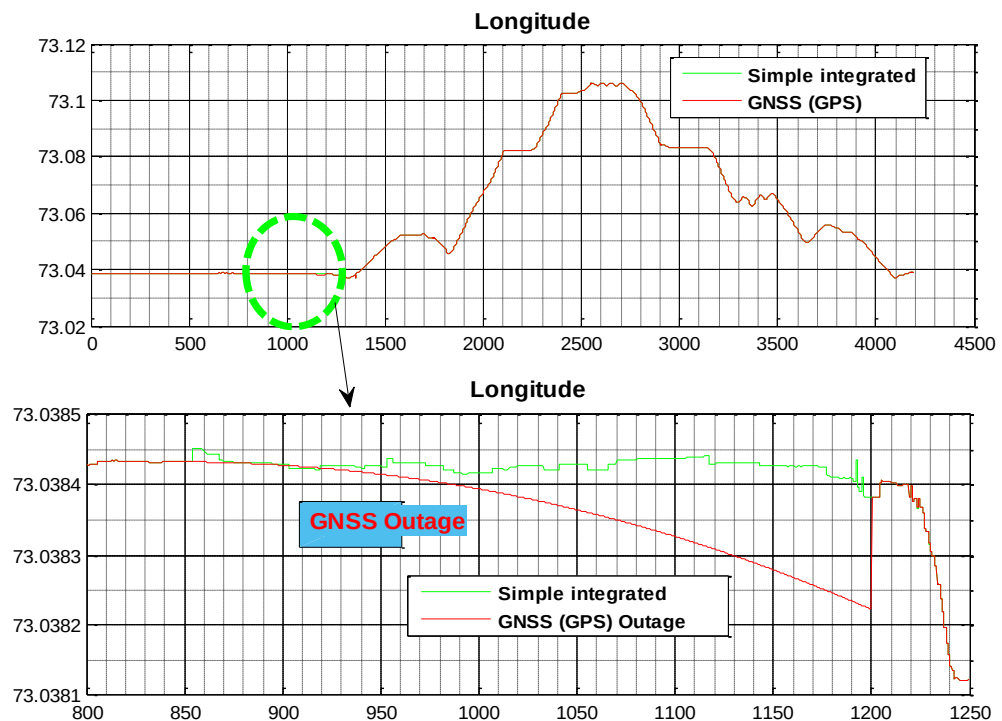


Figure 7.13 Comparison of Longitude error of Simple Integrated/ GNSS outage/ GNSS based solutions

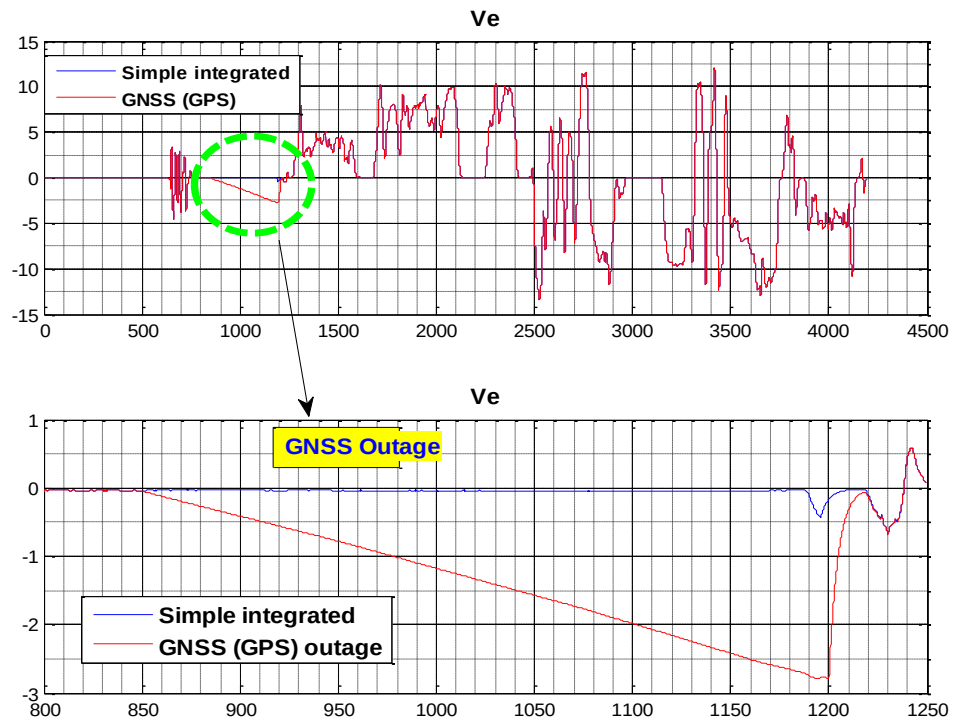


Figure 7.14 Comparison of East Velocity error of Simple Integrated/ GNSS outage/ GNSS based solutions

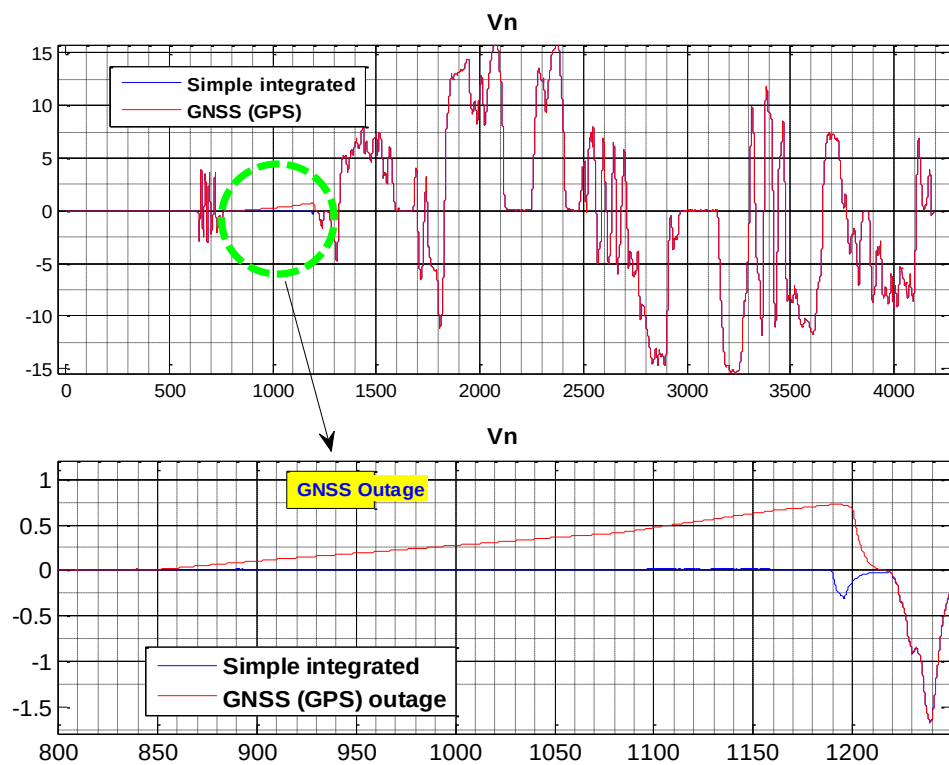


Figure 7.15 Comparison of North Velocity error of Simple Integrated/ GNSS outage/ GNSS based solutions

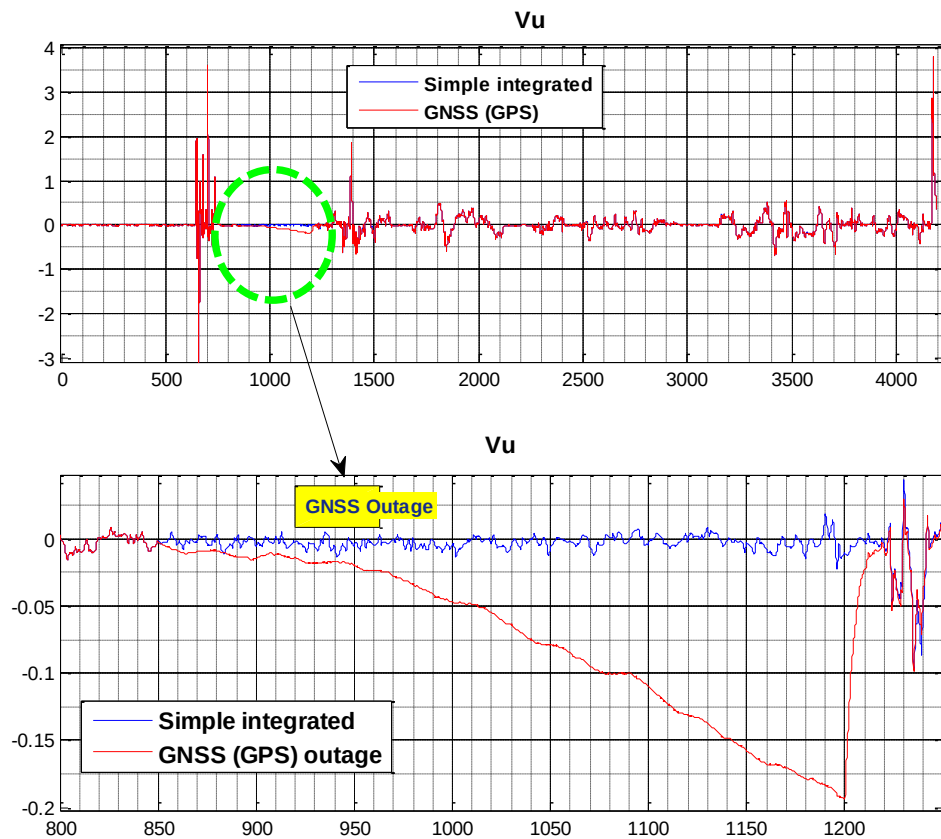


Figure 7.16 Comparison of Up Velocity Error of Simple Integrated/ GNSS outage/ GNSS based solutions

7.2.6 Simple Integrated, GNSS (GPS), GNSS Outage and Compensated with AHRS (Magnetometer) Navigation solution comparison

During integrated navigation, when GNSS signals outages occur, the navigation solution obtained is only due to MEMS INS. This navigation solution degraded due to unbounded errors. For this purpose Magnetometer is used for attitude and heading compensation. Following figures shows that when GNSS signals outages, the compensated navigation solution based on Magnetometer plays an effective role during integrated navigation. The MEMS INS errors are compensated by the using KF during MEMS INS/GNSS/AHRS-Magnetometer integrated navigation. The GNSS outage effect for position (lat & long), velocities and attitude is shown in the figures below.

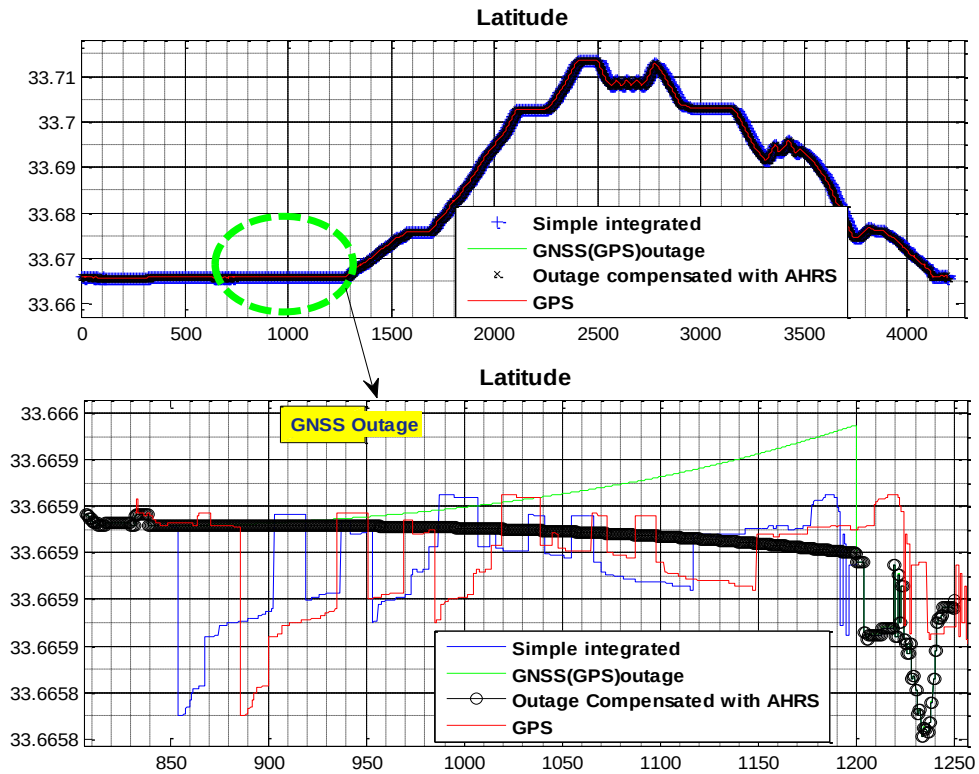


Figure 7.17 Comparison of Latitude errors of Simple integrated/GNSS outage/AHRS compensated solution

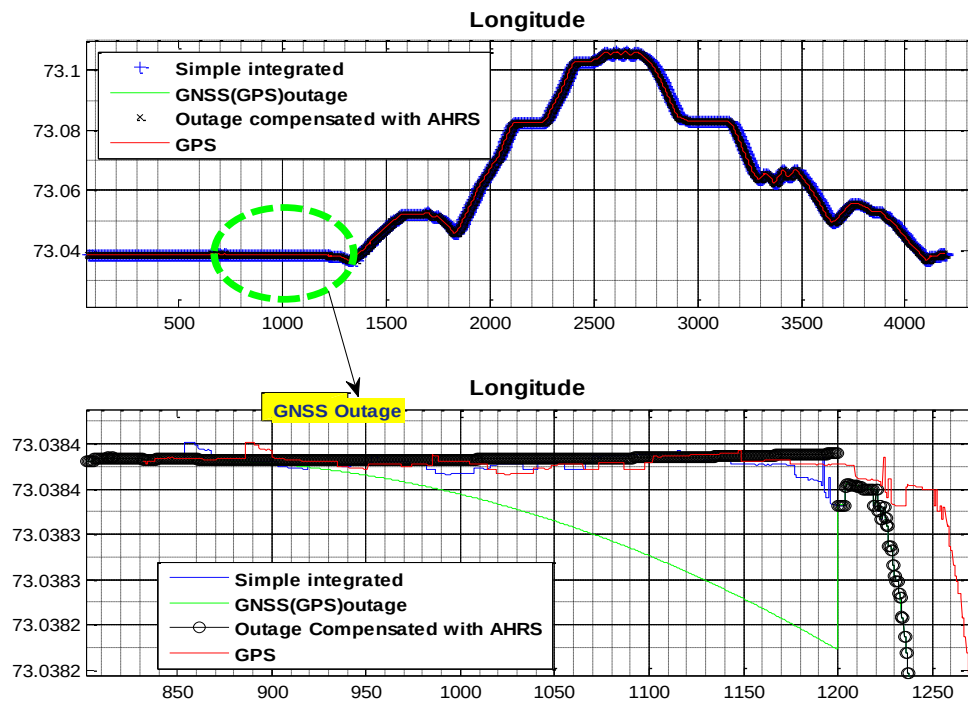


Figure 7.18 Comparison of Longitude errors of Simple integrated/GNSS outage/ AHRS compensated solution

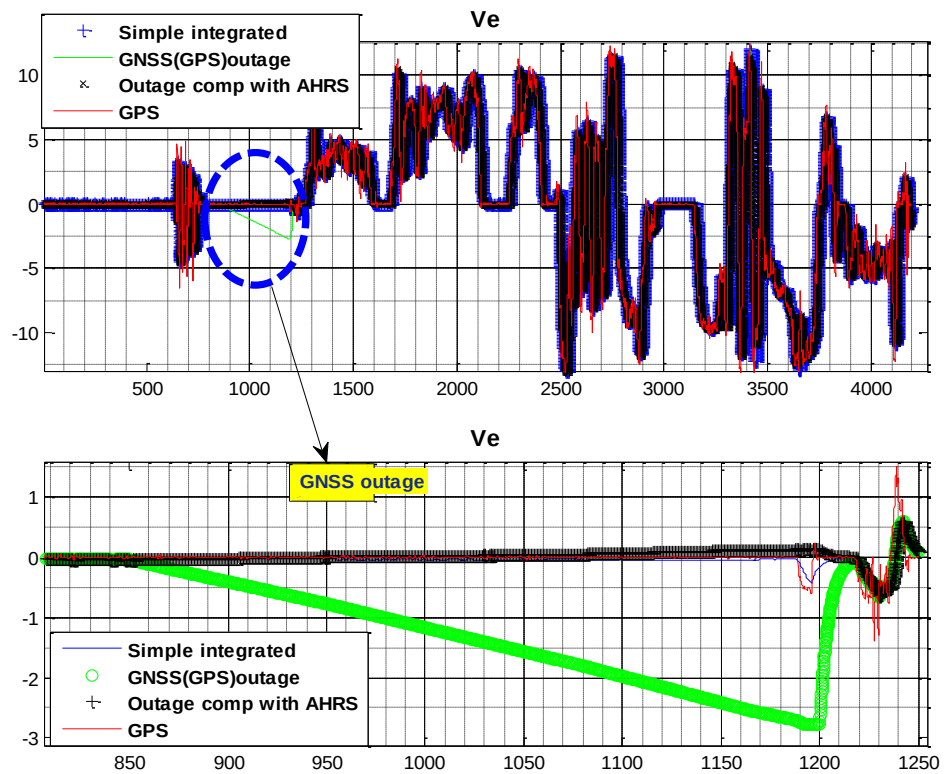


Figure 7.19 Comparison of East Velocity errors of Simple integrated/GNSS outage/AHRS compensated solution

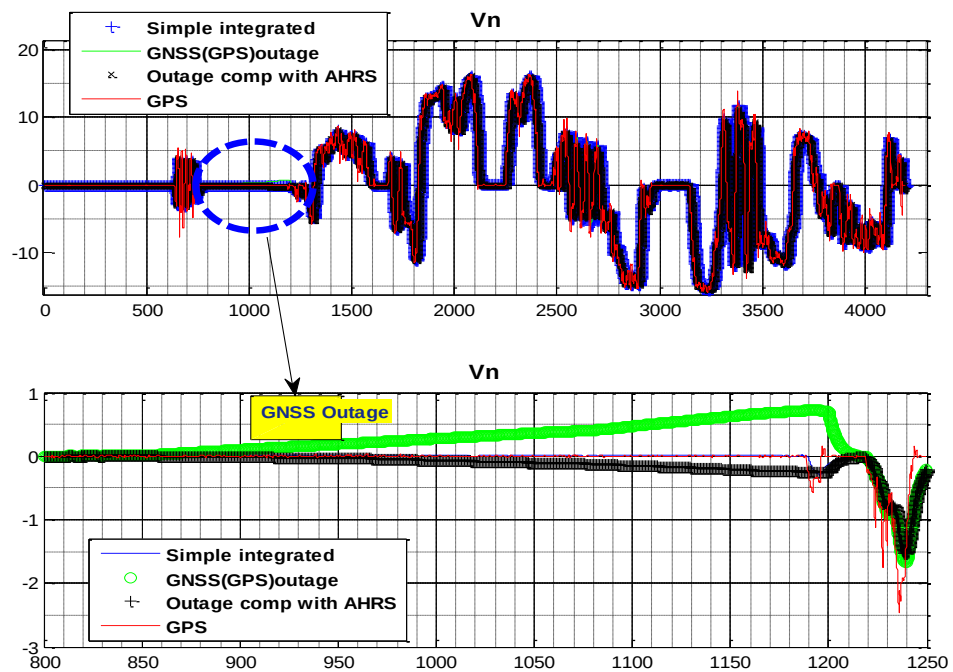


Figure 7.20 Comparison of North Velocity errors of Simple integrated/GNSS outage/AHRS compensated solution

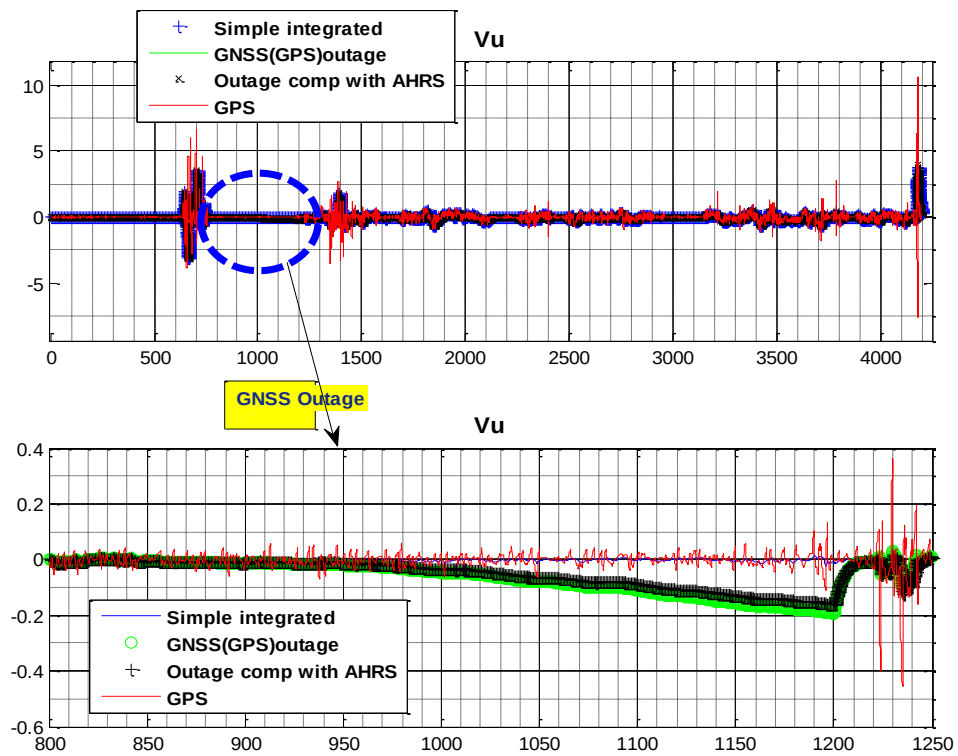


Figure 7.21 Comparison of Up Velocity errors of Simple integrated/GNSS outage/AHRS compensated solution

7.2.7 Attitude Error Comparison of MEMS INS-GNSS integrated and AHRS Compensated Solutions

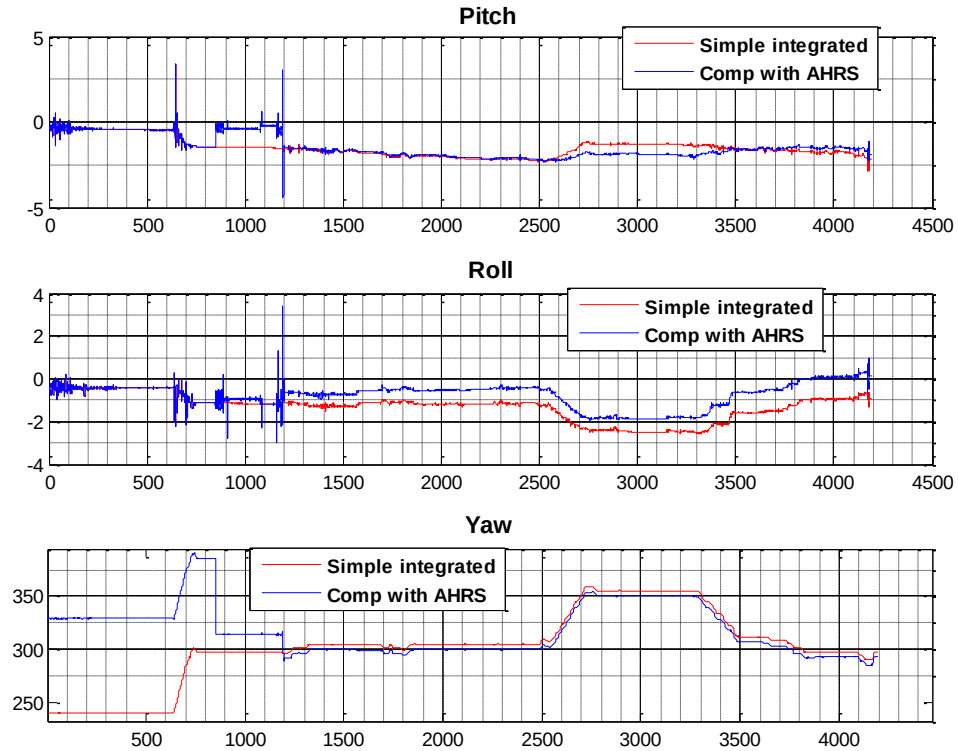


Figure 7.22 Attitude error Comparison of MEMS INS-GNSS integrated/AHRS compensated solutions

From figure 7.22, it can be inferred that the attitude errors of MEMS INS & GNSS integrated navigation solution and AHRS compensated navigation solution are minimal. Initially the pitch and roll error are negligible but slightly diverge with passage of time. In case of yaw the error between simply integrated and AHRS compensated navigation solution is initially diverge but with passage of time it converges.

8 CONCLUSIONS AND RECOMMENDATIONS

This chapter summarizes the research work conducted during the MS study based on MEMS INS/GNSS & AHRS-Magnetometer integrated navigation system for land navigation, draws conclusion from the theoretical development and test results presented in this dissertation.

8.1 Summary

The prime objective of this dissertation is to design and performance assessment of the integrated system comprises of low-cost inertial navigation system i.e. MEMS INS, GNSS and AHRS-Magnetometer for land vehicle navigation (LVN). The motivation behind the design of such a system is the use of MEMS INS & GNSS which gives optimal results in land navigation during GNSS outages, as GNSS may suffers from outages due to signal obstruction, multipath, jamming, spoofing and interference in urban canyon or dense foliage areas.

The main equipment used in this research, is low-price STIM based MEMS INS. Whereas, GNSS receiver can measure velocity of the user with an accuracy of 0.2%. Tactical grade reference Inertial Navigation Systems (Ref. INS) also used for performance comparison purpose. The software for comparison & analysis in post processing is developed in the MATLAB environment.

To assess the MEMS INS/ GNSS /AHRS-Magnetometer integrated performance, this research is done in the way that the performance analysis of MEMS INS/GNSS/AHRS integration for long time period. The error states including velocity, position and attitude are modelled by using Kalman filter. The filter is also augmented with sensor

stochastic errors and AHRS stochastic error. Considering the vehicle motion constraints, the velocity error measurement model for MEMS INS/AHRS integration through Kalman filter, is developed in navigation frame. The primary purpose of this evaluation is the design of integrated system to prevent the error degradation in INS which totally self-contained and can provide continuous navigation solution for land vehicle without external interruption.

In the urban environment, the GNSS signals are frequently interrupted from tall building and tunnels. Due to GNSS signals unavailability, the inertial sensor errors continuous estimation are not possible. Therefore, these errors rapidly grow during GNSS gap (especially for low cost MEMS INS). To limit the error growth during GPS gap for land vehicle, the velocity derived by use of AHRS improve the accuracy of navigation solution. Further, to improve azimuth estimation, the azimuth derived from the AHRS-Magnetometer is used in this research.

8.2 CONCLUSIONS

Following conclusions are drawn from the research presented in this thesis based on the objective set out in chapter 1.

- A medium accuracy MEMS INS/GNSS integrated system that satisfies the requirement of land vehicle navigation application was analysed and designed.
- The performance analysis of MEMS INS & GNSS simple integrated navigation system showing large errors during GNSS outage.
- A Method for Fast online Calibration of Magnetometer with small Space for Magnetometer calibration is implemented in this research.

- The performance of Magnetometer was analysed and its heading information was used during GNSS outages which minimize the navigation errors.
- Algorithms for the use of heading measurement from AHRS-Magnetometer were developed and implemented to achieve the accuracy improvement in position, velocity and heading of MEMS INS.
- The Extended Kalman (EK) filtering technique was used for fusion of data of MEMS INS/ GNSS & AHRS- Magnetometer.
- The attitude errors of MEM INS/ GNSS integrated system solution and AHRS compensated navigation solution were comparable. The AHRS can be used for MEMS INS/GNSS integrated system which improves the performance in term of position, velocity and azimuth during GNSS outages.
- Overall, the designed system increases the potential of using a low-priced MEMS INS/GNSS/AHRS-Magnetometer navigation system to provide the continuous and reliable navigation solution for land vehicle applications.

8.3 Recommendations

The enhancement of navigation accuracy for the low-cost MEMS inertial navigation systems integrated with GNSS in the signal outage/degradation environment is a challenging task. In this work, the use of AHRS-Magnetometer with integrated navigation system comprises of MEMS INS & GNSS to control the error growth during GNSS outage is investigated. Based on the research presented in this dissertation, following are the potential future recommendations.

1. For the integration of low-cost MEMS INS/ GNSS/ AHRS-Magnetometer, the 15 states Kalman filter is designed in which only inertial sensors constant drifts are modelled. In order to improve the systems accuracy, scale factor and misalignment sensor errors can be augmented as extra states to investigate the system performance.
2. AHRS (attitude and heading reference system) can also be utilized as added system during MEMS INS / GNSS integrated navigation. The performance of AHRS was analysed in this research, which improves the performance in term of position, velocity and azimuth during GNSS outages.

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